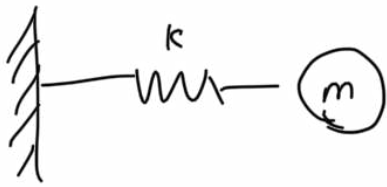


Wave Lecture 1 (20210831)

Lec 1



① Newtonian mechanics. $F = ma \Rightarrow m\ddot{x} + kx = 0$.

② Lagrangian mechanics. based on Hamilton principle

Hamilton principle



[The path of a particle, $x(t)$, is determined such that the function $I(x, \dot{x}) = \int_{t_1}^{t_2} L(t, x, \dot{x}) dt$ has an extremum]

For I to have extremum, $(L = T - V)$.

$$\left[\frac{\partial L}{\partial x} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = 0 \right] : \text{Euler-Lagrange equation}$$

$$\underline{L = T - V} ; T = \frac{1}{2} m (\dot{x})^2, V = \frac{1}{2} k x^2$$

$$L = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} k x^2$$

$$m\ddot{x} + kx = 0$$

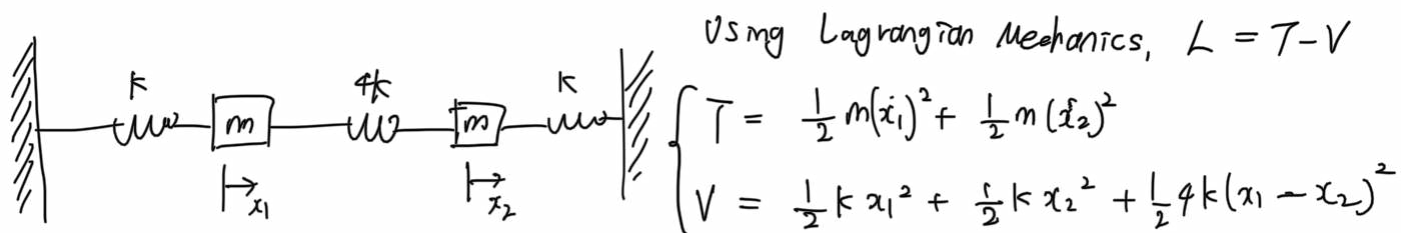
$$- \nabla V = \text{force.}$$

$$\Rightarrow - \frac{dV}{dx} = -kx.$$

$$\Rightarrow V = \frac{1}{2} k x^2 + C.$$

Wave Lecture 2 (20190902)

Lec 2 <Review of "Vibration">



⇒ E.O.M by Lagrangian Mechanics. (Euler-Lagrange Eq.)

$$\left. \begin{aligned} \frac{\partial L}{\partial x_i} - \frac{\partial}{\partial t} \left(\frac{\partial L}{\partial \dot{x}_i} \right) &= 0 \\ i &= 1, 2 \end{aligned} \right\} \Rightarrow \begin{aligned} -kx_1 - 4k(x_1 - x_2) - \frac{d}{dt}(m\dot{x}_1) &= 0 \\ -kx_2 - 4k(x_2 - x_1) - \frac{d}{dt}(m\dot{x}_2) &= 0 \end{aligned}$$

$$\Rightarrow \begin{cases} -5kx_1 + 4kx_2 - m\ddot{x}_1 = 0 \\ -5kx_2 + 4kx_1 - m\ddot{x}_2 = 0 \end{cases} \Rightarrow \ddot{\vec{X}} + \left(\sqrt{\frac{k}{m}} \right)^2 (\sqrt{A})^2 \vec{X} = 0 \quad A = \begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix}$$

? ↓

< Functions of Matrices > ($\sqrt{A}$?)

* Characteristic polynomial of A : $P_A(\lambda) = \det(\lambda I - A)$.

Suppose $\sqrt{\lambda} = f(\lambda) = \underbrace{P_A(\lambda)}_{\text{our Goal}} \cdot \underbrace{g(\lambda)} + \underbrace{r(\lambda)}$

Cayley-Hamilton Theorem : $P_A(A) = \vec{0}$

$$\Rightarrow \sqrt{A} = f(A) = r(A)$$

[E.g.] $A = \begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix}$

$$\begin{aligned} \sqrt{1} &= f(1) = \cancel{P_A(1)}^0 g(1) + (a \cdot \lambda + b) \Big|_{\lambda=1} \\ \sqrt{9} &= f(9) = \cancel{P_A(9)}^0 g(9) + (a \cdot \lambda + b) \Big|_{\lambda=9} \end{aligned}$$

* $r(\lambda)$ is 1st order $\because P_A(\lambda)$ is 2nd order

$$\Rightarrow r(\lambda) = \frac{1}{4}\lambda + \frac{3}{4} \Rightarrow \sqrt{A} = \frac{1}{4}A + \frac{3}{4}I = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

< Homework #1 >

what is $\sin \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ and $\cos \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$? $A = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

① sin

$$P_A(\lambda) = \lambda^2 \Rightarrow \sin \lambda = f(\lambda) = P_A(\lambda)q(\lambda) + r(\lambda) = \lambda^2 q(\lambda) + C\lambda + D$$

$$\Rightarrow \cos \lambda = 2\lambda q(\lambda) + \lambda^2 q'(\lambda) + C$$

$$\Rightarrow D=0, C=1 \Rightarrow f(A) = \sin \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

② cos

$$P_A(\lambda) = \lambda^2 \Rightarrow \cos \lambda = f(\lambda) = P_A(\lambda)q(\lambda) + r(\lambda) = \lambda^2 q(\lambda) + C\lambda + D$$

$$\Rightarrow -\sin \lambda = 2\lambda q(\lambda) + \lambda^2 q'(\lambda) + C$$

$$\Rightarrow C=0, D=1 \Rightarrow f(A) = \cos \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

Solution form of given example. : $\vec{X}(t) = \cos \underbrace{\begin{pmatrix} 2\omega_0 t & -\omega_0 t \\ -\omega_0 t & 2\omega_0 t \end{pmatrix}}_P \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$

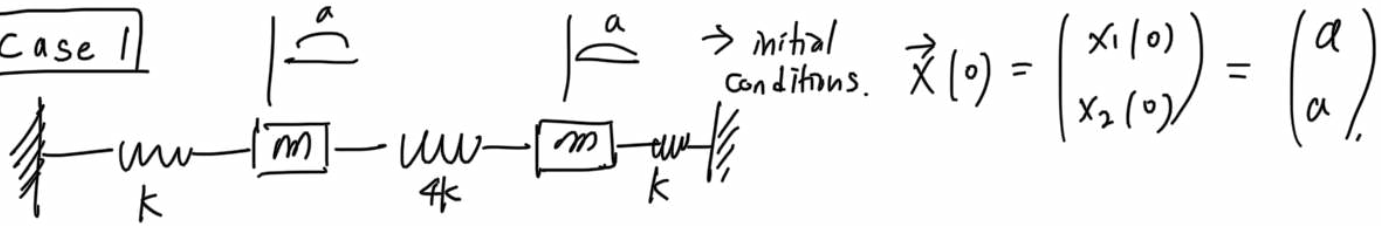
$$p_p(\lambda) = \det(\lambda I - B) = (\lambda - \omega_0 t)(\lambda - 3\omega_0 t)$$

$$\cos \lambda = f(\lambda) = p_p(\lambda) q(\lambda) + r(\lambda) = (\lambda - \omega_0 t)(\lambda - 3\omega_0 t) q(\lambda) + p\lambda + Q$$

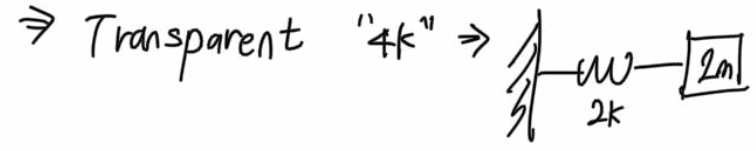
$$\begin{aligned} \Rightarrow \cos \omega_0 t &= \omega_0 t \cdot p + Q & \Rightarrow p &= \frac{\cos 3\omega_0 t - \cos \omega_0 t}{2\omega_0 t} \\ \cos 3\omega_0 t &= 3\omega_0 t \cdot p + Q & Q &= \frac{1}{2} (3\cos \omega_0 t - \cos 3\omega_0 t) \end{aligned}$$

Wave Lecture 3 (20210907)

Case 1



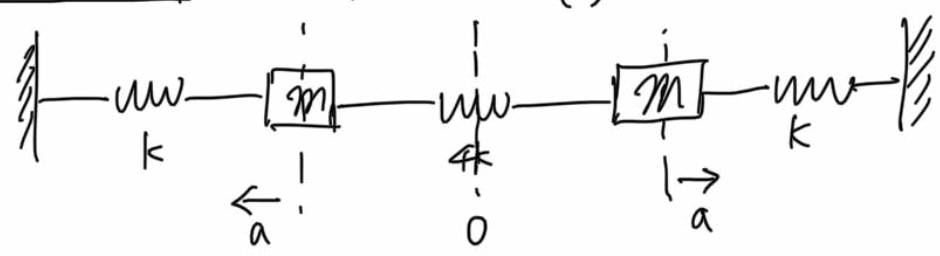
⇒ Intuition: No force by "4k" spring. (two masses synchronized)



< In-phase movement >

⇒ Frequency = $\omega_1 = \sqrt{k/m} = \omega_0$

Case 2 : $x_1(0) = a, x_2(0) = -a$



< out of phase movement >

⇒ Intuition: Symmetric movement via point "0"



⇒ Frequency = $\omega_2 = \sqrt{(4k \cdot 2 + k)/m} = 3\sqrt{k/m} = 3\omega_0$

HW

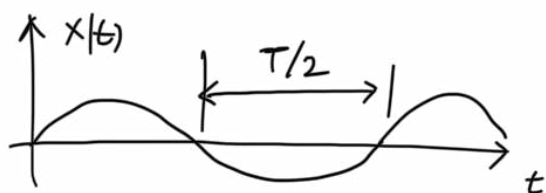
Q) Can we realize higher frequencies just by adjusting Initial conditions?

A) No. Why?

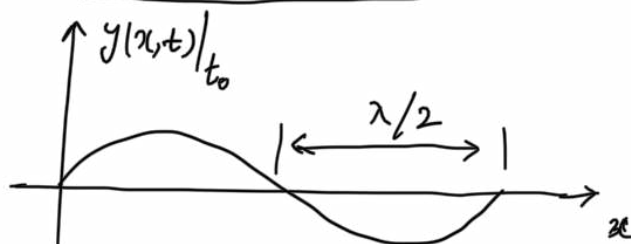
∞ DOF : lattice system.

Terminologies.:

Temporal terms



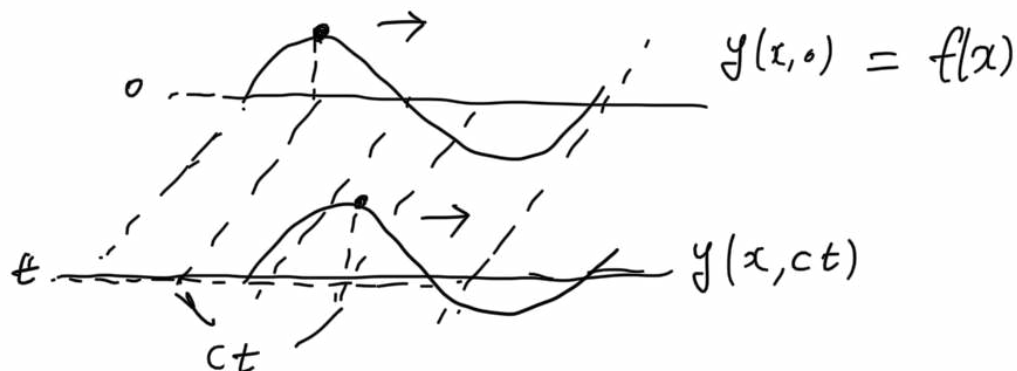
Spatial terms



1) T (period)



λ (wavelength)



$$\Rightarrow y(x,ct) = y(x-ct,0) = \underline{\underline{f(x-ct)}}$$

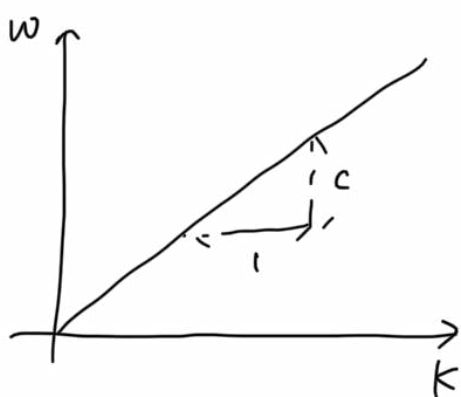
"Phase" $\leftarrow \begin{cases} f(x-ct) : \vec{x} \text{ direction propagation} \\ f(x+ct) : -\vec{x} \text{ " " "} \end{cases}$

$$c = \frac{\lambda}{T} = \frac{\omega}{2\pi} \cdot \lambda = \omega/k$$

2) ω (angular frequency)

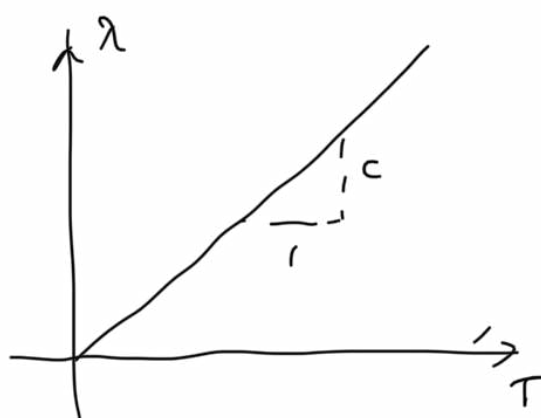


k is # of wave in 2π (unit) length
 $k = 2\pi/\lambda$



" k - ω curve": Dispersion Curve.

$\approx$



Wave Lecture 4 (20210909)

1D Wave Equation

① For $f(kx - \omega t)$: right going

$$\frac{\partial f}{\partial t} = \frac{\partial \phi}{\partial t} \frac{\partial f}{\partial \phi} = (-\omega) f'$$

$$\Rightarrow k f' c - \omega f' = 0$$

$$\frac{\partial f}{\partial x} = \frac{\partial \phi}{\partial x} \frac{\partial f}{\partial \phi} = k f'$$

$$\Rightarrow \left[\frac{\partial}{\partial t} + c \frac{\partial}{\partial x} \right] f = 0$$

② For $f(kx + \omega t)$: left going.

$$\frac{\partial f}{\partial t} = \omega f' \Rightarrow k f' c - \omega f' = 0$$

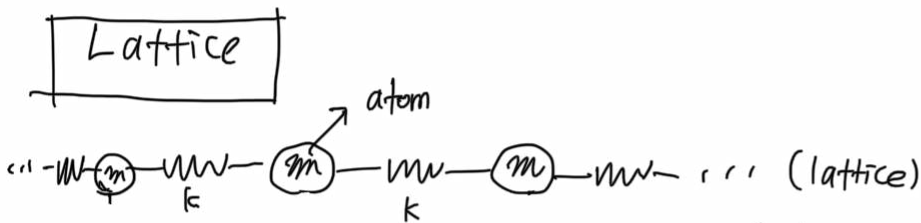
$$\frac{\partial f}{\partial x} = k f' \Rightarrow \left[\frac{\partial}{\partial t} - c \frac{\partial}{\partial x} \right] f = 0$$

From ① and ②,

$$\left(\frac{\partial}{\partial t} + c \frac{\partial}{\partial x} \right) \left(\frac{\partial}{\partial t} - c \frac{\partial}{\partial x} \right) f = 0$$

$$\Rightarrow \left(\frac{\partial^2}{\partial t^2} - c^2 \frac{\partial^2}{\partial x^2} \right) f = 0 \Leftrightarrow \left(\frac{\partial^2}{\partial x^2} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) f = 0$$

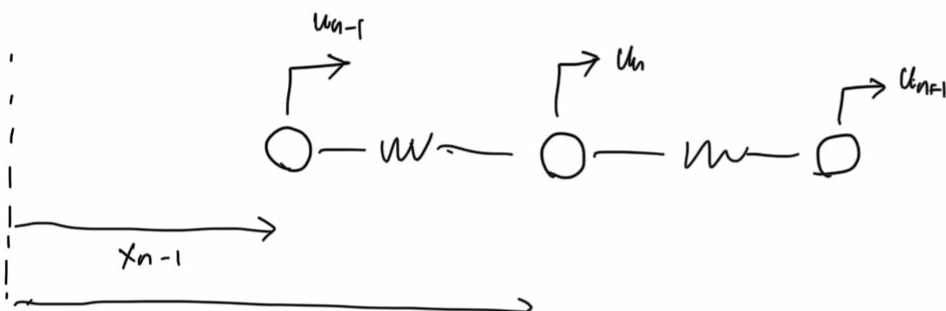
1 Dimensional Wave Equation.



types of mass

1. Mono
2. Di
3. Tri

< Monoatomic lattice >





E.O.M $m \ddot{u}_n = +k(u_{n+1} - u_n) - k(u_n - u_{n-1}) = -k(2u_n - u_{n-1} - u_{n+1})$

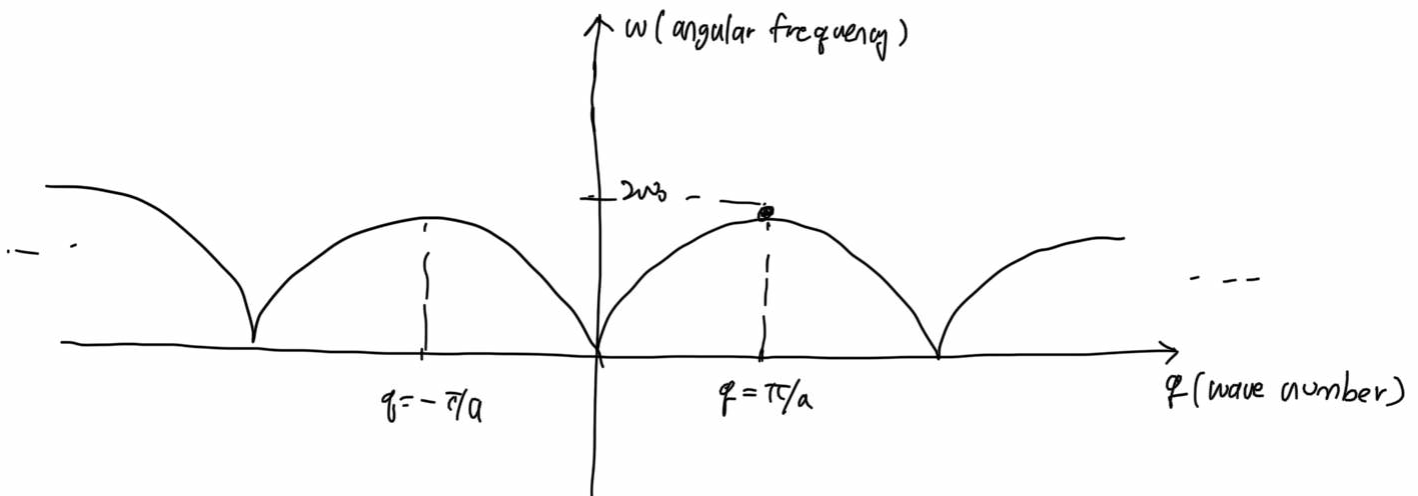
Let $u_n = A e^{i(qx_n - \omega t)}$ (q = wavenumber, A is same for all $\because$ monatomic).

Since $x_n(t) = x_n + u_n(t)$, where x_n is equilibrium constant, $x_n = na$

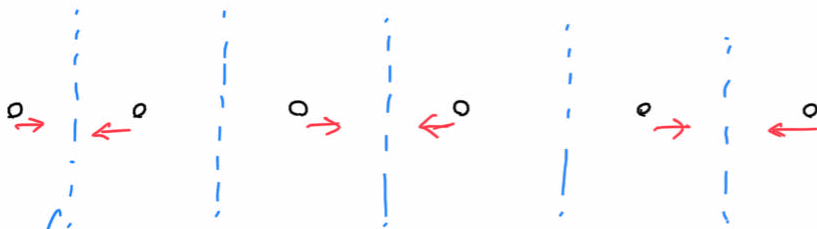
$$\Rightarrow -m\omega^2 e^{i qna} = +k(e^{i qna} \cdot 2 - e^{i q(n-1)a} - e^{i q(n+1)a})$$

$$\Rightarrow m\omega^2 = k(2 - 2\cos qa) = 4k(1 - \cos qa) = 4k \sin^2(qa/2)$$

$$\Rightarrow \omega = 2\sqrt{\frac{k}{m}} \left| \sin\left(\frac{qa}{2}\right) \right| = 2\omega_0 \left| \sin\left(\frac{qa}{2}\right) \right| \quad (\omega_0 = \sqrt{\frac{k}{m}})$$

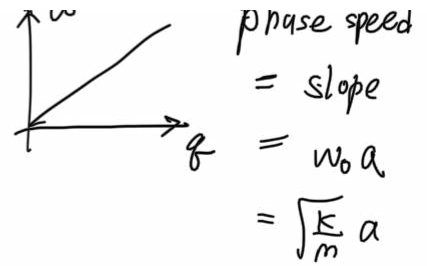


(1) when $q = \pi/a$, $u_n = A e^{i(qna - \omega t)} \rightarrow \phi_n$
 $u_{n+1} = A e^{i(q(n+1)a - \omega t)} \rightarrow \phi_{n+1}$ $\Rightarrow \phi_{n+1} - \phi_n = qa = \pi$
 $\Rightarrow \begin{cases} u_n = -u_{n+1} \\ u_n = -u_{n-1} \end{cases}$



nodal points $\rightarrow$ no transport of wave energy.

$$(2) \quad \frac{\partial \omega}{\partial q} a = 0 \Rightarrow \omega = 2\omega_0 \left| \sin\left(\frac{qa}{2}\right) \right| \approx qa\omega_0.$$



$$(3) \quad \max(\omega) = 2\omega_0 = \sqrt{\frac{4K}{m}} \quad Q) \text{ why } 4K?$$

$$(4) \quad \omega(q) = \omega(-q) \quad (\text{symmetricity}) : \left[\begin{array}{l} \text{right \& left going waves has the} \\ \text{same information} \end{array} \right]$$

Wave Lecture 5 (20210914)

Review of Compressible "Fluid Mechanics"

* Extensive & Intensive properties.

B $\xrightarrow{\text{normalized by mass}}$ β



	B	β
mass	m	1
momentum	$m\vec{v}$	$\vec{v}$
Energy	$E + \frac{1}{2}mV^2$	$e + \frac{1}{2}v^2$
	$\downarrow$ internal $\downarrow$ kinetic.	

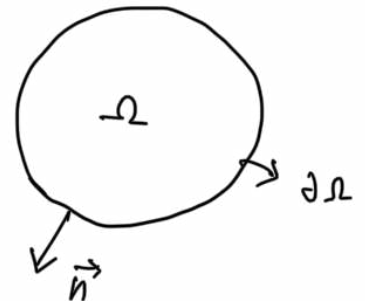
* Conservation Laws.

• Reynolds Transport Theorem (RTT)

The rate of change of an extensive property (B)

= time rate of change of B within C.V. + net rate of flux B through C.S.

$$\Rightarrow \frac{dB}{dt} = \underbrace{\int_{\Omega} \frac{\partial}{\partial t}(\rho\beta) dV}_{\text{control volume}} + \underbrace{\int_{\partial\Omega} (\rho\beta) \vec{v} \cdot \vec{n} dA}_{\text{Control surface.}}$$



$$\Rightarrow \frac{d}{dt} \int_{\Omega} (\rho\beta) dV = \int_{\Omega} \frac{\partial}{\partial t} (\rho\beta) dV + \int_{\partial\Omega} (\rho\beta) \underbrace{\vec{v}_b \cdot \vec{n}}_{\substack{\downarrow \\ \partial\Omega \text{ velocity}}} dA + \int_{\partial\Omega} (\rho\beta) \underbrace{\vec{v}_r \cdot \vec{n}}_{\substack{\downarrow \\ \text{fluid velocity w.r.t } \partial\Omega}} dA$$

$\vec{v}_b$ $\swarrow \searrow$ $\vec{v}_r$
 $\partial\Omega$ velocity fluid velocity w.r.t $\partial\Omega$

① Continuity ($B = m$)

$$\frac{dB}{dt} = \int_{\Omega} \frac{d}{dt}(\rho \cdot 1) dV + \int_{\partial\Omega} (\rho \cdot 1) \vec{u} \cdot \vec{n} dA = \int_{\Omega} \left(\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) \right) dV$$

$$\parallel \frac{dm}{dt} = 0 \quad \left(\because \text{mass of system is conserved} \right) \quad \left(\because \text{Gauss divergence theorem} \right)$$

$$\int_{\Omega} \nabla \cdot \vec{F} dV = \int_{\partial\Omega} \vec{F} \cdot \vec{n} dA$$

$$\Rightarrow \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) = 0 \quad \left(\because \text{if } \int_{\Omega} f dV = 0 \Rightarrow f = 0 \text{ by localization theorem} \right)$$

② Momentum. $B = m\vec{u}$, $\beta = \vec{u}$ $\vec{B} = (B_1, B_2, B_3)$

$$\frac{dB_i}{dt} = \int_{\Omega} \frac{d}{dt}(\rho \cdot u_i) dV + \int_{\partial\Omega} (\rho \cdot u_i) \vec{u} \cdot \vec{n} dA \quad \text{for } i = 1, 2, 3$$

$$\parallel \frac{d\vec{p}}{dt} = \vec{F}$$

(net force acting on $\partial\Omega$) = normal force + shear force

pressure is inwards

$$- \int_{\partial\Omega} p \vec{e}_i \cdot \vec{n} dA$$

$$\begin{cases} \vec{e}_1 = (1, 0, 0) \\ \vec{e}_2 = (0, 1, 0) \\ \vec{e}_3 = (0, 0, 1) \end{cases}$$

$$\Rightarrow \int_{\Omega} \frac{d}{dt}(\rho u_i) dV + \int_{\partial\Omega} [\rho u_i \vec{u} \cdot \vec{n} + p \vec{e}_i \cdot \vec{n}] dA = 0$$

$$\Rightarrow \int_{\Omega} \left[\frac{d}{dt}(\rho u_i) + \nabla \cdot (\rho u_i \vec{u}) + \nabla \cdot (p \vec{e}_i) \right] dV = 0$$

$$\Rightarrow \int_{\Omega} \left[\frac{d}{dt}(\rho u_i) + u_i \nabla \cdot (\rho \vec{u}) + \rho \vec{u} \cdot \nabla u_i + \frac{\partial p}{\partial x_i} \right] dV = 0$$

$$\left(\because \nabla \cdot (p \vec{e}_i) = \nabla \cdot (p, 0, 0) = \frac{\partial p}{\partial x_i} \right)$$

$$\Rightarrow \frac{\partial \rho}{\partial t} u_i + \rho \frac{\partial u_i}{\partial t} + u_i \nabla \cdot (\rho \vec{u}) + \rho \vec{u} \cdot \nabla u_i = - \frac{\partial p}{\partial x_i}$$

$\because$ Continuity equation.

$$\Rightarrow - \frac{\partial p}{\partial x_i} = \rho \frac{\partial u_i}{\partial t} + \rho \vec{u} \cdot \nabla u_i = \rho \left(\frac{\partial u_i}{\partial t} + \vec{u} \cdot \nabla u_i \right)$$

③ State equation. (thermodynamic process)

$$\underline{p = p(\rho, T)} \quad \text{or} \quad \underline{p = p(\rho, s)}$$

$$\underline{p - p_0} = \left. \frac{\partial p}{\partial \rho} \right|_{s=s_i} (\rho - \rho_0) + \cancel{\text{Taylor}} + \left. \frac{\partial p}{\partial s} \right|_{\rho=\rho_i} (s - s_0) + \cancel{\text{Taylor}}$$

||

$$p' = \left. \frac{\partial p}{\partial \rho} \right|_{s_0} \rho' + \underbrace{\left. \frac{\partial p}{\partial s} \right|_{\rho_0}}_{=0 \text{ for isentropic process}} s' \Rightarrow \underline{p' = \left. \frac{\partial p}{\partial \rho} \right|_{s_0} \rho'} \quad \text{①}$$

For adiabatic process,

$$p/\rho^\gamma = p_0/\rho_0^\gamma \Rightarrow \partial p / \partial \rho = \frac{p_0}{\rho_0^\gamma} \gamma \rho^{\gamma-1} = \left(p_0^\gamma \frac{\rho^\gamma}{\rho_0^\gamma} \right)^{1/\gamma} = \frac{\gamma p}{\rho} = \frac{\gamma R T}{\rho}$$

From ① $\Rightarrow \underline{p' = \gamma R T \rho'}$

Note: Isentropic process assumes ideal state from adiabatic process.

Note: $\sqrt{\gamma R T} = \text{speed of sound } (c)$

Wave Lecture 6 (20210916)

• Summary

$$\textcircled{1} \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0$$

$$\textcircled{2} \rho \left(\frac{\partial u}{\partial t} + u \cdot \nabla u \right) = -\nabla p$$

$$\textcircled{3} p' = (\sqrt{\gamma R T})^2 \rho' \quad (\text{isentropic process})$$

$$\begin{cases} \text{Linearization} \\ \rho = \rho_0 + \rho' \\ u = u_0 + u' \Rightarrow \end{cases}$$

<Linearized> • 속제 • 장의 장

$$\frac{\partial \rho'}{\partial t} + \rho_0 \nabla \cdot u' = 0 \quad \textcircled{1}$$

$$\Rightarrow (\rho_0 + \rho') \left(\frac{\partial u'}{\partial t} + u' \cdot \nabla u' \right) \approx \rho_0 \frac{\partial u'}{\partial t} = -\nabla p' \quad \textcircled{2}$$

< speed of sound : Laplace > (Isentropic)

$$\frac{\partial \textcircled{1}}{\partial t} \Rightarrow \frac{\partial^2 \rho'}{\partial t^2} + \rho_0 \frac{\partial}{\partial t} \nabla \cdot u' = 0$$

$$\Rightarrow \nabla^2 p' - \frac{\partial}{\partial t^2} \left(\frac{1}{\sqrt{\gamma R T}} \right)^2 p' = 0$$

$\nabla \cdot \nabla = \nabla^2$
Laplacian.

$$\nabla \cdot \textcircled{2} \Rightarrow \rho_0 \nabla \cdot \frac{\partial}{\partial t} u' = -\nabla \cdot (\nabla p')$$

$$\Rightarrow \nabla^2 p' - \left(\frac{1}{\sqrt{\gamma R T}} \right)^2 \frac{\partial^2}{\partial t^2} p' = 0$$

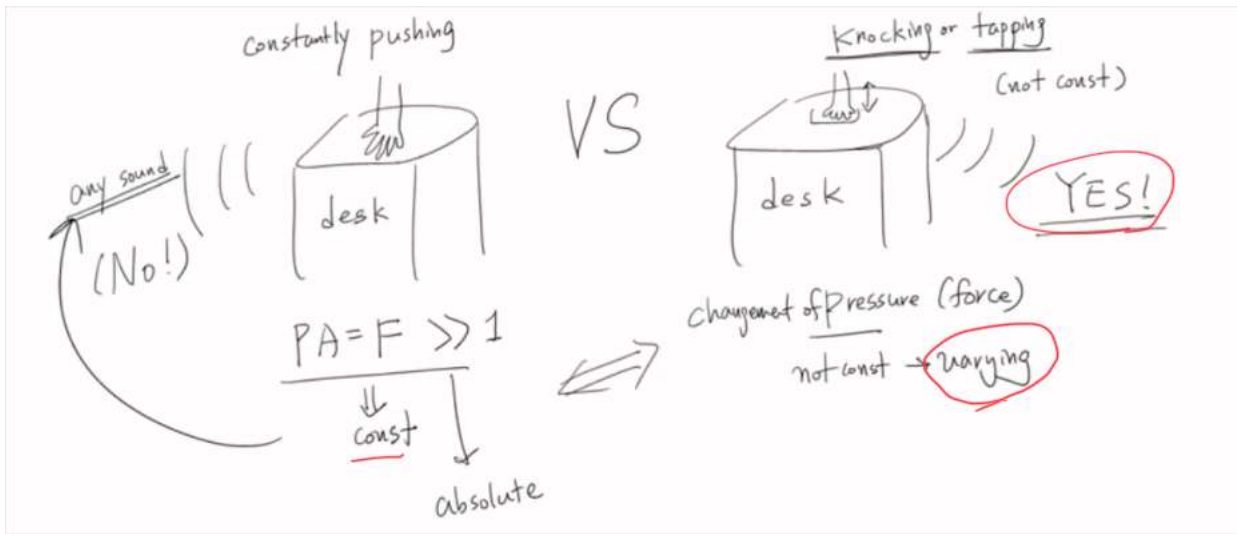
Waves in compressible fluid = "Sound" (Acoustics)

< Speed of sound : Newton > (Isothermal)

$$p = p(\rho, T) \Rightarrow p' = \left. \frac{\partial p}{\partial \rho} \right|_{T_0} \rho' + \left. \frac{\partial p}{\partial T} \right|_{\rho_0} T' \Rightarrow p' = \left. \frac{\partial p}{\partial \rho} \right|_{T_0} \rho'$$

$$\text{Since } p = \rho R T, \quad \partial p / \partial \rho = R T \Rightarrow p' = (\sqrt{R T_0})^2 \rho' \quad \sqrt{R T_0} \approx 290 \rightarrow \text{Wrong!}$$

Q1. What is sound?

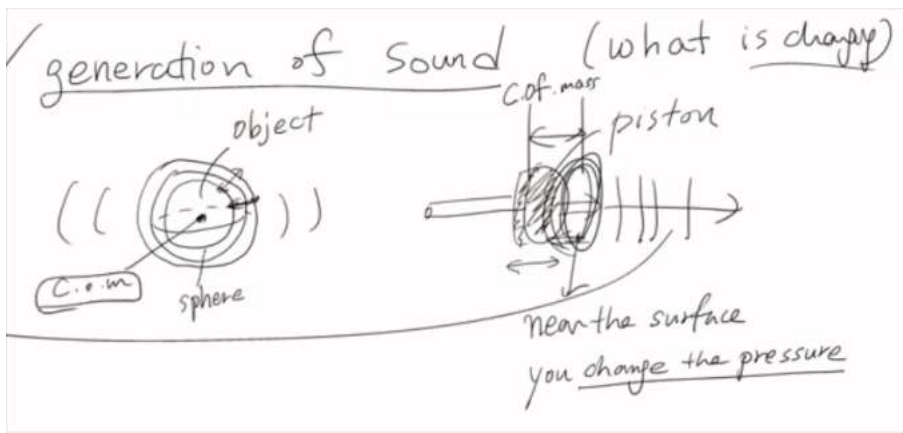


$$p = \underbrace{p_0}_{\text{mean}} + \underbrace{p'}_{\text{perturbed}} \rightarrow \text{Sound} = \text{Acoustic wave}$$

Q2. What Differential Equation governs the propagation of sound?

$$\left. \begin{array}{l} \text{Lin Cont.} \\ \text{Lin Euler} \\ \text{State Eq. (isentropic)} \end{array} \right\} \Rightarrow \underline{\nabla^2 p' - \frac{1}{c^2} \frac{\partial^2 p'}{\partial t^2} = 0} \quad \text{--- Eq (1)}$$

Q3. How does the sound propagate? : change.



Alternative way of expression.

$$\xi = x - ct \quad \text{and} \quad \eta = x + ct \quad (\text{for 1 dimension : X axis}).$$

$$\frac{\partial}{\partial x} = \frac{\partial \xi}{\partial x} \frac{\partial}{\partial \xi} + \frac{\partial \eta}{\partial x} \frac{\partial}{\partial \eta} = \frac{\partial}{\partial \xi} + \frac{\partial}{\partial \eta} \Rightarrow \frac{\partial^2}{\partial x^2} = \left(\frac{\partial}{\partial \xi} + \frac{\partial}{\partial \eta} \right) \left(\frac{\partial}{\partial \xi} + \frac{\partial}{\partial \eta} \right)$$

$$\frac{\partial}{\partial t} = \frac{\partial \xi}{\partial t} \frac{\partial}{\partial \xi} + \frac{\partial \eta}{\partial t} \frac{\partial}{\partial \eta} = c \left(\frac{\partial}{\partial \eta} - \frac{\partial}{\partial \xi} \right) \Rightarrow \frac{\partial^2}{\partial t^2} = c \left(\frac{\partial}{\partial \eta} - \frac{\partial}{\partial \xi} \right) c \left(\frac{\partial}{\partial \eta} - \frac{\partial}{\partial \xi} \right)$$

$$\Rightarrow \frac{\partial^2}{\partial x^2} = \frac{\partial^2}{\partial \xi^2} + 2 \frac{\partial^2}{\partial \xi \partial \eta} + \frac{\partial^2}{\partial \eta^2} \quad \text{and} \quad \frac{\partial^2}{\partial t^2} = c^2 \left(\frac{\partial^2}{\partial \eta^2} - 2 \frac{\partial^2}{\partial \xi \partial \eta} + \frac{\partial^2}{\partial \xi^2} \right)$$

$\Rightarrow$ From equation ①, assuming that $p'(x, t) = \tilde{p}(\xi, \eta)$,

$$\text{for 1 dimensional case, } \frac{\partial^2}{\partial x^2} p' = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} p'$$

$$\Rightarrow \left(\frac{\partial^2}{\partial \xi^2} + 2 \frac{\partial^2}{\partial \xi \partial \eta} + \frac{\partial^2}{\partial \eta^2} \right) \tilde{p}(\xi, \eta) = \frac{1}{c^2} \cdot c^2 \left(\frac{\partial^2}{\partial \eta^2} - 2 \frac{\partial^2}{\partial \xi \partial \eta} + \frac{\partial^2}{\partial \xi^2} \right) \tilde{p}(\xi, \eta)$$

$$\Rightarrow 4 \frac{\partial^2}{\partial \xi \partial \eta} \tilde{p}(\xi, \eta) = 0 \Rightarrow \frac{\partial^2}{\partial \xi \partial \eta} \tilde{p}(\xi, \eta) = 0$$

$\rightarrow$ Canonical form of wave equation
 $\Rightarrow$ Easy to solve PDE.

General solution form is, $\tilde{p}(\xi, \eta) = f(\xi) + g(\eta)$ (D'Alembert Solution).

Wave Lecture 7 (20210928)

Lec 07. Waves in Compressible Fluids (air)

Acoustic waves in the air
(sound)

D'Alembert solution of wave equation
in canonical form

original (1D)	canonical (1D)
$\frac{\partial^2}{\partial x^2} p' - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} p' = 0$	$\frac{\partial^2}{\partial \xi \partial \eta} \tilde{p}(\xi, \eta) = 0$ (PDE)

where $p' = p'(x, t)$

$$(x, y) \rightarrow (\xi, \eta)$$

$$\begin{matrix} x - ct & x + ct \\ (x, t) \end{matrix}$$

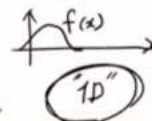
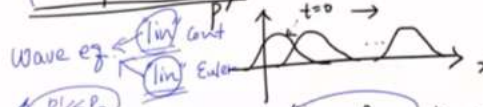
benefit

$$\tilde{p}(\xi, \eta) = f(\xi) + g(\eta)$$

for smooth (differentiable) functions f & g .

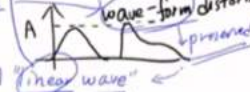
$$p'(x, t) = f(x - ct) + g(x + ct)$$

interpretation (i) — for $f(\cdot)$



same amplitude the wave-form is not changed in 1D propagation
(lossless medium)

1D "nonlinear" wave
beyond the scope
of ME119 [Note]



the amplitude of wave is not changing. vs lossy
[Note] Is the amplitude always preserved in 1D prop?
No, not always. "loss in the medium"

interpretation (2) D'Alembert sol'n

preparation ① "acoustic potential" ϕ

Recall the velocity potential Φ .

For an irrotational velocity field, $\nabla \times \mathbf{u} \equiv 0$ (irrotational)

$$\mathbf{u} = \mathbf{u}_{\text{irrot}} + \mathbf{u}_{\text{rot}}$$

There exists a scalar fn Φ such that $\mathbf{u}_{\text{irrot}} = \nabla \Phi$ scalar fn.

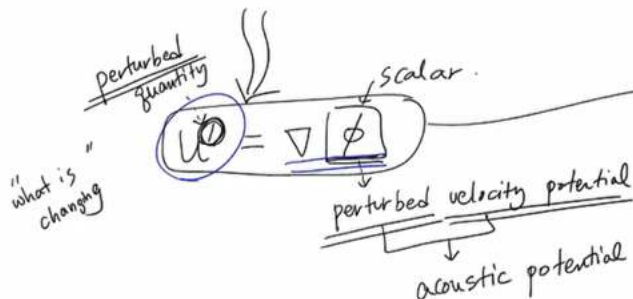
By using a vector identity,

$$\nabla \times \nabla \Phi \equiv 0$$

scalar fn
vector ∇ -vector

$$0 \equiv \nabla \times \mathbf{u}_{\text{irrot}} = \nabla \times \nabla \Phi \equiv 0$$

single scalar fn $\rightarrow$ express a vector field.



preparation ② "wave eq" in terms of ϕ not p'

(i) Linearized Euler eq.

$$\rho_0 \frac{\partial \mathbf{u}'}{\partial t} = -\nabla p' \quad (\mathbf{u}_0 = 0)$$

$$\mathbf{u}' = \nabla \phi$$

$$\rho_0 \frac{\partial}{\partial t} (\nabla \phi) = -\nabla p'$$

$$\nabla \left(\rho_0 \frac{\partial \phi}{\partial t} \right) = \nabla (-p')$$

$$\Rightarrow p' = -\rho_0 \frac{\partial \phi}{\partial t}$$

relation btw p' & ϕ

$$\mathbf{u}' = \nabla \phi$$

relation btw $\mathbf{u}'$ & ϕ

solution ϕ

(ii) Linearized continuity eq.

$$\frac{\partial p'}{\partial t} + \rho_0 \nabla \cdot \mathbf{u}' = 0$$

state eq.

Wave eq.

$$\nabla^2 p' - \frac{1}{c^2} \frac{\partial^2 p'}{\partial t^2} = 0$$

$$\rightarrow \frac{1}{c^2} \frac{\partial}{\partial t} \left(-\rho_0 \frac{\partial \phi}{\partial t} \right) + \rho_0 \nabla \cdot \nabla \phi = 0$$

$$\rightarrow \nabla^2 \phi - \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} = 0$$

wave eq. in terms of ϕ

interpretation of D'Alembert solution (1D)

$$u' = \nabla \phi$$

$$p' = -\rho_0 \frac{\partial \phi}{\partial t}$$

"1D"

$$u' = \frac{\partial \phi}{\partial x} = \frac{\partial f(x-ct)}{\partial x}$$

$$p' = -\rho_0 \frac{\partial \phi}{\partial t} = -\rho_0 \frac{\partial f(x-ct)}{\partial t}$$

Combining

$$- \rho_0 \frac{\partial f(x-ct)}{\partial t} = -\rho_0 \frac{\partial f}{\partial \xi} \frac{\partial \xi}{\partial t} = -\rho_0 \frac{\partial f}{\partial \xi} (-c) = \rho_0 c \frac{\partial f}{\partial \xi}$$

$$\frac{\partial \xi}{\partial t} = -c$$

$$\frac{\partial \xi}{\partial x} \frac{\partial f(\xi)}{\partial \xi} = 1 \cdot f'(\xi)$$

$$\phi = \frac{1}{2} (f(x-ct) + g(x+ct))$$

"right-going" wave

left-going

$$u' = f'(\xi)$$

$$p' = \rho_0 c f'(\xi)$$

HW: Find relation btw p', u' for leftgoing wave.

$$u' = \frac{\partial \phi}{\partial x} = \frac{\partial g}{\partial x}(x+ct)$$

$$p' = -\rho_0 \frac{\partial \phi}{\partial t} = -\rho_0 c g'$$

$$\left. \begin{array}{l} u' = \frac{\partial \phi}{\partial x} \\ p' = -\rho_0 \frac{\partial \phi}{\partial t} \end{array} \right\} \frac{\partial \eta}{\partial t} = c$$

$$\underline{Z = -\rho_0 c.}$$

$$\Rightarrow \frac{p'}{u'} = \rho_0 c$$

phy. var. "material" properties

area

$$\frac{P'}{U} = \text{acoustic impedance}$$

recall $\frac{F}{U} = \text{"mechanical impedance"}$

low resistive

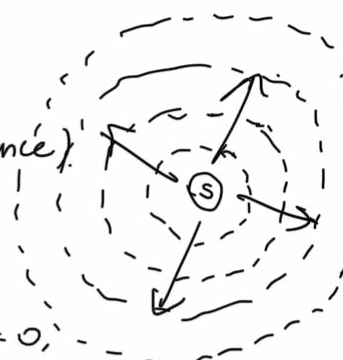
< Concentrate on sound propagation. >

① 1D propagation.

$$p'(x,t) = f(x-ct) + g(x+ct) \quad \text{and} \quad \phi(x,t) = f(x-ct) + g(x+ct)$$

② 3D propagation.

$$p'(r, \theta, \phi, t) = p'(r, t) : \text{function of } r, t \text{ (radial distance)}$$



$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) p' = 0 \xrightarrow{\text{Cartesian}} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) p' = 0$$

$$\xrightarrow{\text{spherical}} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} p' = 0$$

< Homework > : derivation.

w.e in 3 dim : $\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) p'(r,t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} p'(r,t) = 0$

$$\Rightarrow \frac{1}{r^2} \left(2r \frac{\partial}{\partial r} \right) p' + \frac{\partial^2}{\partial r^2} p' - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} p' = 0$$

$$\Rightarrow \frac{\partial^2}{\partial r^2} p' + \frac{2}{r} \frac{\partial}{\partial r} p' - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} p' = 0$$

$$\Rightarrow \frac{1}{r} \left(r \frac{\partial^2}{\partial r^2} + 2 \frac{\partial}{\partial r} \right) p' - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} p' = 0$$

$$\Rightarrow \frac{1}{r} \frac{\partial}{\partial r} \left(\frac{\partial}{\partial r} (r p') \right) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} p' = 0 \Rightarrow \frac{\partial^2}{\partial r^2} (r p') - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} (r p') = 0$$

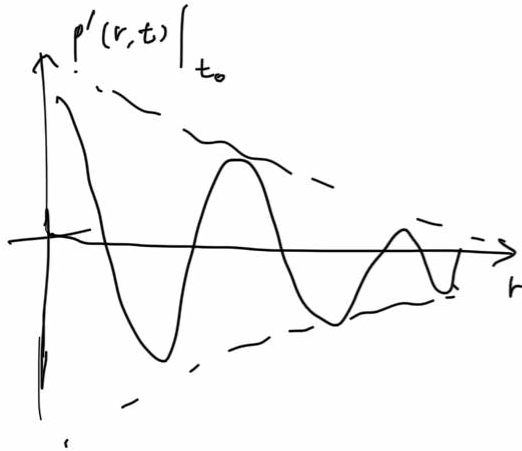
1 dimensional equation respect to $r p'$ D'Alembert

$$\Rightarrow r p' = f(r-ct) + g(r+ct)$$

$$\Rightarrow p' = \frac{f(r-ct)}{r} + \frac{g(r+ct)}{r}$$

far away from origin
 $\Rightarrow$ outgoing wave

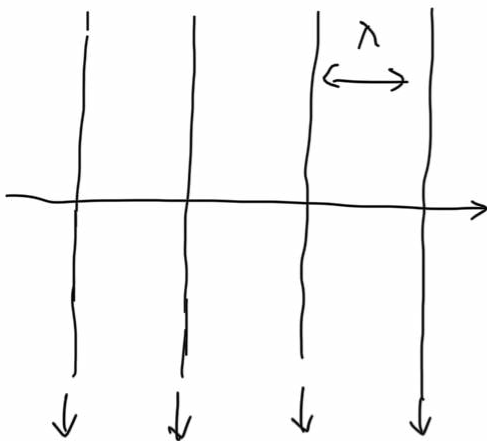
closer to origin.
 $\Rightarrow$ in-coming wave.



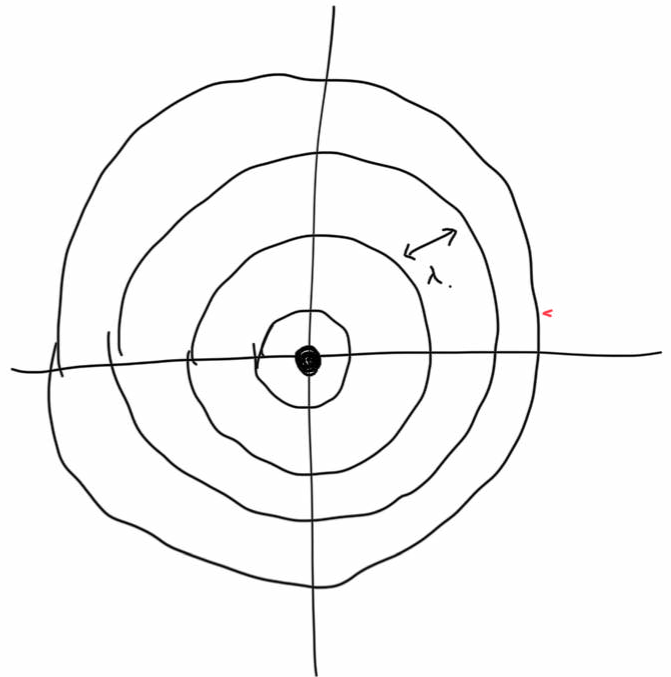
1. Wave-form is preserved but scaled ($1/r$)
2. Amplitude is changing depending on "r"

Problem

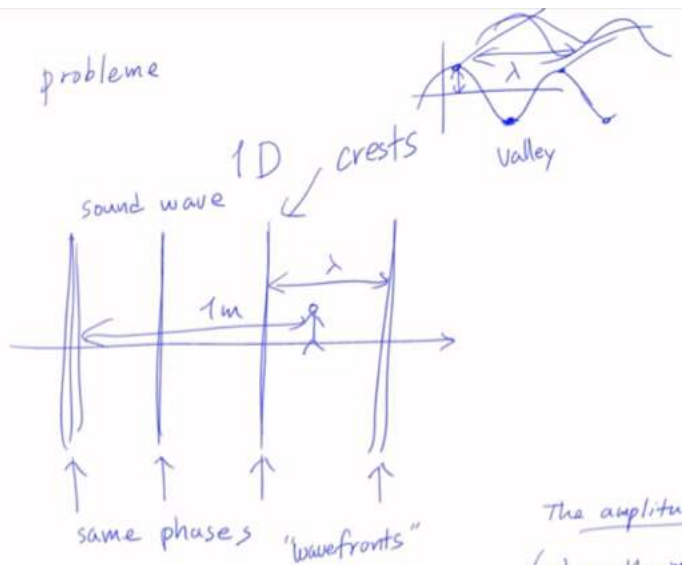
1D
 sound wave.



same phases : wave fronts.



probleme



The amplitude of pressure perturbation at a distance 1m from a sound source (where the medium is air) was 1 N/m^2 and the frequency of the wave was 500 Hz.

Calculate the amplitude of particle velocity at 0.1m from the source

$$\phi = \frac{f(r-ct)}{r} + \frac{g(r+ct)}{r}$$

$$p' = \frac{h(r-ct)}{r} \quad \text{form.}$$

Linearized equations

$$\frac{\partial \phi'}{\partial t} + \rho_0 \nabla \cdot u = 0 \quad (\text{continuity})$$

$$\begin{cases} p' = -\rho_0 \frac{\partial \phi}{\partial t} \\ u' = \nabla \phi \end{cases}$$

$$\Rightarrow \frac{1}{c^2} \frac{\partial p'}{\partial t} + \rho_0 \nabla \cdot (\nabla \phi) = 0$$

$$\Rightarrow \underline{\nabla^2 \phi - \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} = 0}$$

$$u' = \nabla \left(\frac{f(r-ct)}{r} \right) \quad \text{and} \quad p' = -\rho_0 \frac{\partial}{\partial t} \left(\frac{f(r-ct)}{r} \right) \quad (\text{assume right-going wave})$$

$$\Rightarrow u' = \frac{\partial}{\partial r} \left(\frac{f(r-ct)}{r} \right) \hat{r} \quad (\text{no components in } \theta, \varphi).$$

$$= \frac{f'(r-ct) \cdot r - f(r-ct)}{r^2} \hat{r} = \frac{r}{r} \frac{\partial f}{\partial r}(r-ct) - \frac{1}{r^2} f(r-ct).$$

$$p' = -\rho_0 \frac{1}{r} (-c) f'(r-ct) = \frac{\rho_0 c}{r} \frac{\partial f}{\partial t}(r-ct)$$

① Since $f(x) = e^{jkx}$ so that $f(r-ct) = \underline{e^{jk(r-ct) - \omega t}}$ $kc = \omega$

$$\Rightarrow u' = \frac{1}{r} jk f(r-ct) - \frac{1}{r^2} f(r-ct) \quad \text{and} \quad p' = \frac{\rho_0 c}{r} jk f(r-ct)$$

$$\Rightarrow \underset{\substack{\downarrow \\ \text{Acoustic} \\ \text{impedance}}}{Z} = \frac{p'}{u'} = \frac{\frac{1}{r} \cdot k \cdot \rho_0 c j}{-\frac{1}{r^2} + \frac{1}{r} \cdot k \cdot j} \cdot \frac{\cancel{f(r-ct)}}{\cancel{f(r-ct)}} = \left\{ \frac{rk j}{-1 + rk \cdot j} \right\} \rho_0 c$$

$$\Rightarrow |Z| = \frac{|p'|}{|u'|} = \frac{rk}{\sqrt{1 + (rk)^2}} \rho_0 c = \frac{r\omega}{\sqrt{c^2 + (r\omega)^2}} \rho_0 c$$

since $|p'| = 1 \text{ N/m}^2$ and $f = 500 \text{ Hz}$

since waveform does not change, $|p'|$ at $0.1 \text{ m} = 10 \text{ N/m}^2$ with same angular frequency $f = 500 \text{ Hz} \rightarrow \omega = 2\pi \cdot 500 \text{ rad/s}$.

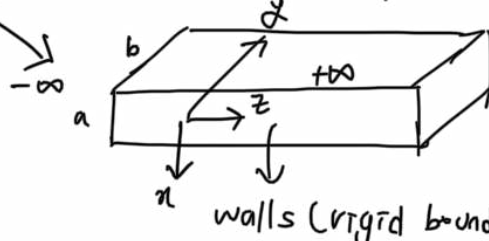
$$\Rightarrow |Z| = \frac{|10|}{|u'|_{r=0.1\text{m}}} = \frac{0.1 \cdot 2\pi \cdot 500}{\sqrt{(343 \text{ m/s})^2 + (0.1 \cdot 2\pi \cdot 500)^2}} \cdot \underset{\text{kg/m}^3}{(1.225)} \underset{\text{m/s}}{(343)} = 283.796$$

$$\Rightarrow |u'|_{r=0.1\text{m}} = 0.035 \text{ (m/s)} = \underline{\underline{35.24 \text{ (mm/s)}}}$$

Wave Lecture 8 (20210930)

3D sound propagation

free-space (spherical symmetric) $\Rightarrow p'(r,t)$



E.g. Rectangular, uniform cross section,

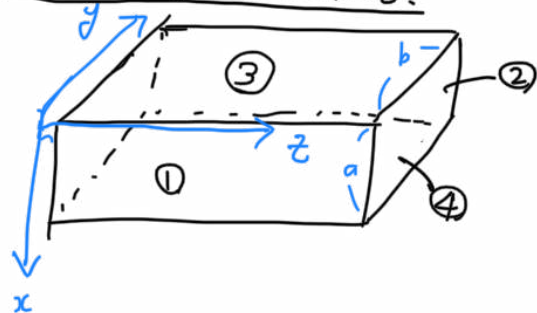
walls (rigid boundary): non-penetration condition

$$u_n = 0 \cdot \downarrow = 0$$

① G.E. (Governing Equation)

$$\nabla^2 \phi - \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} = 0 \quad (\vec{u}' = \nabla \phi)$$

② Boundary Conditions.



$$\left. \begin{array}{l} \textcircled{1} u'_y|_{y=0} = 0 \\ \textcircled{2} u'_y|_{y=b} = 0 \end{array} \right\} \begin{array}{l} 0 \leq x \leq a \\ -\infty < z < \infty \end{array} \quad \left. \begin{array}{l} \textcircled{3} u'_x|_{x=0} = 0 \\ \textcircled{4} u'_x|_{x=a} = 0 \end{array} \right\} \begin{array}{l} 0 \leq y \leq b \\ -\infty < z < \infty \end{array}$$

Note: $u'_x = \frac{\partial \phi}{\partial x}$
 $u'_y = \frac{\partial \phi}{\partial y}$

$$\Rightarrow \left. \begin{array}{l} \textcircled{1} \frac{\partial \phi}{\partial y} \Big|_{y=0,b} = 0 \quad (0 \leq x \leq a) \\ \textcircled{2} \frac{\partial \phi}{\partial x} \Big|_{x=0,a} = 0 \quad (0 \leq y \leq b) \end{array} \right\} 4 \text{ B.C.s.}$$

Solve

$$\phi(x,y,z,t) = X(x)Y(y)Z(z)e^{-j\omega t}$$

$$\left\{ X''YZ + XY''Z + XYZ'' - \frac{1}{c^2} (-j\omega)^2 XYZ \right\} e^{-j\omega t} = 0$$

$\frac{\omega^2}{c^2} = k^2$

$$\Rightarrow \frac{X''}{X} + \frac{Y''}{Y} + \frac{Z''}{Z} + k^2 = 0$$

$$(1) \frac{X''}{X} = -\frac{Y''}{Y} - \frac{Z''}{Z} - k^2 = -\mu^2 \quad (\because \text{to get sinusoidal form, we set negative})$$

negative!

x-only y, z only $\Rightarrow X'' + \mu^2 X = 0 \Rightarrow X(x) = A \cos(\mu x) + B \sin(\mu x)$

$X'' = 0 \rightarrow \text{linear}$ $B=0$ or $\mu=0$

$\mu a = n\pi \quad (n=0,1,2,\dots)$
 $\mu_n = \frac{n\pi}{a}$

$B\mu \cdot 1 = 0 \Rightarrow B=0$
 $-A\mu \sin \mu a = 0$

$\frac{\partial X}{\partial x} \Big|_{x=0,a} = 0 \Rightarrow -A\mu \sin \mu x + B\mu \cos \mu x \Big|_{x=0} = 0$
 $-A\mu \sin \mu x + B\mu \cos \mu x \Big|_{x=a} = 0$

$\frac{\partial \phi}{\partial x} \Big|_{x=0,a} = 0$ for any $0 \leq y \leq b$

$$\therefore X(x) = \sum_m A_m \cos\left(\frac{m\pi}{a}x\right) \quad \therefore \left.\frac{\partial \phi}{\partial x}\right|_{x=0,a} = 0$$

$$(2) \quad \frac{Y''}{Y} = -\frac{Z''}{Z} + \mu_m^2 - k^2 = -\nu^2 \Rightarrow Y(y) = \sum_n B_n \cos\left(\frac{n\pi}{b}y\right)$$

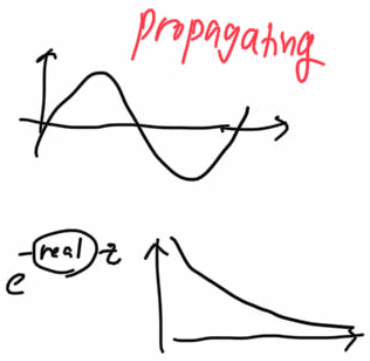
$$(3) \quad \frac{Z''}{Z} = \mu_m^2 + \nu_n^2 - k^2 = -\alpha_{mn}^2 \Rightarrow Z(z) = \sum_m \sum_n \left[C_{mn} e^{j\alpha_{mn}z} + D_{mn} e^{-j\alpha_{mn}z} \right]$$

From (1), (2), (3),

$$\phi(x, y, z, t) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \cos\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) \left[C_{mn} e^{j\alpha_{mn}z} + D_{mn} e^{-j\alpha_{mn}z} \right]$$

* Physical Interpretation.

$$\alpha_{mn} = \pm \sqrt{\left(\frac{\omega}{c}\right)^2 - \pi^2 \left(\frac{m^2}{a^2} + \frac{n^2}{b^2}\right)}$$

$$\begin{aligned} \sqrt{\text{///}} > 0 &\Rightarrow e^{j(\text{real})z} \Rightarrow \text{Propagating} \\ \sqrt{\text{///}} < 0 &\Rightarrow e^{j(j(\text{real}))z} = e^{-(\text{real})z} \Rightarrow \text{Evanescent} \end{aligned}$$


$$\left(\frac{\omega}{c}\right)^2 - \pi^2 \left(\frac{m^2}{a^2} + \frac{n^2}{b^2}\right) > 0 : \text{Propagating}$$

$$< 0 : \text{Evanescent}$$

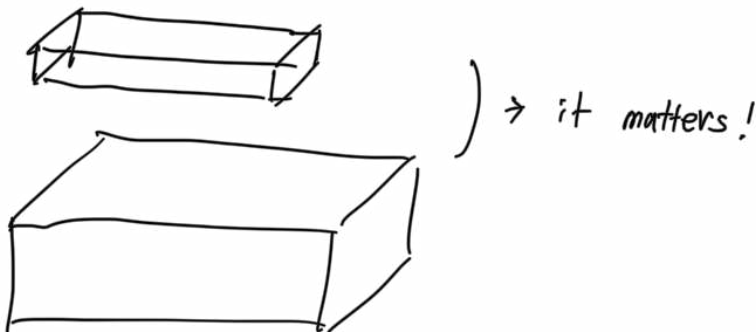
① $(m, n) = (0, 0)$: plane wave mode



② Plane wave $\rightarrow$ always propagation inside waveguide.

For higher modes, it can become evanescent waves.

③ $(a, b) \rightarrow$ (size) of duct.



Note: These are all $\checkmark$ compressible fluids.

Wave Lecture 9 (20211005)

• Review of Incompressible Fluid Mechanics → Water waves.

$$\textcircled{1} \text{ Cont: } \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) = 0 \xrightarrow{\rho = \rho_0 \text{ (constant)}} \underline{\nabla \cdot \vec{u} = 0}$$

if irrotational flow, $\vec{u} = \nabla \phi \Rightarrow \underline{\nabla^2 \phi = 0}$

$$\textcircled{2} \text{ Euler Eq.: } \rho \left(\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} \right) = -\nabla p - \rho g \vec{e}_z$$

$$\Rightarrow \frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla) \vec{u} = -\nabla \left(\frac{p}{\rho} \right) - g \vec{e}_z$$

Since $\nabla (\vec{A} \cdot \vec{B}) = (\vec{A} \cdot \nabla) \vec{B} + (\vec{B} \cdot \nabla) \vec{A} + \vec{A} \times (\nabla \times \vec{B}) + \vec{B} \times (\nabla \times \vec{A})$
 $(\vec{A} = \vec{B} = \vec{u} = \nabla \phi) \rightarrow$

$$\Rightarrow \nabla (\nabla \phi \cdot \nabla \phi) = 2 \cdot (\nabla \phi \cdot \nabla) \nabla \phi + 2 \cdot \nabla \phi \times (\cancel{\nabla \times \nabla \phi}) \rightarrow 0$$

$$\Rightarrow \nabla (\vec{u} \cdot \vec{u}) = 2 (\vec{u} \cdot \nabla) \vec{u}$$

$$\Rightarrow \frac{\partial u}{\partial t} + \frac{1}{2} \nabla |\vec{u}|^2 + \nabla \left(\frac{p}{\rho} \right) + g \vec{e}_z = 0$$

$$\Rightarrow \nabla \left(\frac{\partial \phi}{\partial t} + \frac{1}{2} |\nabla \phi|^2 + \frac{p}{\rho} \right) = -\nabla (gz)$$

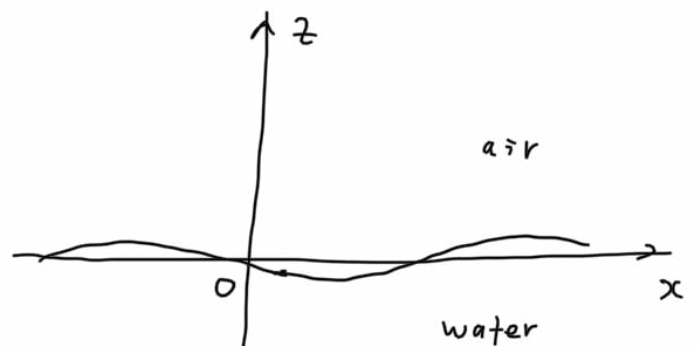
$$\Rightarrow \nabla \left(\frac{\partial \phi}{\partial t} + \frac{1}{2} |\nabla \phi|^2 + \frac{p}{\rho} + gz \right) = 0$$

HW #2. $\nabla \times \nabla \phi = 0$ prove it!

$$\nabla \times \nabla \phi = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix}$$

$$= 0 \hat{i} + 0 \hat{j} + 0 \hat{k} = \underline{\underline{\vec{0}}}$$

$$\Rightarrow \underline{\underline{\frac{\partial \phi}{\partial t} + \frac{1}{2} |\nabla \phi|^2 + \frac{p}{\rho} + gz = \text{Const.}}}$$



At $p = p_{\text{ambient}}$, undisturbed, neglect slope $\Rightarrow \underline{\underline{\frac{p_{\text{amb}}}{\rho} = \text{Constant.}}}$, $|\nabla \phi|^2 \sim 0$

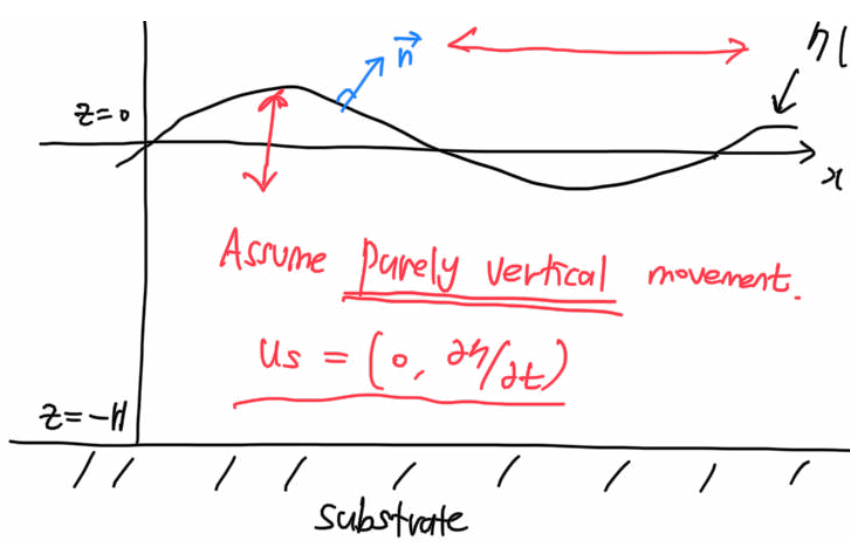
$$\Rightarrow \frac{\partial \phi}{\partial t} + \cancel{\frac{1}{2} |\nabla \phi|^2} + \frac{p_{\text{absolute}}}{\rho} + gz = \frac{p_{\text{ambient}}}{\rho}$$

$$\Rightarrow \underline{\underline{\frac{\partial \phi}{\partial t} + \frac{p_{\text{gauge}}}{\rho} + gz = 0}}$$

Surface gravity wave.



Long wavelength $\rightarrow \frac{\partial h}{\partial x} \ll 1$



$$G.E: \frac{\partial \phi}{\partial t} + \frac{p}{\rho} + gz = 0$$

$$B.C: \textcircled{1} u_z(z=-H) = 0 \Rightarrow \left. \frac{\partial \phi}{\partial z} \right|_{z=-H} = 0$$

$$\textcircled{2} \vec{n} \cdot \vec{u}_p \Big|_{z=h} = \vec{n} \cdot \vec{u}_s$$

fluid particle velocity normal to the surface velocity of surface normal to it

$$\textcircled{2} \Rightarrow \text{Surface } f(x, z, t) = 0 \Rightarrow z - \eta(x, t) = 0$$

$$\left(\vec{n} = \frac{\nabla f}{\|\nabla f\|} = \frac{(-\frac{\partial \eta}{\partial x}, 1)}{\|\nabla f\|} \right) \Rightarrow \vec{n} \cdot \vec{u} \Big|_{z=h} = \vec{n} \cdot \vec{u}_s$$

$$\Rightarrow \left(-\frac{\partial \eta}{\partial x}, 1 \right) \cdot (u, w) = \left(-\frac{\partial \eta}{\partial x}, 1 \right) \cdot \left(0, \frac{\partial \eta}{\partial t} \right)$$

Q) what if we don't neglect x direction of surface?

$$\Rightarrow -\frac{\partial \eta}{\partial x} u + w = \frac{\partial \eta}{\partial t} \quad \left(w = \frac{\partial \phi}{\partial z} \Big|_{z=h} = \frac{\partial \phi}{\partial z} \Big|_{z=0} + \eta \frac{\partial^2 \phi}{\partial z^2} + \dots \text{Higher order} \dots \right)$$

Neglect $\rightarrow 0$

$$\Rightarrow \frac{\partial \phi}{\partial z} \Big|_{z=0} = \frac{\partial \eta}{\partial t} + \frac{\partial \eta}{\partial x} \cdot \frac{\partial \phi}{\partial x} \Rightarrow \frac{\partial \phi}{\partial z} \Big|_{z=0} = \frac{\partial \eta}{\partial t}$$

$\ll 1$

Q) what if we include all Higher Order Terms?

Q) what if λ is small?

$$\textcircled{3} \text{ at } z=h, \text{ gauge pressure at interface } = 0 \quad (\text{surface tension } \times)$$

$$\frac{\partial \phi}{\partial t} + \frac{p}{\rho} + gz = 0 \Rightarrow \frac{\partial \phi}{\partial t} \Big|_{z=h} = \frac{\partial \phi}{\partial t} \Big|_{z=0} = -gh$$

$$\Rightarrow \frac{\partial \phi}{\partial t} \Big|_{z=0} = -gh$$

Problem: $\phi(x, z, t) = \underbrace{f(z)}_{\text{depth term}} \sin(kx - \omega(k) \cdot t)$

Initial condition: $\eta(x, t=0) = a \cos kx$

Q From continuity, $\nabla^2 \phi = 0 \Rightarrow \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial z^2} = 0 \Rightarrow \frac{\partial^2 f}{\partial z^2} - k^2 f = 0$

$$\Rightarrow \underline{f(z) = A e^{kz} + B e^{-kz}}$$

$$\Rightarrow \phi(x, z, t) = (Ae^{kz} + Be^{-kz}) \sin(kx - \omega(k)t)$$

$$\textcircled{1} \left. \frac{\partial \phi}{\partial z} \right|_{z=-H} = 0 \Rightarrow Ak e^{-kH} - Bk e^{+kH} = 0 \Rightarrow A = B e^{2kH} \rightarrow \textcircled{1}$$

$$\textcircled{2} \left. \frac{\partial \phi}{\partial z} \right|_{z=0} = \frac{\partial \eta}{\partial t} \Rightarrow (A k e^{kz} - B k e^{-kz}) \Big|_{z=0} \sin(kx - \omega(k)t) = \frac{A+B}{g} \omega(k)^2 \sin(kx - \omega(k)t) \quad (\text{by } \textcircled{1})$$

$$\Rightarrow (A-B)k = \frac{A+B}{g} \{\omega(k)\}^2$$

$$\Rightarrow \omega(k) = \sqrt{gk \frac{A-B}{A+B}} = gk \frac{e^{2kH} - 1}{e^{2kH} + 1} = \sqrt{gk \frac{e^{kH} - e^{-kH}}{e^{kH} + e^{-kH}}}$$

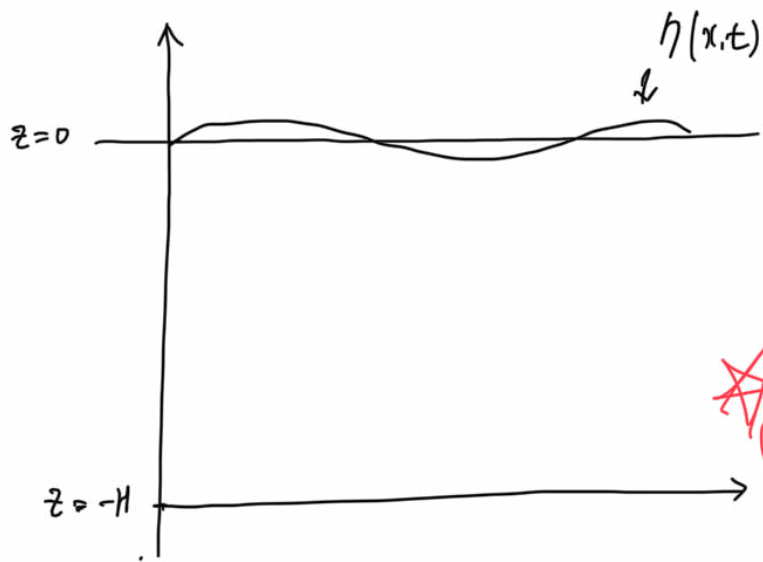
$$\textcircled{3} \left. \frac{\partial \phi}{\partial t} \right|_{z=0} = -g\eta \Rightarrow (Ae^{kz} + Be^{-kz}) \{-\omega(k)\} \cos(kx - \omega(k)t) \Big|_{z=0} = -g\eta$$

$$\Rightarrow g\eta = (A+B) \omega(k) \cos(kx - \omega(k)t)$$

$$\Rightarrow \eta = \frac{A+B}{g} \omega(k) \cos(kx - \omega(k)t) \rightarrow \textcircled{3}$$

From $\textcircled{2}$, $\omega(k) = \sqrt{gk \tanh(kH)}$





at $t=0$ $\eta(x,t=0) = a \cos kx$ (I.C.)

Incompressible $\rightarrow \nabla \cdot \vec{u} = 0$

Irrrotational $\rightarrow$ there exists $\nabla \phi = \vec{u} \Rightarrow \nabla^2 \phi = 0$

$\phi(x,z,t) = f(z) \sin(kx - \omega t)$

$\nabla^2 \phi = 0 \Rightarrow \frac{d^2 f}{dz^2} - k^2 f = 0 \Rightarrow \underline{Ae^{kz} + Be^{-kz} = f(z)}$

① $z = -H, \frac{\partial \phi}{\partial z} \Big|_{z=-H} = 0$ ($\because u_z \Big|_{\text{bottom}} = 0$).

$\Rightarrow \frac{\partial f}{\partial z} \sin(kx - \omega t) = 0 \Rightarrow \frac{\partial f}{\partial z} \Big|_{z=-H} = 0 \Rightarrow k(Ae^{-kH} - Be^{kH}) = 0 \Rightarrow \underline{A = Be^{2kH}}$

② $z = 0, \frac{\partial \phi}{\partial z} \Big|_{z=0} = \frac{\partial \eta}{\partial t}$, Note that $\eta(x,t) = a \cos(kx - \omega t)$ — ①

$\Rightarrow \frac{\partial \eta}{\partial t} = (Ak - Bk) \sin(kx - \omega t) \Rightarrow a(-\sin(kx - \omega t))(-\omega) = (Ak - Bk) \sin(kx - \omega t)$

$\Rightarrow \underline{k(A - B) = a\omega} \Rightarrow \underline{A = \frac{a\omega}{k} \frac{1}{1 - e^{-2kH}}}, \underline{B = \frac{a\omega}{k} \frac{e^{-2kH}}{1 - e^{-2kH}}}$

$\Rightarrow A = \frac{a\omega}{k} \frac{e^{kH}}{e^{kH} - e^{-kH}} = \frac{a\omega}{k} \frac{e^{kH}}{2 \sinh kH}$

$B = \frac{a\omega}{k} \frac{e^{-kH}}{e^{kH} - e^{-kH}} = \frac{a\omega}{k} \frac{e^{-kH}}{2 \sinh kH}$

$\Rightarrow f(z) = e^{kz} \frac{a\omega}{k} \frac{e^{kH}}{2 \sinh kH} + e^{-kz} \frac{a\omega}{k} \frac{e^{-kH}}{2 \sinh kH}$

$= \frac{a\omega}{k} \frac{e^{k(z+H)} + e^{-k(z+H)}}{\sinh(kH) \cdot 2} = \frac{a\omega}{k} \frac{\cosh k(z+H)}{\sinh kH}$

$\Rightarrow \underline{\phi(x,z,t) = \frac{a\omega}{k} \frac{\cosh k(z+H)}{\sinh kH} \sin(kx - \omega t)}$ — ②

③ $\frac{\partial \phi}{\partial t} \Big|_{z=0} = -g\eta$ from ① and ②,

$\Rightarrow -\frac{a\omega^2}{k} \frac{\cosh k(0+H)}{\sinh kH} \cos(kx - \omega t) = -ga \cos(kx - \omega t)$

$\Rightarrow \omega^2 = gk \frac{\sinh kH}{\cosh kH} \Rightarrow \omega = \sqrt{gk \tanh(kH)}$

* Properties.

$$\begin{cases} v_p = \omega/k = \sqrt{\frac{g}{k}} \sqrt{\tanh(kH)} = \sqrt{gH \left[\frac{\tanh(kH)}{kH} \right]} \\ v_g = \partial\omega/\partial k = \frac{1}{2} \sqrt{\frac{g}{k}} \sqrt{\tanh(kH)} = \frac{1}{2} \sqrt{gH \left[\frac{\tanh(kH)}{kH} \right]} \end{cases}$$

$\left(\frac{H}{\lambda} \gg 1 \Rightarrow kH \gg 1 \right)$ Cp for deep water ($H \gg \lambda$)	$\left(\frac{H}{\lambda} \ll 1 \Rightarrow kH \ll 1 \right)$ Cp for shallow water ($H \ll \lambda$)
$\lim_{kH \rightarrow \infty} \tanh kH = 1 \Rightarrow C_p = \sqrt{\frac{g}{k}}$	$\lim_{kH \rightarrow 0} \frac{\tanh kH}{kH} = 1 \Rightarrow C_p = \sqrt{gh}$
$\lambda \uparrow \quad k \downarrow \quad C_p \uparrow$	$H \uparrow \quad C_p \uparrow$
	

* Trajectory of particles.

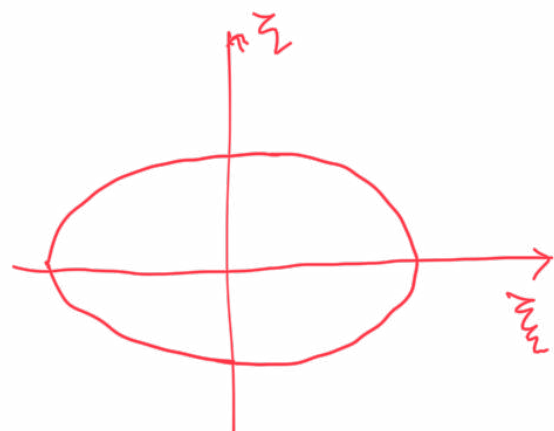
$$\vec{x}_p(t) = x_p(t) \vec{a}_x + z_p(t) \vec{a}_z$$

$$\begin{aligned} x_p(t) &= x_0 + \xi(t) & \Rightarrow \partial x_p / \partial t &= \partial \xi / \partial t = \partial \phi / \partial x = \frac{a\omega}{k} \frac{\cosh k(z+H)}{\sinh kH} \cos(kx - \omega t) \\ z_p(t) &= z_0 + \zeta(t) & \partial z_p / \partial t &= \partial \zeta / \partial t = \partial \phi / \partial z = \frac{a\omega}{k} \end{aligned}$$

$$\begin{cases} \xi(t) = -a \frac{\cosh k(z+H)}{\sinh kH} \sin(kx - \omega t) + \text{"0"} \\ \zeta(t) = a \frac{\sinh k(z+H)}{\sinh kH} \cos(kx - \omega t) \end{cases} \rightarrow \because \xi \text{ and } \zeta \text{ means perturbation.}$$

$$\Rightarrow \left[\frac{\xi(t)}{-a \frac{\cosh k(z+H)}{\sinh kH}} \right]^2 + \left[\frac{\zeta(t)}{+a \frac{\sinh k(z+H)}{\sinh kH}} \right]^2 = 1$$

Ellipse!



Note $-H < z < 0 \Rightarrow \cosh k(z+H) \propto e^{\square} + e^{-\square} > \sinh k(z+H) \propto e^{\square} - e^{-\square}$
↓
 strict!

HW #2 prob 5.

Find the orientation. (direction of rotation)

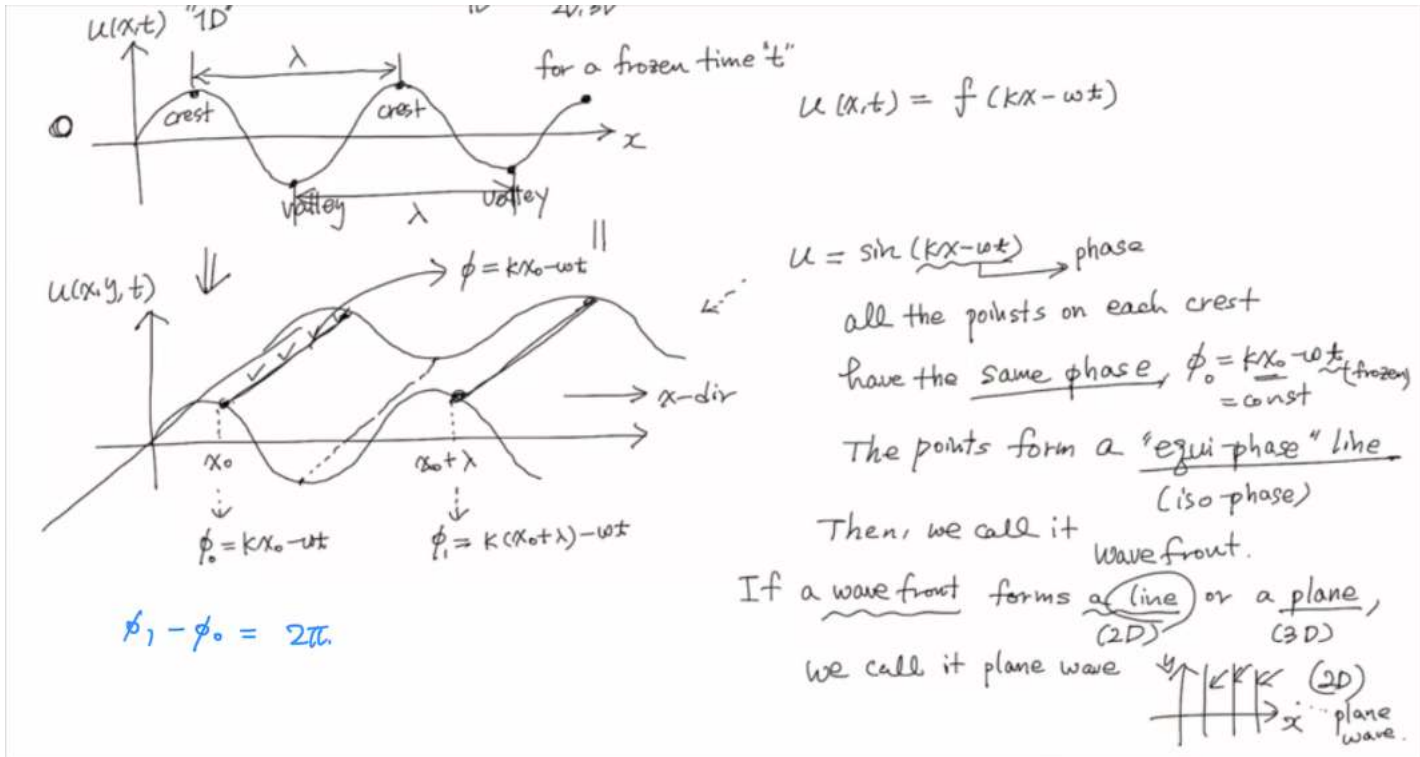


Trajectory	✓ deep water $H \gg \lambda$	intermediate depth $\lambda \sim H$	shallow water $\lambda \gg H$
	$kH \gg 1$ 		

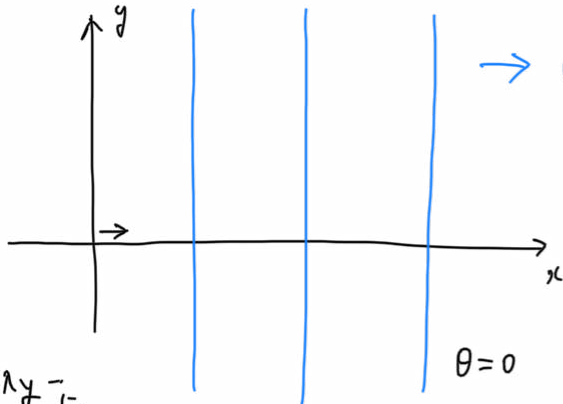
recall $\left[\frac{\xi}{a \frac{\cosh K(z+H)}{\sinh KH}} \right]^2 + \left[\frac{\zeta}{a \frac{\sinh K(z+H)}{\sinh KH}} \right]^2$

Wave Lecture 11 (20211012)

- Wavefronts, Wavenumber, wave packets.



< Top view of 2D wave p.p >

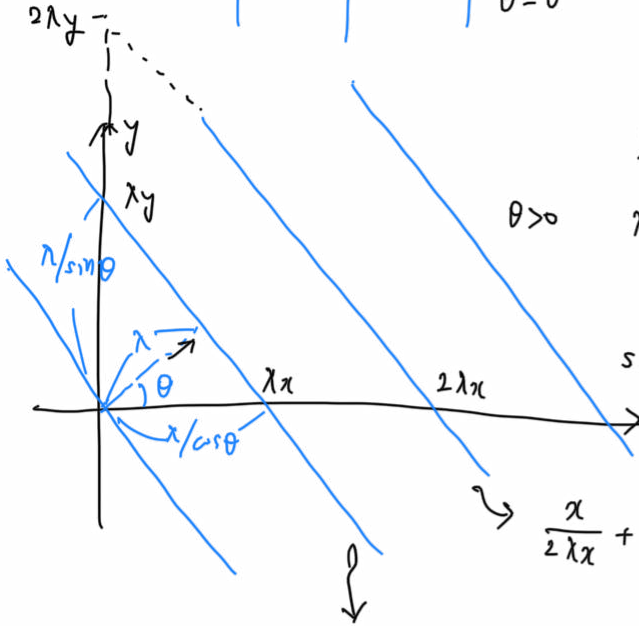


→ wave fronts (= line)

⇒ plane wave $\vec{r} = (x, y)$

$$\Rightarrow \sin((k, 0) \cdot (x, y) - \omega t)$$

$\theta = 0$



$$\lambda_x = \frac{\lambda}{\cos \theta} \rightarrow k_x = k \cos \theta$$

$$\theta > 0 \quad \lambda_y = \frac{\lambda}{\sin \theta} \rightarrow k_y = k \sin \theta$$

$$\sin((k_x, k_y)(x, y) - \omega t) \quad ?$$

$$\frac{x}{2\lambda_x} + \frac{y}{2\lambda_y} = 1 \Rightarrow k_x \cdot x + k_y \cdot y = 4\pi$$

$$\frac{x}{\lambda / \cos \theta} + \frac{y}{\lambda / \sin \theta} = 1 \Rightarrow k_x \cdot x + k_y \cdot y = 2\pi$$

↗ 2π phase diff

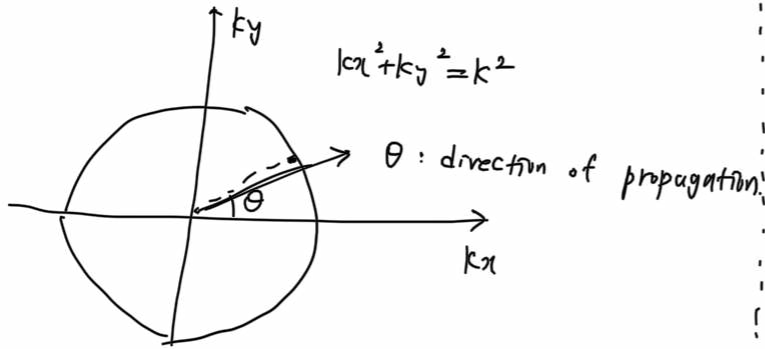
⇒ For all dimensions, $u(\vec{k} \cdot \vec{r} - \omega t)$ (Note: wavefront $\perp \vec{k}$)

E.g.) cylindrical wave

$$\vec{r} = (r \cos \theta, r \sin \theta), \quad \vec{k} = (k \cos \theta, k \sin \theta)$$

$$u(\vec{k} \cdot \vec{r} - \omega t) = u(kr(\cos^2 + \sin^2) - \omega t) = u(kr - \omega t)$$

< Wavenumber plane. (2p) >



what if.

Q) $k_x > k$?

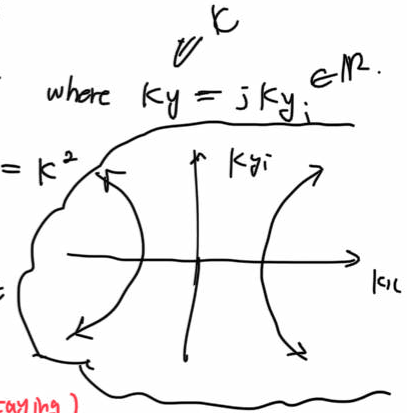
$$k_x^2 + k_y^2 = k^2 \quad \text{where } k_y = j k_{yi} \in \mathbb{R}.$$

$$\Rightarrow k_x^2 - k_{yi}^2 = k^2$$

Note: $e^{j(\mathbf{k} \cdot \mathbf{r} - \omega t)}$

$$= e^{j k_x x} e^{-k_{yi} y} e^{-j \omega t}$$

Evanescent (Decaying).



< Wave packets > Superposition, phase velocity, Group velocity.

$$\left. \begin{array}{l} f_1 = A \cos(k_1 x - \omega_1 t) \\ f_2 = A \cos(k_2 x - \omega_2 t) \end{array} \right\} \begin{array}{l} \alpha \\ \beta \end{array} \oplus \Rightarrow f_1 + f_2 = A \left[2 \cos\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right) \right]$$

$$\Rightarrow f_1 + f_2 = 2A \cos\left(\frac{k_1 + k_2}{2} x - \frac{\omega_1 + \omega_2}{2} t\right) \cos\left(\frac{k_1 - k_2}{2} x - \frac{\omega_1 - \omega_2}{2} t\right)$$

$$= 2A \cos(\bar{k} x - \bar{\omega} t) \cos(\Delta k x - \Delta \omega t)$$

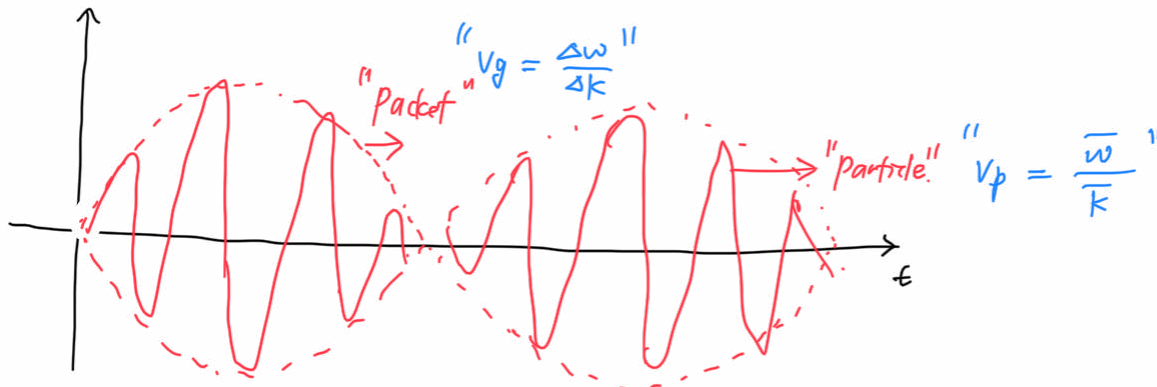
Fastly oscillating

slowly oscillating.

$$\Delta \omega \ll \bar{\omega}$$

$\therefore \omega_1 \approx \omega_2$

$$k_1 \approx k_2$$



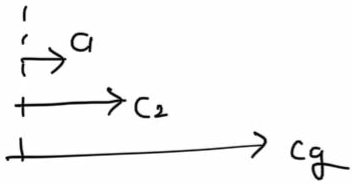
$$c_g = \frac{\Delta \omega}{\Delta k} \rightarrow \lim_{\substack{\Delta k \rightarrow 0 \\ \Delta \omega \rightarrow 0}} \frac{\Delta \omega}{\Delta k} \Rightarrow \left(\frac{\partial \omega}{\partial k} \right), \quad c_p = \left(\frac{\omega}{k} \right)$$

Q1. $k_1 \neq k_2$, $\omega_1 \neq \omega_2$, what's different?

Q2. $\frac{\omega_1}{k_1} < \frac{\omega_2}{k_2}$ and $k_1 < k_2$

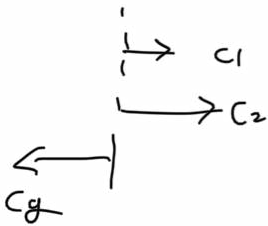
$$c_g = \frac{\omega_2 - \omega_1}{k_2 - k_1} > \frac{\omega_2 - k_1/k_2 \cdot \omega_2}{k_2 - k_1} = \frac{\omega_2}{k_2} = c_2$$

$\Rightarrow c_g > c_2 > c_1$ (Group velocity is the fastest among c_1 & c_2)



Q3. $\frac{\omega_1}{k_1} > \frac{\omega_2}{k_2}$ and $k_1 < k_2$ and $\omega_1 > \omega_2$

$$c_g = \frac{\omega_2 - \omega_1}{k_2 - k_1} < 0 \text{ (wave packets move in the opposite direction)}$$



Moving medium wave equation. $(\rho = \rho_0 + \rho', u = u_0 + u', p = p_0 + p')$

$$\textcircled{1} \text{ cont: } \partial \rho / \partial t + \nabla \cdot (\rho \vec{u}) = 0 \Rightarrow \partial \rho' / \partial t + \nabla \cdot (\rho_0 u' + \rho' u_0) = 0$$

$$\Rightarrow \partial \rho' / \partial t + \rho_0 \nabla \cdot u' + u_0 \cdot \nabla \rho' = 0.$$

Stationary ($u_0 = 0$)

Euler.

$$\textcircled{2} \rho \left(\partial u / \partial t + u \cdot \nabla u \right) = -\nabla p \Rightarrow (\rho_0 + \rho') \left(\partial u' / \partial t + \underbrace{(u_0 + u') \cdot \nabla (u_0 + u')}_{\text{H.O.T} \rightarrow 0} \right) = -\nabla p'$$

$$\Rightarrow \rho_0 \left(\partial u' / \partial t + u_0 \cdot \nabla u' \right) = -\nabla p'$$

Stationary ($u_0 = 0$)

$\textcircled{3}$ Isentropic ($S=0$) state.

$$p' = c^2 \rho' = (\sqrt{\gamma R T})^2 \rho'$$

$\star$ let $\frac{\partial}{\partial t} + u \cdot \nabla = \frac{D}{Dt}$
 $\frac{\partial}{\partial t} + u_0 \cdot \nabla = \frac{D_0}{Dt}$

$$\Rightarrow \frac{D_0}{Dt} \rho' + \rho_0 \nabla \cdot u' = 0 \quad \text{--- ①}$$

$$\rho_0 - \frac{D_0}{Dt} u' = -\nabla p' \quad \text{--- ②}$$

$$\Rightarrow \frac{D_0}{Dt} \textcircled{1} - \nabla \cdot \textcircled{2}$$

$$\Rightarrow \nabla^2 p' - \frac{1}{c^2} \frac{D_0^2}{Dt^2} p' = 0$$

$$\nabla^2 p' - \frac{1}{c^2} \left(\frac{D_0^2}{Dt^2} p' \right) = 0 \Rightarrow \nabla^2 p' - \frac{1}{c^2} \left(\underbrace{\frac{\partial^2}{\partial t^2}}_{\textcircled{1}} + \underbrace{2 \vec{u}_0 \cdot \nabla \frac{\partial}{\partial t}}_{\textcircled{2}} + \underbrace{(\vec{u}_0 \cdot \nabla)^2}_{\textcircled{3}} \right) p' = 0.$$

$$\underline{p'(x, t) = \tilde{p}(\vec{x}) e^{-j\omega t}}$$

Note that $\vec{x}$ is vector ($\vec{x} = (x, y)$)

$$\textcircled{1} \frac{\partial^2 p'}{\partial t^2} = (-j\omega)^2 \tilde{p}(\vec{x}) e^{-j\omega t} = \underline{-\omega^2 \tilde{p}(\vec{x}) e^{-j\omega t}}$$

$$\textcircled{2} \quad 2\vec{u}_0 \cdot \nabla \partial/\partial t p' = 2\vec{u}_0 \cdot \nabla (-j\omega) \tilde{p}(x) e^{-j\omega t} = \underline{-2j\omega u_\infty \frac{\partial \tilde{p}}{\partial x} e^{-j\omega t}}$$

$$\underline{-2j\omega (u_x, u_y) \cdot \left(\frac{\partial \tilde{p}}{\partial x}, \frac{\partial \tilde{p}}{\partial y} \right) e^{-j\omega t}}$$

(for $u_x, u_y, u_x + u_\infty$)

$$\therefore \text{Assume } \vec{u}_0 = (u_\infty, 0) \text{ in } \mathbb{R}^2$$

$$\Rightarrow \vec{u}_0 \cdot \nabla = (u_\infty, 0) \cdot \left(\partial/\partial x, \partial/\partial y \right) = u_\infty \partial/\partial x$$

$$\textcircled{3} \quad u_0 \cdot \nabla = u_\infty \frac{\partial}{\partial x} \Rightarrow \left(u_\infty \frac{\partial}{\partial x} \right)^2 p' = \underline{u_\infty^2 \frac{\partial^2 \tilde{p}}{\partial x^2}}$$

$$u_0 \cdot \nabla = (u_x, u_y) \cdot \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right) = u_x \frac{\partial}{\partial x} + u_y \frac{\partial}{\partial y} \Rightarrow \text{square} \Rightarrow \underline{u_x^2 \frac{\partial^2 \tilde{p}}{\partial x^2} + 2u_x u_y \frac{\partial^2 \tilde{p}}{\partial x \partial y} + u_y^2 \frac{\partial^2 \tilde{p}}{\partial y^2}}$$

$$\Rightarrow \left[\nabla^2 \tilde{p}(\vec{r}) - \frac{1}{c^2} \left(-\omega^2 \tilde{p}(\vec{r}) - 2j\omega u_\infty \frac{\partial \tilde{p}}{\partial x} + u_\infty^2 \frac{\partial^2 \tilde{p}}{\partial x^2} \right) \right] e^{-j\omega t} = 0$$

$$\Rightarrow \frac{\partial^2 \tilde{p}}{\partial x^2} + \frac{\partial^2 \tilde{p}}{\partial y^2} + \left(\frac{\omega}{c} \right)^2 \tilde{p} + 2j \left(\frac{\omega}{c} \right) \left(\frac{u_\infty}{c} \right) \frac{\partial \tilde{p}}{\partial x} - \left(\frac{u_\infty}{c} \right)^2 \frac{\partial^2 \tilde{p}}{\partial x^2} = 0$$

$$\Rightarrow \frac{\partial^2 \tilde{p}}{\partial x^2} + \frac{\partial^2 \tilde{p}}{\partial y^2} + k^2 \tilde{p} + 2jkM \frac{\partial \tilde{p}}{\partial x} - M^2 \frac{\partial^2 \tilde{p}}{\partial x^2} = 0$$

$$\left(\begin{array}{l} M = \frac{u_\infty}{c} : \text{Mach \#} \\ k = \frac{\omega}{c} : \text{wave \#} \end{array} \right) \quad \begin{array}{l} M=1 : \text{Sonic} \\ M>1 : \text{super Sonic} \end{array}$$

$$\Rightarrow \boxed{(1-M^2) \frac{\partial^2 \tilde{p}}{\partial x^2} + \frac{\partial^2 \tilde{p}}{\partial y^2} + 2jkM \frac{\partial \tilde{p}}{\partial x} + k^2 \tilde{p} = 0}$$

$$\text{if } M=0, \quad \nabla^2 \tilde{p} + k^2 \tilde{p} = 0$$

$$\Rightarrow \text{Helmholtz Eq.}$$

< In 1-Dimension >

$$\Rightarrow (1-M^2) \frac{\partial^2 \tilde{p}}{\partial x^2} + 2jkM \frac{\partial \tilde{p}}{\partial x} + k^2 \tilde{p} = 0 \quad \text{set } \tilde{p} = e^{jk_m x}$$

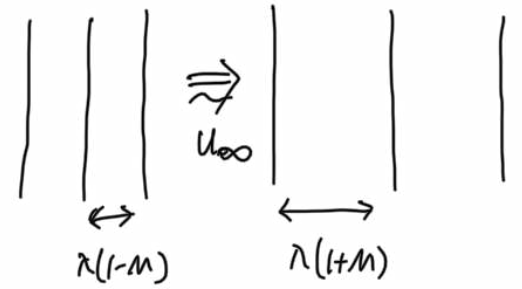
(Note: $k \neq k_m$)

$$\Rightarrow (1-M^2)(-k_m^2) + 2jk \cdot M(jk_m) + k^2 = 0$$

$$\Rightarrow \underline{(1-M^2)k_m^2 + 2k k_m \cdot M - k^2 = 0} \quad \Rightarrow k_m = \frac{-kM \pm \sqrt{k^2 M^2 + k^2(1-M^2)}}{1-M^2}$$

$$\Rightarrow k_m = \frac{-kM \pm k}{(1-M)(1+M)} \begin{cases} \frac{k}{1+M} \\ \frac{-k}{1-M} \end{cases}$$

$$\Rightarrow \tilde{p}(x) = \begin{cases} e^{j \frac{k}{1+M} x} & : \text{right-going } \lambda_r = (1+M) \lambda \\ e^{-j \frac{k}{1-M} x} & : \text{left-going } \lambda_l = (1-M) \lambda \end{cases}$$



⇒ Modulation of wavelength due to the flow.

< Prandtl - Glauert Transformation (p-G trans) >

$$(1-M^2) \frac{\partial^2 \tilde{p}}{\partial x^2} + \frac{\partial^2 \tilde{p}}{\partial y^2} + 2jkM \frac{\partial \tilde{p}}{\partial x} + k^2 \tilde{p} = 0$$

$$\xi = \frac{x}{\sqrt{1-M^2}} \quad \text{and} \quad \eta = y \quad \text{and} \quad \psi = \tilde{p} e^{-j\alpha \xi} \quad \text{? Cf. 2.5.1 Calibration.}$$

$$\hookrightarrow (1-M^2) \frac{\partial^2 \psi e^{j\alpha \xi}}{\partial \xi^2} + \frac{\partial^2 \psi e^{j\alpha \xi}}{\partial \eta^2} + 2jkM \frac{\partial \psi e^{j\alpha \xi}}{\partial \xi} + k^2 \psi e^{j\alpha \xi} = 0$$

$$\Rightarrow (1-M^2) \frac{1}{1-M^2} \frac{\partial^2}{\partial \xi^2} (\psi e^{j\alpha \xi}) + \frac{\partial^2}{\partial \eta^2} (\psi e^{j\alpha \xi}) + 2jkM \frac{\partial}{\partial \xi} (\psi e^{j\alpha \xi}) + k^2 \psi e^{j\alpha \xi} = 0$$

$$\Rightarrow \frac{\partial^2 \psi}{\partial \xi^2} e^{j\alpha \xi} + 2(j\alpha) \frac{\partial \psi}{\partial \xi} e^{j\alpha \xi} + (j\alpha)^2 \psi e^{j\alpha \xi} + \frac{\partial^2 \psi}{\partial \eta^2} e^{j\alpha \xi} + \frac{2jkM}{\sqrt{1-M^2}} \left(\frac{\partial \psi}{\partial \xi} + j\alpha \psi \right) e^{j\alpha \xi} + k^2 \psi e^{j\alpha \xi} = 0$$

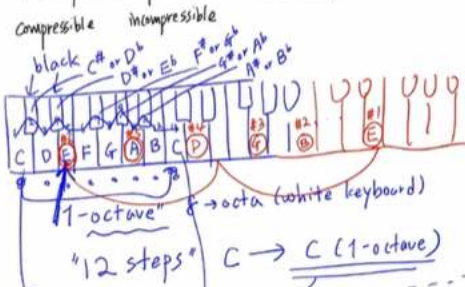
$$\Rightarrow \frac{\partial^2 \psi}{\partial \xi^2} + \frac{\partial^2 \psi}{\partial \eta^2} + 2j \left(\alpha + \frac{kM}{\sqrt{1-M^2}} \right) \frac{\partial \psi}{\partial \xi} + \psi \left(k^2 - \alpha^2 - \frac{2kM\alpha}{\sqrt{1-M^2}} \right)$$

Set $\alpha = \frac{-kM}{\sqrt{1-M^2}}$ to eliminate $\partial \psi / \partial \xi$ term!

$$\Rightarrow \nabla^2 \psi + \psi \left(k^2 - \frac{k^2 M^2}{1-M^2} + \frac{2k^2 M^2}{1-M^2} \right) \Rightarrow \boxed{\nabla^2 \psi + \psi \left(\frac{k^2}{1-M^2} \right) = 0}$$

Wave Lecture 13 (20211026)

Lec. 'Part II' - (D)



$$f(C) = f_0$$

$$f(C_1) = 2f_0$$

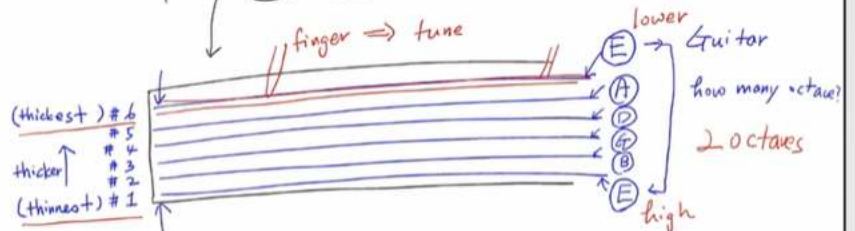
$$(12 \text{ steps}) \quad (x \cdot 2^{\frac{1}{12}})^{12} = 2^1 = 2$$

$$f(C^{\#}) = 2^{\frac{1}{12}} \cdot f_0$$

$$f(D) = (2^{\frac{1}{12}})^2 f_0 = 2^{\frac{2}{12}} f_0$$

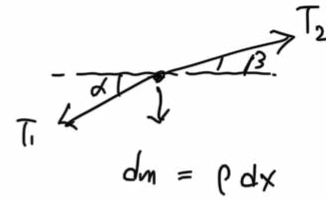
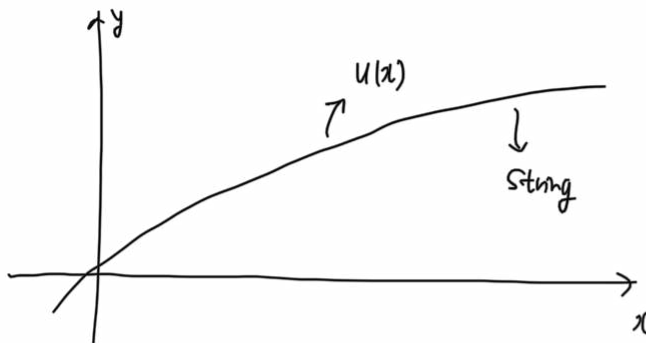
Waves in String (1D)

Math & Phys of Guitar (violin, cello, etc)



why the thin string have
does
high tone?

Waves in String



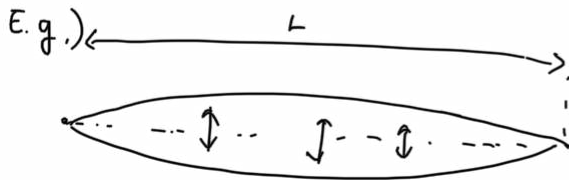
$$1) x: -T_1 \cos \alpha + T_2 \cos \beta = 0 \Rightarrow T_1 \approx T_2 = T$$

$$2) y: T_2 \sin \beta - T_1 \sin \alpha = dm \ddot{u} \Rightarrow T(\tan \beta - \tan \alpha) = dm \ddot{u} = \rho \Delta x \ddot{u}$$

$$\Rightarrow \tan \beta = \left. \frac{\partial u}{\partial x} \right|_{x=x_0+\Delta x} \text{ and } \tan \alpha = \left. \frac{\partial u}{\partial x} \right|_{x=x_0}$$

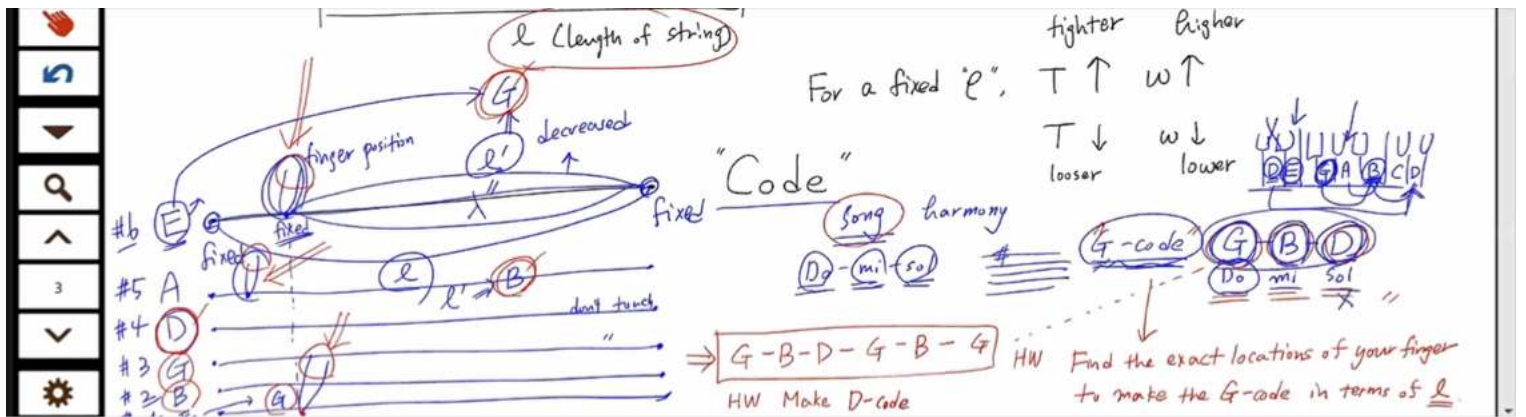
$$\Rightarrow T \left(\left. \frac{\partial u}{\partial x} \right|_{x=x_0+\Delta x} - \left. \frac{\partial u}{\partial x} \right|_{x=x_0} \right) = T \left(\cancel{\left. \frac{\partial u}{\partial x} \right|_{x_0}} + \frac{\partial}{\partial x} \left. \frac{\partial u}{\partial x} \right|_{x_0} \cdot \Delta x + \cancel{H.o.T - \left. \frac{\partial u}{\partial x} \right|_{x_0}} \right)$$

$$\Rightarrow T \frac{\partial^2 u}{\partial x^2} \cancel{\Delta x} = \rho \cdot \cancel{\Delta x} \frac{\partial^2 u}{\partial t^2} \Rightarrow \boxed{\frac{\partial^2 u}{\partial x^2} - \frac{1}{c_s^2} \frac{\partial^2 u}{\partial t^2} = 0 \quad (c_s = \sqrt{T/\rho})}$$



$$\lambda = 2L \Rightarrow k = \frac{2\pi}{\lambda} = \frac{\pi}{L}$$

$$\omega = kc = \frac{\pi}{L} \cdot \sqrt{\frac{T}{\rho}}$$



D : D - F# - A } , 7. GDC c-de $\frac{1}{2} \frac{1}{2} \frac{1}{2}$

A : A - C# - E }

Next Class : String G.E using Lagrangian Mechanics . (continuum) System

Not late!

Waves in Strings (Lagrangian)

• $T = \int_{x_1}^{x_2} \frac{1}{2} (\rho dx) \left(\frac{du}{dt} \right)^2$, $V =$ Work done in "stretching" the string away from equilibrium.

Tension: $k \Rightarrow \Delta V = k \cdot \Delta l \Rightarrow \Delta l = \sqrt{(dx)^2 + (du)^2} - dx$

$\Rightarrow \Delta l = dx \left(\sqrt{1 + \left(\frac{du}{dx} \right)^2} - 1 \right) \approx dx \left(1 + \frac{1}{2} \left(\frac{du}{dx} \right)^2 - 1 \right) = \frac{1}{2} \left(\frac{du}{dx} \right)^2 dx$ ($\because u \ll 1$, $u_x, u_t \ll 1$)



$V = \int_{x_1}^{x_2} \Delta V = \int_{x_1}^{x_2} k \frac{1}{2} \left(\frac{du}{dx} \right)^2 dx$

$\mathcal{L} = T - V = \int_{x_1}^{x_2} \left[\frac{1}{2} \rho \left(\frac{\partial u}{\partial t} \right)^2 - \frac{1}{2} k \left(\frac{\partial u}{\partial x} \right)^2 \right] dx$ $\mathcal{L} = \mathcal{L}(t, x, u, u_t, u_x)$

$\Rightarrow$ Euler-Lagrange Eq. can't be used.

$\Rightarrow$ Generalized Euler-Lagrange Eq.: $\frac{\partial \mathcal{L}}{\partial u} - \frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial u_t} \right) - \frac{\partial}{\partial x} \left(\frac{\partial \mathcal{L}}{\partial u_x} \right) = 0$

① $\frac{\partial \mathcal{L}}{\partial u} = 0$
 ② $\frac{\partial \mathcal{L}}{\partial u_t} = \rho \frac{\partial u}{\partial t}$
 ③ $\frac{\partial \mathcal{L}}{\partial u_x} = -k \frac{\partial u}{\partial x}$

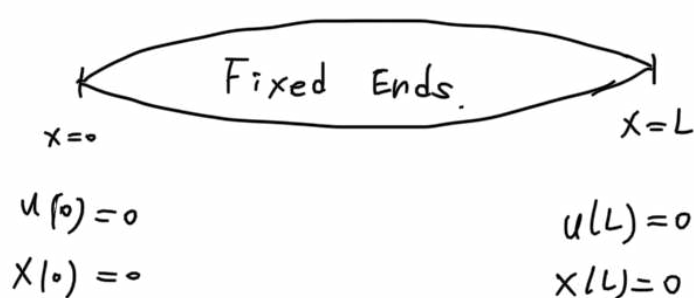
$\Rightarrow -\rho \frac{\partial^2 u}{\partial t^2} + k \frac{\partial^2 u}{\partial x^2} = 0$

$\Rightarrow \frac{\partial^2 u}{\partial t^2} - \frac{k}{\rho} \frac{\partial^2 u}{\partial x^2} = 0$

Wave Eq. for string.

Solving Equation.

$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0 \Rightarrow u(x, t) = X(x)T(t) \Rightarrow \ddot{T}X - c^2 T\ddot{X} = 0 \Rightarrow \frac{1}{c^2} \frac{\ddot{T}}{T} = \frac{\ddot{X}}{X} = -\mu^2$



$X(x) = A \cos \mu x + B \sin \mu x$

$A=0, \mu L = m\pi \Rightarrow \mu_m = \frac{m\pi}{L} \quad (m=1, 2, \dots)$

$X(x) = \sum_{m=1}^{\infty} B_m \sin \left(\frac{m\pi}{L} x \right)$

$$T(t) = A' \cos(c\mu t) + B' \sin(c\mu t) \Rightarrow u(x, t) = \sum_{m=1}^{\infty} \sin\left(\frac{m\pi}{L}x\right) \left[C_m \cos\left(c \cdot \frac{m\pi}{L}t\right) + D_m \sin\left(c \cdot \frac{m\pi}{L}t\right) \right]$$

C_m, D_m ?

$$\left. \begin{aligned} u(x, 0) &= f(x) \\ u_t(x, 0) &= g(x) \end{aligned} \right\} \Rightarrow \sum_{m=1}^{\infty} \sin\left(\frac{m\pi}{L}x\right) C_m = f(x) \quad \text{--- 1.}$$

$$\Rightarrow \sum_{m=1}^{\infty} \sin\left(\frac{m\pi}{L}x\right) \cdot c \cdot \frac{m\pi}{L} \cdot D_m = g(x) \quad \text{--- 2.}$$

$$\textcircled{1} \int_0^{2L/m} \sin\left(\frac{m\pi}{L}x\right) \cdot \sum_{m=1}^{\infty} \sin\left(\frac{m\pi}{L}x\right) \cdot C_m dx = \int_0^{2L/m} \sin\left(\frac{m\pi}{L}x\right) \cdot f(x) dx$$

② In the same way as ①.

physics

$$: u(x, t) = \sum_{m=1}^{\infty} \sin\left(\frac{m\pi}{L}x\right) \left(\text{Temporal terms} \right)$$

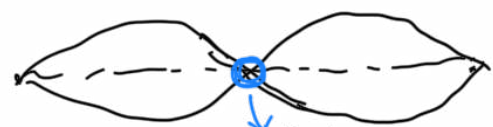
$$\therefore \text{---} = \sum \textcircled{1} + \textcircled{2} + \textcircled{3} + \dots$$

① $m=1$



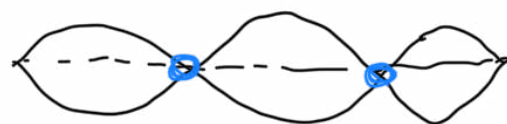
1st mode / fundamental mode.

② $m=2$



2nd mode.

③ $m=3$



$m=1 \rightarrow$ no node

$m \geq 2 \rightarrow$ always node,

Physical meanings of
the General Solution (fixed ends)

String (1D)

$$\underline{u(x,t)} = \sum_{m=1}^{\infty} \sin\left(\frac{m\pi x}{L}\right) \{ \text{temporal terms} \}$$

temporal terms

mode

$$\cdot \underline{m=1} \quad \sinh\left(\frac{\pi x}{L}\right)$$

1st mode (fundamental mode)

$$\bullet m = 2 \quad \sin\left(\frac{2\pi x}{L}\right)$$

2nd mode

$$m = 3 \quad \sin\left(\frac{3\pi x}{L}\right)$$

2nd mode

$$\underline{m=4}$$

$m=1 \rightarrow$ there's no node

$m \geq 2 \rightarrow$ there's always nodes

HW. Derive the G.E. for waves in
membrane by using Newtonian/
Lagrangian
mechanics.

Find the general sol'n (up to you)

$$= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \sinh\left(\frac{m\pi x}{a}\right) \sinh\left(\frac{n\pi y}{b}\right) \{ \text{temp} \}$$

1D string

$$\sin\left(\frac{m\pi}{L}x\right) \left[\cos\left(c\frac{m\pi}{L}t\right) + \sin\left(c\frac{m\pi}{L}t\right) \right]$$

↑
period

$$m=1 \rightarrow \frac{C_{MTL}}{1} = C$$

$$m=2 \rightarrow \frac{20\pi}{1} \cdot t$$

$$C \cdot \frac{m\pi}{T} \cdot T_m = 2\pi \quad \text{one cycle} \quad (1.1)$$

$$T_m = \frac{2\pi L}{c \cdot m} \quad (m) \uparrow T \downarrow \text{farthly}$$

$(m, n) = (\underline{1}, \underline{2})$

(m, n) "pairs"

to couple with temporal terms

nodal line

(2,2)

2D
general sol'n

temporal mode "2D"

$$\cos \left(\sqrt{\left(\frac{m\pi L}{a} \right)^2 + \left(\frac{n\pi L}{b} \right)^2} t \right) \Rightarrow \underline{C \cos} \left(\sqrt{\left(\frac{m}{a} \right)^2 + \left(\frac{n}{b} \right)^2} t \right)$$

temporal mode

parts

temporal model

C. F. Gauss

of math

[Number Theory]

* multiplication of two prime numbers

of $(4n+1)$ can be expressed by

"Summation" between two squares

$$p^2 + q^2$$

$$\text{ex. } (5) \cdot (13) = 65 = 1^2 + 8^2$$

$$= 8^2 + 1^2$$

$$= 4^2 + 7^2$$

$$= 7^2 + 4^2$$

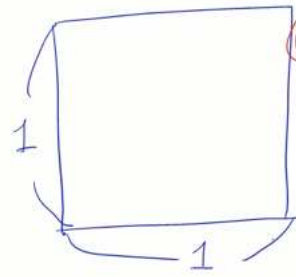
$$\begin{cases} 5 = 4 \cdot 1 + 1 \\ 13 = 4 \cdot 3 + 1 \\ 17 = 4 \cdot 4 + 1 \\ \cancel{21 = 4 \cdot 5 + 1} \end{cases}$$

String membrane

membrane motion

$$(m, n) \text{ pairs} \rightarrow C \pi \sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}}$$

Let $a=b=1$. (square membrane)



Q Find all pairs of (m, n) corresponding to

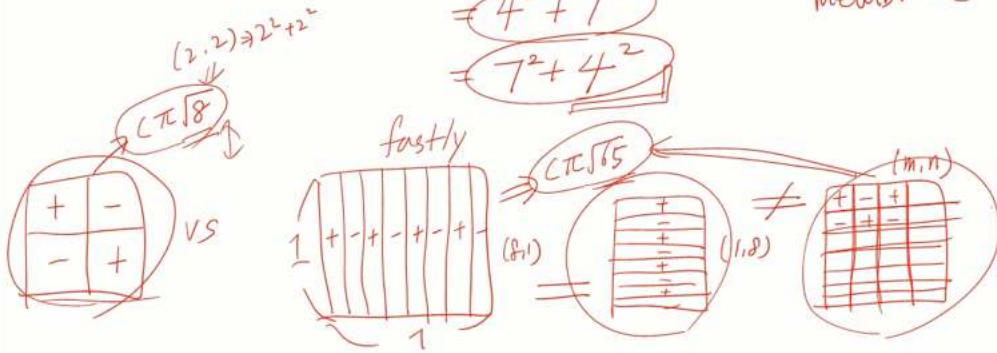
$$\omega = C \pi \sqrt{65}$$

$$\cos(\omega t)$$

$$\sin(\omega t)$$

$$C \pi \sqrt{m^2 + n^2} = C \pi \sqrt{65}$$

$$m^2 + n^2 = 65 \Leftrightarrow (2, 1) \text{ or } (1, 2) \Leftrightarrow (4, 7) \text{ or } (7, 4)$$



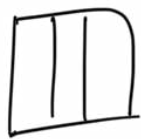
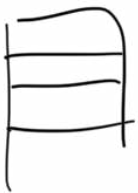
$$\textcircled{1} \sin(\pi x) \sin(2\pi y) + \sin(2\pi x) \sin(\pi y)$$



$$\textcircled{2} \frac{1}{2} \sin(\pi x) \sin(2\pi y) + \sin(2\pi x) \sin(\pi y)$$



$$\textcircled{3} \sin(\pi x) \sin(3\pi y) + \sin(3\pi x) \sin(\pi y)$$



$$\begin{aligned} & 2 \sin \pi x \sin \pi y \cos \pi y + 2 \sin \pi x \cos \pi x \sin \pi y \\ &= 2 \sin \pi x \sin \pi y (\cos \pi x + \cos \pi y) = 0 \end{aligned}$$

$$\begin{aligned} & \sin \pi x \sin \pi y \cos \pi y + 2 \sin \pi x \cos \pi x \sin \pi y \\ &= \sin \pi x \sin \pi y (\cos \pi y + 2 \cos \pi x) = 0 \end{aligned}$$

$$\begin{aligned} & \sin \pi x (3 \sin \pi y - 4 \sin^3 \pi y) \\ &+ \sin \pi y (3 \sin \pi x - 4 \sin^3 \pi x) \\ &= 3 \sin \pi x \sin \pi y (6 - 4 \sin^2 \pi x - 4 \sin^2 \pi y) = 0 \end{aligned}$$

$$\textcircled{4} \cos \pi x + \cos \pi y = 0$$

$$\cos \pi x + \frac{1}{2} \cos \pi y = 0$$

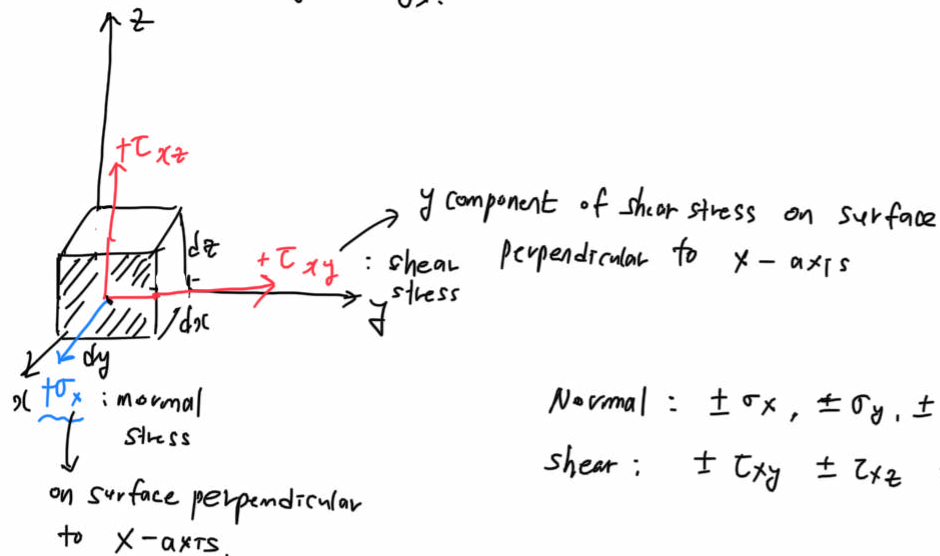
Wave Lecture 16 (20211104)

Elastodynamics : Dynamic theory of elasticity.

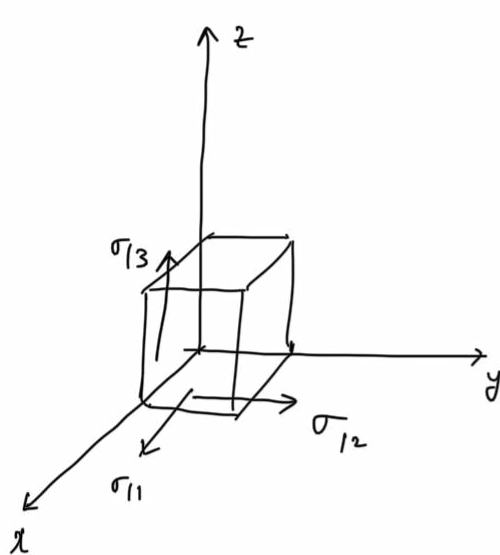
- Basic background.

- 1) stress
- 2) strain
- 3) Displacement
- 4) modulus $\begin{cases} E \text{ (Young's)} \\ G \text{ (Shear)} \end{cases}$
- 5) Poisson ratio ν
- 6) Hooke's law $\sigma_x = E \epsilon_x$
- 7) Cauchy, Lamé

① stress : contact force / unit area of an infinitesimal element.
 $\neq$ body force (e.g. gravity).



$$\left. \begin{array}{l} \text{Normal : } \pm \sigma_x, \pm \sigma_y, \pm \sigma_z \\ \text{shear : } \pm \tau_{xy}, \pm \tau_{xz}, \pm \tau_{yz} \end{array} \right\}$$



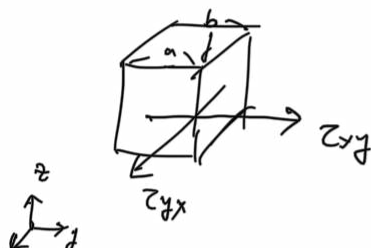
$$\left(\begin{array}{ccc} \sigma_x = \sigma_{11} & \sigma_y = \sigma_{22} & \sigma_z = \sigma_{33} \\ \tau_{xy} = \sigma_{12} & \tau_{yz} = \sigma_{23} & \tau_{zx} = \sigma_{31} \\ \tau_{xz} = \sigma_{13} & \tau_{yx} = \sigma_{21} & \tau_{zy} = \sigma_{32} \end{array} \right)$$

$$\Rightarrow \begin{pmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{pmatrix} = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{pmatrix}$$

σ_{ij} = normal ($i = j$)
 shear ($i \neq j$)

stress tensor.

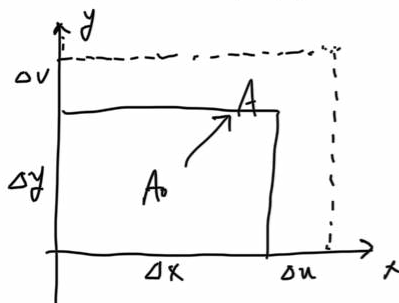
HW) prove $\tau_{xy} = \tau_{yx}$ (state condition for equality).



$$\tau_z = I_z \cdot \alpha = 0 = a \tau_{zy} - b \tau_{yx}$$

$$\Rightarrow \tau_{xy} = \tau_{yx} \quad (\underbrace{a = b}_{\text{condition}})$$

② Dilatation (2D)



$$A_0 = \Delta x \Delta y$$

$$A = (\Delta x + \Delta u)(\Delta y + \Delta v) \approx \Delta x \Delta y + \Delta x \Delta v + \Delta y \Delta u$$

$$\Rightarrow A - A_0 = \Delta x \Delta v + \Delta y \Delta u$$

$$\text{Relative change: } \frac{A - A_0}{A_0} = \frac{\Delta u}{\Delta x} + \frac{\Delta v}{\Delta y}$$

$$\epsilon = \lim_{\substack{\Delta x \Delta y \rightarrow 0 \\ \Delta u \Delta v \rightarrow 0}} R = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right) \cdot \underbrace{\begin{pmatrix} u \\ v \end{pmatrix}}_{\substack{\text{displacement} \\ \text{vector } \vec{d}}} = \underbrace{\nabla \cdot \vec{d}}_{\text{Divergence of displacement.}}$$

Dilatation (Scalar)

Note that $\partial u / \partial x = \epsilon_x$ and $\partial v / \partial y = \epsilon_y$ (strain)

$$\partial w / \partial z = \epsilon_z \Rightarrow \epsilon_{2D} = \epsilon_x + \epsilon_y \rightarrow \underline{\epsilon_{3D} = \epsilon_x + \epsilon_y + \epsilon_z}$$

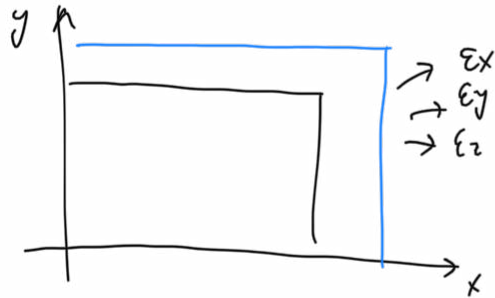
Strain: Tensor

$$\begin{pmatrix} \epsilon_x & \gamma_{xy} & \gamma_{xz} \\ \gamma_{yx} & \epsilon_y & \gamma_{yz} \\ \gamma_{zx} & \gamma_{zy} & \epsilon_z \end{pmatrix}$$

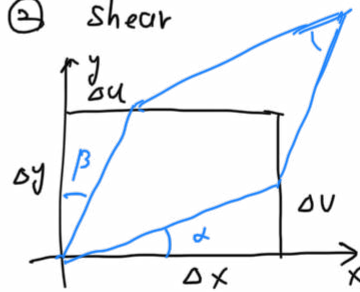
③ Strain - Displacement Relation.

$$\vec{d} = (u, v, w) = u \hat{i} + v \hat{j} + w \hat{k} = (u_1, u_2, u_3)$$

① Normal



② Shear



$$\gamma_{xy} = \alpha + \beta \approx \tan \alpha + \tan \beta$$

$$= \lim \frac{\Delta v}{\Delta x} + \frac{\Delta u}{\Delta y} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}$$

$$\text{Suppose } ()_{,i} = \frac{\partial}{\partial x_i} ()$$

$$\Rightarrow \gamma_{xy} = u_{2,1} + u_{1,2}$$

$$\begin{aligned} & \gamma_{12} = u_{1,2} + u_{2,1} \\ & \gamma_{21} = u_{2,1} + u_{1,2} \end{aligned} \Rightarrow \gamma_{12} = \gamma_{21} \quad (\gamma_{xy} = \gamma_{yx})$$

French mathematician.

Cauchy) let $\epsilon_{ij} = \frac{1}{2} (u_{i,j} + u_{j,i}) \rightarrow \epsilon_{11} = \frac{1}{2} \cdot 2 u_{1,1} = \epsilon_x$

$\gamma_{12} = u_{1,2} + u_{2,1} = 2 \epsilon_{12}$

$$\begin{pmatrix} \epsilon_x & \gamma_{12} & \gamma_{13} \\ \gamma_{21} & \epsilon_y & \gamma_{23} \\ \gamma_{31} & \gamma_{32} & \epsilon_z \end{pmatrix} \Rightarrow \begin{pmatrix} \epsilon_{11} & 2\epsilon_{12} & 2\epsilon_{13} \\ 2\epsilon_{21} & \epsilon_{22} & 2\epsilon_{23} \\ 2\epsilon_{31} & 2\epsilon_{32} & \epsilon_{33} \end{pmatrix} \rightarrow \text{Strain tensor.}$$

Summary :

6 stresses	} 15 unknown
6 strains	
3 displacements	

$$\epsilon_{ij} = \frac{1}{2} (u_{i,j} + u_{j,i}) \quad \} 6 \text{ Equations.}$$

9 more equations needed!

~ Previous class.

① stress $\rightarrow$ 3 normal, 3 shear $\rightarrow$ (6) unknowns.

② dilation $\rightarrow$ 3 " 3 " $\rightarrow$ (6) "

③ relation ($\sigma \sim \epsilon$) $\rightarrow$ 6 relations.

$\nabla \cdot \vec{d}$ = dilation 15 unknowns!
③ unknowns

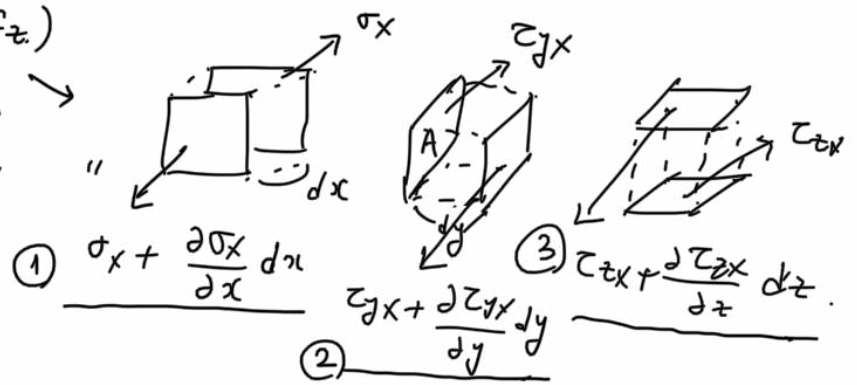
Equation of motion

$$\vec{f} = d\vec{F} = dm \cdot \vec{a} = \rho dx dy dz \cdot \frac{d^2 \vec{d}}{dt^2}$$

($\vec{d}$: displacement vector)

$$f = (f_x, f_y, f_z)$$

1 normal
2 shear



From ①, ②, ③, we get.

$$f_x = \left(\sigma_x + \frac{\partial \sigma_x}{\partial x} dx - \sigma_x \right) dy dz + \left(\tau_{yx} + \frac{\partial \tau_{yx}}{\partial y} dy - \tau_{yx} \right) dx dz + \left(\tau_{zx} + \frac{\partial \tau_{zx}}{\partial z} dz - \tau_{zx} \right) dx dy$$

$$\Rightarrow f_x = \left(\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right) dx dy dz$$

i) x direction.

$$f_x = \left(\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right) dx dy dz = \rho dx dy dz \cdot \frac{d^2 u}{dt^2} \quad (\because \vec{d} = u, v, w)$$

$$\Rightarrow \frac{\partial \sigma_{11}}{\partial x_1} + \frac{\partial \sigma_{21}}{\partial x_2} + \frac{\partial \sigma_{31}}{\partial x_3} = \sigma_{11,1} + \sigma_{21,2} + \sigma_{31,3} = \rho \ddot{u}_1$$

ii) y direction $\Rightarrow \sigma_{12,1} + \sigma_{22,2} + \sigma_{32,3} = \rho \ddot{u}_2$

iii) z direction $\Rightarrow \sigma_{13,1} + \sigma_{23,2} + \sigma_{33,3} = \rho \ddot{u}_3$

$$\sigma_{ji,j} = \rho \ddot{u}_i$$

for $i=1,2,3$

j : summation index

Hw) Derive (ii) and (iii). for exercise!

Cauchy : $\epsilon_{ij} = \frac{1}{2} (u_{i,j} + u_{j,i})$ $i, j = 1, 2, 3$. 6 Equations.
 → strain-displacement

Navier : $\sigma_{ji,j} = \rho \ddot{u}_i$ for $i = 1, 2, 3$. 3 Equations.
 → stress-displacement.

Lame' : strain-stress via Hooke's law ($\sigma_x = E \epsilon_x$) $\langle 1D \rangle$ 6 Equations.

Note : $\sigma_x = \frac{E}{(1+\nu)(1-2\nu)} [\epsilon_x(1-\nu) + (\epsilon_y + \epsilon_z)\nu]$ $\langle 3D \rangle$

$\Rightarrow \sigma_x = \frac{E}{(1+\nu)(1-2\nu)} [\epsilon_x(1-2\nu) + (\epsilon_x + \epsilon_y + \epsilon_z)\nu]$
 $= \epsilon$.

$= \frac{E\nu}{(1+\nu)(1-2\nu)} \epsilon + \frac{E}{1+\nu} \epsilon_x$ and $\tau_{xy} = \frac{E}{2(1+\nu)} \gamma_{xy}$
 Independent of direction.

Can we unify these two?

$\epsilon_x = \epsilon_{11}$ Let $\frac{E}{2(1+\nu)} = \mu$
 $\gamma_{xy} = 2\epsilon_{12}$ Lame's 2nd const

and $\frac{E\nu}{(1+\nu)(1-2\nu)} = \lambda$.
 Lame's 1st const

Rewriting equations,

$\sigma_{11} = \lambda \epsilon + 2\mu \epsilon_{11}$ $\sigma_{12} = \mu 2\epsilon_{12}$
 $\sigma_{22} = \lambda \epsilon + 2\mu \epsilon_{22}$ $\sigma_{23} = \mu 2\epsilon_{23}$
 $\sigma_{33} = \lambda \epsilon + 2\mu \epsilon_{33}$ $\sigma_{31} = \mu 2\epsilon_{31}$
 $\Rightarrow \sigma_{ij} = 2\mu \epsilon_{ij} + \delta_{ij} \lambda \epsilon$
 where $\delta_{ij} \rightarrow$ Kronecker delta. func.
 $\delta_{ij} = \begin{pmatrix} 0 & j \neq i \\ 1 & i = j \end{pmatrix}$

Since $\epsilon = \epsilon_{11} + \epsilon_{22} + \epsilon_{33} = \epsilon_{kk}$ (ϵ_{kk} : Einstein summation tensor)

$\Rightarrow \sigma_{ij} = \lambda \epsilon_{kk} \delta_{ij} + 2\mu \epsilon_{ij}$
 Lame' Einstein Kronecker Cauchy.

Conclusion

① $\epsilon_{ij} = \frac{1}{2} (u_{i,j} + u_{j,i})$ $i, j = 1, 2, 3$

6 strain-displacement

② $\sigma_{ji,j} = \rho \ddot{u}_i$ $i = 1, 2, 3$

3 stress-displacement

$$(3) \quad \sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2\mu \varepsilon_{ij} \quad i, j = 1, 2, 3$$

6 stress-strain

1)	$\varepsilon_{ij} \ (i, j = 1, 2, 3)$	6	strain
2)	$\sigma_{ij} \ (i, j = 1, 2, 3)$	6	stress
3)	$u_i \ (i = 1, 2, 3)$	3	displacement

$$\textcircled{1} \sigma_{ji,j} = \rho \ddot{u}_i \quad \textcircled{2} \epsilon_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i}) \quad \textcircled{3} \sigma_{ij} = \lambda \epsilon_{kk} \delta_{ij} + 2\mu \epsilon_{ij}$$

$$\rho \ddot{u}_i = \sigma_{ij,j} = (\lambda \epsilon_{kk} \delta_{ij} + 2\mu \epsilon_{ij})_{,j} = \left\{ \lambda \cdot \frac{1}{2} (u_{k,k} + u_{k,k}) \delta_{ij} + 2\mu \epsilon_{ij} \right\}_{,j}$$

$$= \left\{ \lambda u_{k,k} \delta_{ij} + \mu (u_{i,j} + u_{j,i}) \right\}_{,j} = \lambda u_{k,k,i} + \mu (u_{i,j,j} + u_{j,i,j})$$

$$(\lambda u_{k,k} \delta_{ij})_{,j} = \lambda u_{k,k,j} = \lambda u_{k,k,i} \quad \left(\because \delta_{ij} = 0 \text{ if } i \neq j \Rightarrow \text{meaningless.} \right. \\ \left. \text{Consider only } i=j \text{ case} \Rightarrow \delta_{ij} = 1 \right)$$

$$\Rightarrow \rho \ddot{u}_i = \lambda u_{k,k,i} + \mu u_{i,j,j} + \mu u_{j,i,j} = \lambda u_{j,j,i} + \mu u_{i,j,j} + \underbrace{\mu u_{j,i,j}}_{\mu u_{j,j,i}}$$

$$\Rightarrow \rho \ddot{u}_i = \mu u_{i,j,j} + (\mu + \lambda) u_{j,j,i} \quad \textcircled{1} u_{j,j} = \nabla \cdot \vec{d} \Rightarrow u_{j,j,i} = \nabla(\nabla \cdot \vec{d})$$

$$\textcircled{2} u_{i,j,j} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) u_i \\ = \nabla^2 u_i$$

$$\Rightarrow \underline{\rho \ddot{\vec{d}}} = \mu \nabla^2 \vec{d} + (\mu + \lambda) \nabla(\nabla \cdot \vec{d}) \quad \left(\text{Note that } \nabla^2 f = \nabla \cdot (\nabla f) \text{ is} \right. \\ \left. \text{conventionally used for scalar} \right. \\ \left. \text{field } f \text{ only.} \right)$$

$\Rightarrow$ Here Laplacian ∇^2 is for each component u_1, u_2, u_3 .

$\textcircled{1} (\nabla \cdot)$ operator

$\textcircled{2} (\nabla \times)$ operator.

$$\rho \nabla \cdot \ddot{\vec{d}} = \mu \nabla^2 (\nabla \cdot \vec{d}) + (\mu + \lambda) \nabla^2 (\nabla \cdot \vec{d})$$

$$\rho (\nabla \times \ddot{\vec{d}}) = \mu \nabla^2 (\nabla \times \vec{d}) + (\mu + \lambda) \nabla \times \nabla (\nabla \cdot \vec{d})$$

$$\Rightarrow \frac{\partial^2 w}{\partial t^2} - \frac{2\mu + \lambda}{\rho} \nabla^2 w = 0$$

$$\Rightarrow \frac{\partial^2 \Omega}{\partial t^2} - \frac{\mu}{\rho} \nabla^2 \Omega = 0$$

$$c_p = \sqrt{\frac{2\mu + \lambda}{\rho}}$$

$$c_s = \sqrt{\frac{\mu}{\rho}}$$

$\rightarrow$ Bulk modulus

$\nabla \cdot \vec{d}$: compression : P wave

$\nabla \times \vec{d}$: spin/twist : S wave

$\square \square \dots$

$\nabla \nabla \dots$

Seismic waves.

Note: These results are for infinitely long solids.

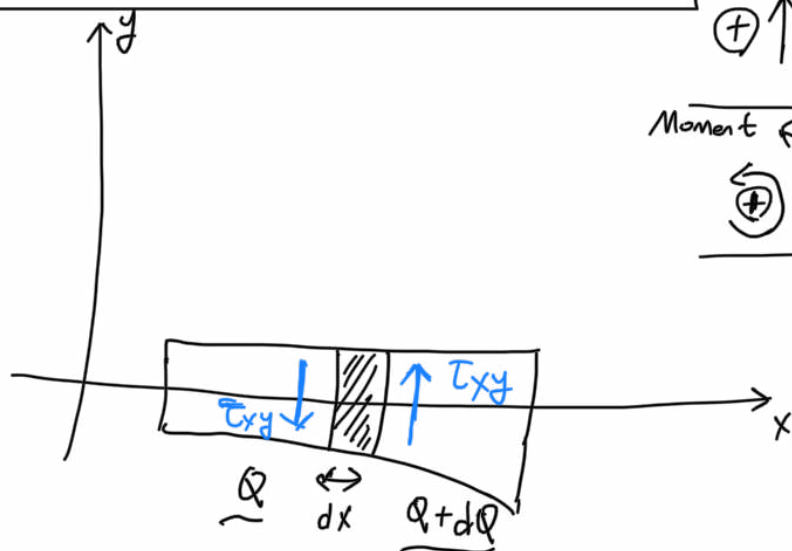
String/membrane
Infinite elastic solids.

Beams. (Euler-Bernoulli $\Rightarrow$ Timoshenko)
Plate (Kirchhoff $\Rightarrow$ Mindlin)

$\Rightarrow$ Approximate theory of elastic wave motions

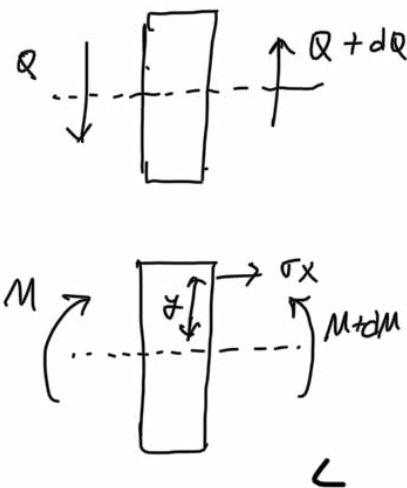
$\therefore$ Elastic wave motions related to "Vibrations" $\rightarrow$ noise
heat
...

1. Euler-Bernoulli beam theory



$$\begin{aligned} \text{Force } \uparrow \quad Q &= \int_A \tau_{xy} dA \\ \text{Moment } \leftarrow \quad M &= \int_A -\sigma_x \cdot y dA \end{aligned}$$

Effect of τ_{xy} to M
will be concerned later.



* Nomenclature (명명법)

- ① $w(x, t)$: transverse deflection of an infinitesimal element.
- ② $\theta(x, t)$: rotation angle / slope of centroidal axis. (CCW: +)
- ③ $Q(x, t)$: shear force
 $M(x, t)$: bending moment.
- ④ ρ : density / E : Young's modulus.
- ⑤ $A(x)$: cross-sectional area / $I(x)$: moment of inertia of area = $\int y^2 dA$.
 $J(x)$: rotational inertia = $\int r^2 dm$.



Assumptions

- ① Beam has a straight centroidal axis. and cross-section has axis of symmetry.
(x-axis) (y-axis)

Timoshenko.

- ② Material is elastic / isotropic / homogeneous / continuous
 $E_x = E_y = E$ = no holes.

③ Transverse deflections are small ($w \ll 1$) $\rightarrow$ Linearization!

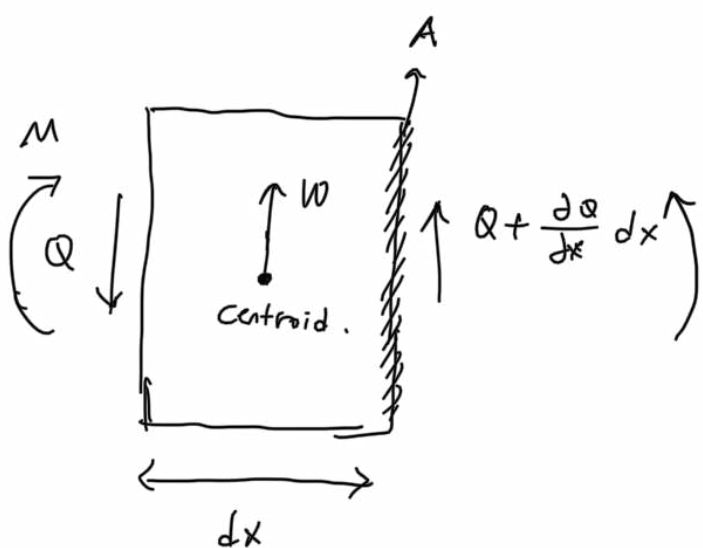
E.B. { ④ Transverse deflections are close to purely translational motion in y direction.
(Thus, rotational inertia can be neglected)

$$J \ddot{\theta} = 0$$

⑤ Cross sections remain plane/perpendicular to x -axis. during/after deformation.
(not distorted) (Neglect shear deformation.)

Discard ④, ⑤ $\Rightarrow$ Timoshenko Beam

Equations



$$\textcircled{1} \quad \int F_y = d m \quad \partial^2 w / \partial t^2$$

$$\Rightarrow Q + \frac{\partial Q}{\partial x} dx - Q = \rho A dx \quad \partial^2 w / \partial t^2$$

$$\Rightarrow \frac{\partial Q}{\partial x} = \rho A \frac{\partial^2 w}{\partial t^2}$$

$$\textcircled{2} \quad \int M = J \partial^2 \theta / \partial t^2 = 0. \quad (\text{for E.B. beams}).$$

$$\Rightarrow M + \frac{\partial M}{\partial x} dx - M + \left(Q + \frac{\partial Q}{\partial x} dx \right) \frac{dx}{2} + Q \cdot \frac{dx}{2} = 0$$

$$\Rightarrow \frac{\partial M}{\partial x} = -Q \quad (\because (dx)^2 \ll 1).$$

Recall that $M = E \cdot I \frac{\partial^2 w}{\partial x^2}$

$$\Rightarrow \rho A \frac{\partial^2 w}{\partial t^2} = \frac{\partial Q}{\partial x} = - \frac{\partial^2 M}{\partial x^2} = - \frac{\partial^2}{\partial x^2} \left(E \cdot I \frac{\partial^2 w}{\partial x^2} \right).$$

$$\Rightarrow \underline{\underline{\rho A \frac{\partial^2 w}{\partial t^2} + \frac{\partial^2}{\partial x^2} \left(E \cdot I \cdot \frac{\partial^2 w}{\partial x^2} \right) = 0}} \quad \left(\text{Euler-Bernoulli beam equation.} \right)$$

Boundary conditions. (w, θ, Q, M) at $x=0$.

① Simply supported. (SS)  $w=0, M=0$

② Free (F)  $\alpha = 0, M = 0$

③ Clamped (C)  $w = 0, \theta = 0$

④ Sliding (S)  $\theta = 0, \alpha = 0$

choose. $\begin{pmatrix} 4 \\ 2 \end{pmatrix} = 6 \rightarrow 4 \text{ B.C.s} / 1 \text{ beam.}$

Waves in Solids (E-B beam) : $\frac{\partial^2 w}{\partial t^2} + \frac{EI}{\rho A} \frac{\partial^4 w}{\partial x^4} = 0$ ✓

For prismatic beam)

Constant for prismatic beam.

B.C.s : $\begin{cases} \text{SS} : w=0, M=0 \rightarrow w=0, w''=0 \\ \text{F} : Q=0, M=0 \rightarrow w'''=0, w''=0 \\ \text{C} : w=0, \theta=0 \rightarrow w=0, w'=0 \\ \text{S} : \theta=0, Q=0 \rightarrow w'=0, w'''=0 \end{cases}$

$M = EI w''$

$Q = -\frac{\partial M}{\partial x} = -EI w'''$

Note that for acoustic perturbation, $\frac{\partial^2 p'}{\partial t^2} - c^2 \frac{\partial^2 p'}{\partial x^2} = 0 \Leftrightarrow \frac{\partial^2 w}{\partial t^2} + \frac{EI}{\rho A} \frac{\partial^4 w}{\partial x^4} = 0$

1) $p' = A e^{j(kx - \omega t)} \Rightarrow A e^{j(kx - \omega t)} \{ (-j\omega)^2 - c^2 (jk)^2 \} = 0 \Rightarrow \omega = ck \Rightarrow c_p = c$

2) $w = W e^{j(kx - \omega t)} \Rightarrow (-j\omega)^2 + \frac{EI}{\rho A} (jk)^4 = 0 \Rightarrow \omega = \sqrt{\frac{EI}{\rho A}} k^2$

[Constant in all frequencies (ω)]

< Dispersion Relation in E.B. beam >

$\Rightarrow c_B = \sqrt[4]{\frac{EI}{\rho A}} \cdot \sqrt{\omega} \quad c_B \sim \sqrt{\omega}$

$\omega \uparrow : c_B \uparrow \quad \text{mv} \longrightarrow$

$\omega \downarrow : c_B \downarrow \quad \sim \longrightarrow$

[phase speed of elastic waves in E-B is depending on frequency.]

$k^4 = \frac{\rho A}{EI} \omega^2 \Rightarrow k = \pm k_B, \pm j k_B \Rightarrow w(x,t) = \{ c_1 e^{jk_B x} + c_2 e^{-jk_B x} + c_3 e^{-k_B x} + c_4 e^{k_B x} \} (e^{-j\omega t})$

k_B is called "Bending wavenumber" General solution

* General solution.



$w(x,t) = \left\{ \underbrace{c_1 e^{jk_B x} + c_2 e^{-jk_B x}}_{\text{Travelling (propagating) waves}} + \underbrace{c_3 e^{-k_B x} + c_4 e^{k_B x}}_{\text{Evanescent wave}} \right\} e^{-j\omega t}$

✗ No exponentially "growing" allowed

Recall speeds in elastic solids $\left(c_1 = \sqrt{\frac{\lambda + 2\mu}{\rho}}, c_2 = \sqrt{\frac{\mu}{\rho}}, c_B = \sqrt[4]{\frac{EI}{\rho A}} \omega \right)$

Express in E, ν

P wave $\rightarrow c_1 = \sqrt{\frac{E(1-\nu)}{\rho(1+\nu)(2-\nu)}}$

S wave $\rightarrow c_2 = \sqrt{\frac{E}{2\rho(1+\nu)}}$

* Phase speeds in elastic solids

Navier's
 $C_1 = \sqrt{\frac{\lambda + 2\mu}{\rho}}$

$C_2 = \sqrt{\frac{\mu}{\rho}}$

$C_B = \sqrt{\frac{EI}{\rho A}}$

λ, μ : Lamé

E, ν

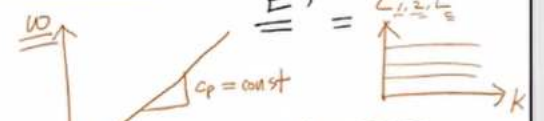
P-wave
 dilatational wave

S-wave
 (equivol.)

$C_1 = \sqrt{\frac{E(1-\nu)}{\rho(1+\nu)(2-\nu)}}$

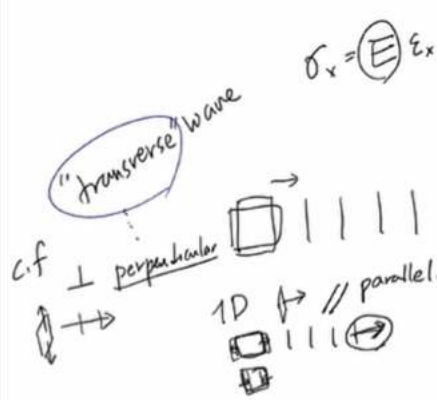
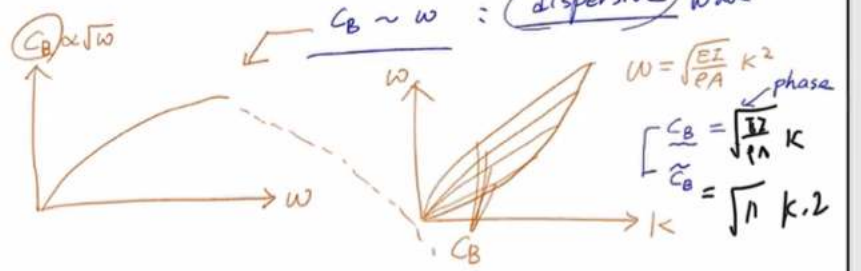
$C_2 = \sqrt{\frac{E}{2\rho(1+\nu)}}$

$C_L = \sqrt{\frac{E}{\rho}}$

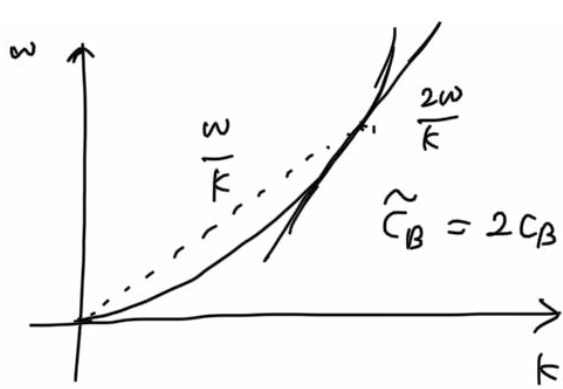


$C_1, C_2, C_L \propto \omega$ non-dispersive wave

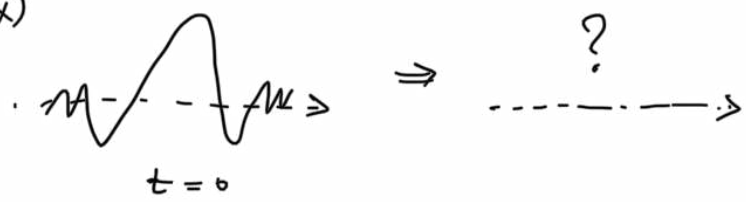
$C_B \sim \omega^2$ dispersive wave



phase speed of
 longitudinal wave
 in rod



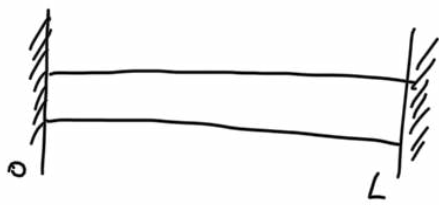
EX)



$$\because c_B \sim \omega \quad \tilde{c}_B \sim 2\omega \Rightarrow \text{Wave form not preserved}$$

Dispersive

• Prismatic beam of length "L"



$$G, E, \frac{\partial^2 w}{\partial t^2} + \frac{EI}{\rho A} \frac{\partial^4 w}{\partial x^4} = 0 \Rightarrow \begin{cases} c_1 e^{jk_B x} + c_2 e^{-jk_B x} + c_3 e^{k_B x} + c_4 e^{-k_B x} \\ b_1 \cos k_B x + b_2 \sin k_B x + b_3 \cosh k_B x + b_4 \sinh k_B x \end{cases}$$

$$\text{B.C. } w(0, t) = 0 \quad w'(0, t) = 0 \\ w(L, t) = 0 \quad w'(L, t) = 0$$

$$\Rightarrow \left. \begin{aligned} w(0) &= b_1 + b_3 = 0 \\ w'(0) &= k_B b_2 + k_B b_4 = 0 \\ w(L) &= \text{---} = 0 \\ w'(L) &= \text{---} = 0 \end{aligned} \right\}$$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ \cos k_B L & \sin k_B L & \cosh k_B L & \sinh k_B L \\ -\sin k_B L & \cos k_B L & \sinh k_B L & \cosh k_B L \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix} = \vec{0}$$

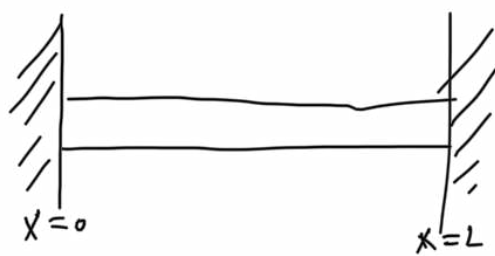
$A \quad \quad \quad \vec{B}$

Since $b_1 = b_2 = b_3 = b_4 = 0$ is not considered, $\det(A) = 0$

$$\det(A) = \left\{ (\cosh^2 k_B L - \sinh^2 k_B L) + \sin k_B L \sinh k_B L - \cos k_B L \cosh k_B L \right\} \cdot 1 \\ + \left\{ \cos^2 k_B L + \sin^2 k_B L - \sin k_B L \sinh k_B L - \cos k_B L \cosh k_B L \right\} \cdot 1 = 0$$

$$\Rightarrow 2 - 2 \cos k_B L \cosh k_B L = 0 \Rightarrow \boxed{\cos k_B L \cosh k_B L = 1} \quad \text{Characteristic Equation}$$

Waves in E-B beam.



C-C prismatic beam.

clamped.

$$\cos k_B L \cosh k_D L = 1$$

$$\text{where } k_B = \sqrt[4]{\frac{\rho A}{EI}} \sqrt{\omega}$$

$$\Rightarrow k_B L = 4.73 \dots$$

$$7.85 \dots$$

$$10.99 \dots$$

$$14.13 \dots$$

From previous.
Equations of motion.

$$b_1 + b_3 = 0, \quad b_2 + b_4 = 0$$

$$b_1 (\cos k_B L - \cosh k_D L) + b_2 (\sin k_B L - \sinh k_D L) = 0$$

$$\Rightarrow b_3 = -b_1, \quad b_4 = -b_2$$

$$b_2 = \frac{\cos k_B L - \cosh k_D L}{\sin k_B L - \sinh k_D L} b_1$$

$$\Rightarrow W(x,t) = b_1 \int \cos k_B L - \sigma \sin k_B L - \cosh k_D L + \sigma \sinh k_D L \} e^{-j\omega t}$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 $b_{1,n}$ $k_{B,n}$ $k_{D,n}$ $k_{D,n}$ $k_{B,n}$

$$\Rightarrow W(x,t) = \sum_{n=1}^{\infty} W_n(x,t)$$

"Temporal" Eigenfrequency

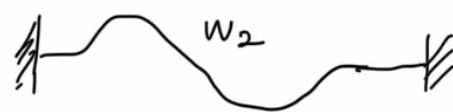
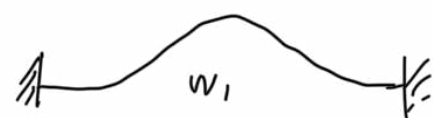
$$f_1 = \frac{4.73^2}{2\pi} \sqrt{\frac{EI}{\rho A}}$$

$$f_2 = \frac{7.85^2}{2\pi} \sqrt{\frac{EI}{\rho A}}$$

$$f_3 = \frac{10.99^2}{2\pi} \sqrt{\frac{EI}{\rho A}}$$

geometric features
material properties $\Rightarrow f$

"Spectral" mode shape



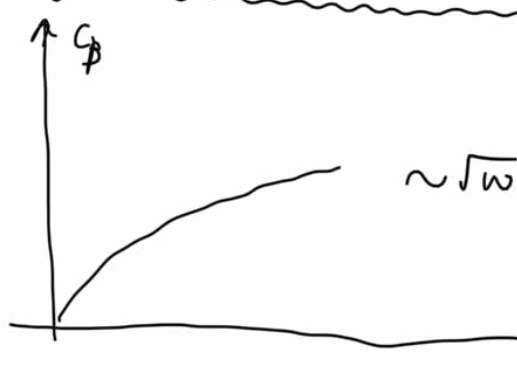
string (1D) $\neq$ Beam (1D)

$$k_{B,n} = \sqrt[4]{\frac{\rho A}{EI}} \sqrt{\omega}$$

HW) compare eigenfrequencies of string & beam. for same length

$$n=1 \rightarrow f_{\text{beam}} = \textcircled{111} \quad f_{\text{string}} = \frac{1}{2L} \cdot c = \frac{1}{2L} \sqrt{T/\mu}$$

Questions! for E-B beams.



No upper bound $\lim_{\omega \rightarrow \infty} C_p = \infty$?

Limitation of E-B beam theory

From the assumptions of

- ✓ 4) rotational inertia neglected
- ✓ 5) shear deformation neglected. $J\ddot{\theta} = 0$

Timoshenko theory

$$\Sigma F_y = \rho A \cancel{dx} \frac{d^2 w}{dt^2} = \cancel{Q} + \frac{dQ}{dx} \cancel{dx} - \cancel{Q} \Rightarrow \underline{\frac{dQ}{dx} = \rho A \frac{d^2 w}{dt^2}} \quad (1)$$

$$\Sigma M = \frac{dM}{dx} dx + Q dx = \underbrace{\rho I dx}_{=J} \frac{d^2 \psi}{dt^2} \quad \left\{ \begin{array}{l} \psi : \text{total angle} / \gamma_0 = \text{shear angle} (=0 \text{ for E.B. beam}) \\ \psi = \frac{dw}{dx} - \gamma_0 \end{array} \right. \quad (2) \quad (4)$$

$$M = EI \frac{d\psi}{dx} \quad (3)$$

$$Q = \int_A \tau_{xy} dA = G \int_A \gamma_{xy} dA = kGA \gamma_0$$

correction factor

$$\left\{ \begin{array}{l} k(\text{rectangular}) : \frac{10(1+\nu)}{12+11\nu} \\ k(\text{circular}) : \frac{6(1+\nu)}{7+6\nu} \end{array} \right.$$

$$\Rightarrow Q = kGA (dw/dx - \psi) \quad (5)$$

$$(1) \quad dQ/dx = \rho A d^2 w/dt^2$$

$$(2) \quad dM/dx + Q = \rho I d^2 \psi/dt^2$$

$$(3) \quad M = EI d\psi/dx$$

$$(4) \quad \psi = dw/dx - \gamma_0$$

$$(5) \quad Q = kGA \underbrace{(dw/dx - \psi)}_{\gamma_0}$$

$$(1), (5) \quad \frac{d}{dx} (kGA (dw/dx - \psi)) = \rho A d^2 w/dt^2 \quad (1)$$

$$(2), (3), (5) \quad \frac{d}{dx} (EI d\psi/dx) + kGA (dw/dx - \psi) = \rho I \frac{d^2 \psi}{dt^2} \quad (2)$$

"Timoshenko" beam equation (2 vars, 2 eqns.)
 $\downarrow$
 w, ψ

HW) ① Set $\rho I d^2 \psi/dt^2 \rightarrow 0$ Derive E.B. beam.
 ② Set $\gamma_0 \rightarrow 0$

Prismatic Timoshenko beam theory ($EI = \text{const.}, A = \text{const.}$)

$$\textcircled{1} \quad \underline{d\psi/dx = d^2 w/dx^2 - \frac{1}{kG} d^2 w/dt^2}$$

$$\textcircled{2} \quad EI \cdot d^2 \psi/dx^2 + kGA (dw/dx - \psi) = \rho I \frac{d^2 \psi}{dt^2}$$

$$\Rightarrow EI \frac{d^3 \psi}{dx^3} + kGA \left(\underline{d^2 w/dx^2 - d\psi/dx} \right) = \rho I \frac{d}{dx} \frac{d^2 \psi}{dt^2}$$

$\Rightarrow \psi$ or ① or ② $\rightarrow$ Calculations ...

$$\Rightarrow \underbrace{\frac{EI}{\rho A} \frac{d^4 w}{dx^4} + \frac{d^2 w}{dt^2}}_{\text{E.B. beam}} - \underbrace{\frac{I}{A} \left(1 + \frac{E}{Gk} \right) \frac{d^4 w}{dx^2 dt^2}}_{\substack{\uparrow \\ \text{Rotational} \\ \text{inertia}}} + \underbrace{\frac{\rho I}{kAG} \frac{d^4 w}{dt^4}}_{\substack{\uparrow \\ \text{shear} \\ \text{deformation}}} = 0$$

Q) Which is more important?

Dispersion relation $w = A e^{i(\gamma x - \omega t)}$

$$\frac{EI}{\rho A} \gamma^4 - \omega^2 - \frac{I}{A} \left(1 + \frac{E}{Gk}\right) \gamma^2 \omega^2 + \frac{\rho I}{kAG} \omega^4 = 0$$

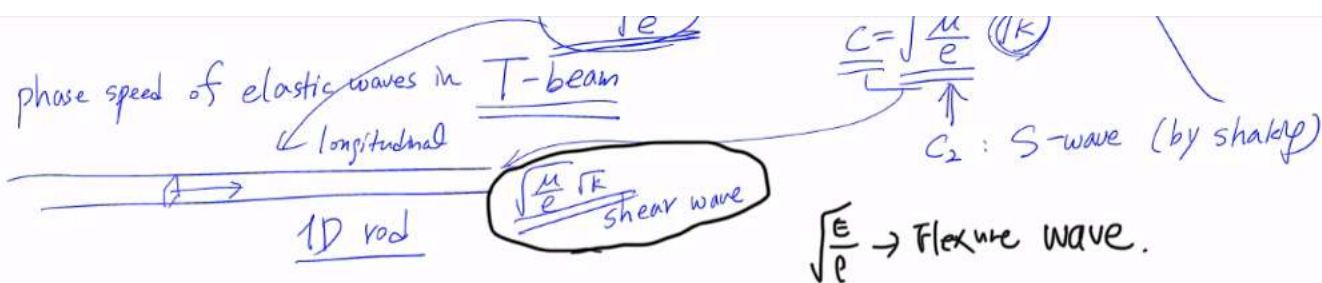
$c_p \sim \sqrt{\omega}$ (Using $\omega = c\gamma$)

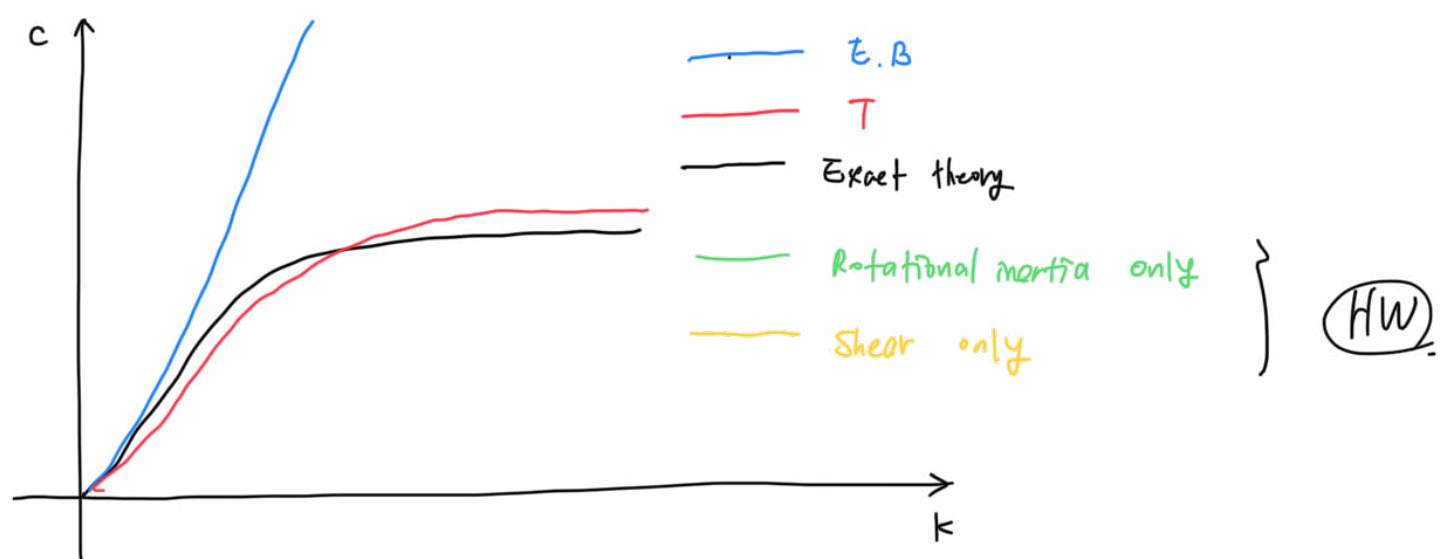
Tedious derivation
(math)

$$\Rightarrow \bar{c}^4 - \left(1 + \frac{Gk}{E}\right) \bar{c}^2 + \frac{Gk}{E} = 0 \quad \left(\bar{c} = \frac{c}{c_L}, \quad c_L = \sqrt{\frac{E}{\rho}}\right)$$

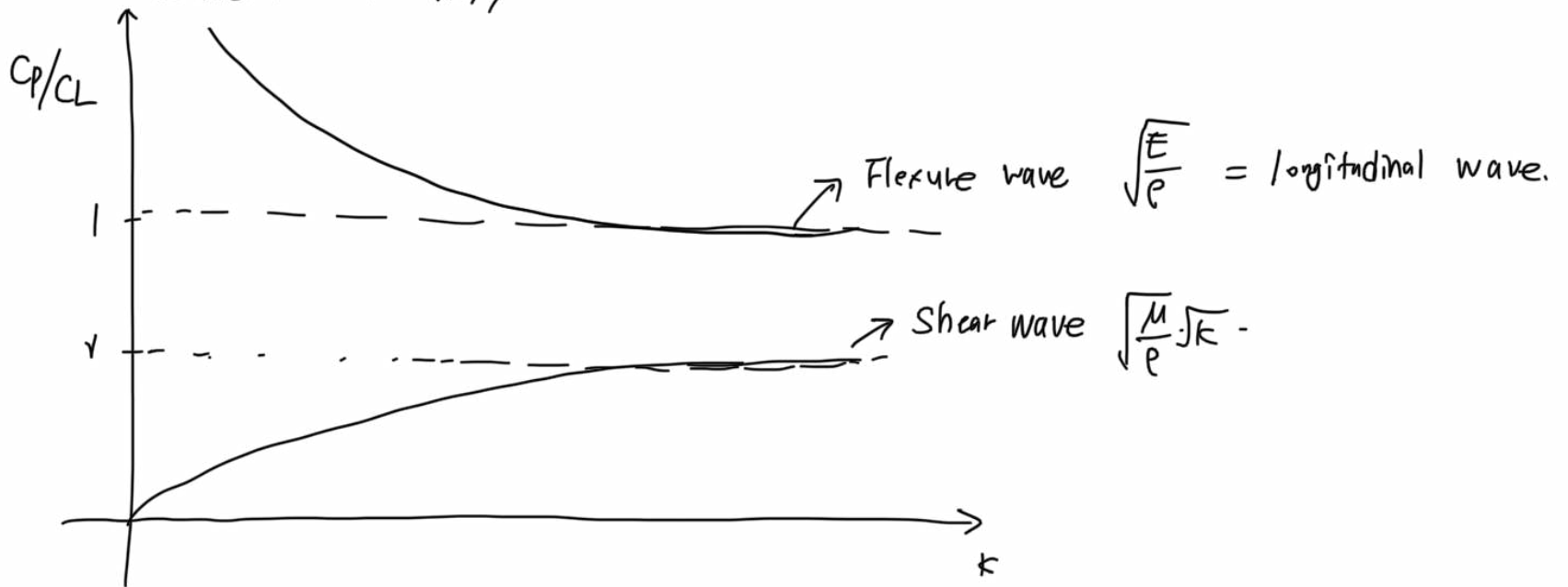
$$\Rightarrow \bar{c} = 1 \quad \text{or} \quad \bar{c} = \sqrt{\frac{Gk}{E}} \Rightarrow c = c_L \quad \text{or} \quad c = \sqrt{\frac{Gk}{E}} \sqrt{\frac{E}{\rho}} = \sqrt{\frac{Gk}{\rho}}$$

$\parallel$
 $\sqrt{E/\rho}$





< Timoshenko Beam >



< Rayleigh Beam > : only considers rotational inertia (not shear strain)

$$1) \rho I \frac{\partial^2 \psi}{\partial t^2}$$

$$2) \psi = \frac{\partial w}{\partial x} - v_0 = \frac{\partial w}{\partial x}$$

$$\textcircled{1} - \frac{\partial}{\partial x} \textcircled{2} \Rightarrow \frac{\partial^2}{\partial x^2} \left(EI \frac{\partial^2 w}{\partial x^2} \right) + \rho A \frac{\partial^2 w}{\partial t^2} - \frac{\partial}{\partial x} \left(\rho I \frac{\partial^2}{\partial t^2} \frac{\partial w}{\partial x} \right) = 0$$

For prismatic Rayleigh beam. EI , ρA , ρI are constants.

$$\Rightarrow EI \frac{\partial^4 w}{\partial x^4} - \rho I \frac{\partial^4 w}{\partial x^2 \partial t^2} + \rho A \frac{\partial^2 w}{\partial t^2} = 0 \rightarrow w = b e^{i(kx - \omega t)}$$

$\rightarrow$ Solve!

$$\Rightarrow \tilde{\omega}^2 = \frac{\tilde{k}^4}{1 + \tilde{k}^2} \quad \left(\text{where } \tilde{\omega} = \frac{\omega \cdot \sqrt{I/A}}{\sqrt{E/\rho}}, \tilde{k} = \sqrt{I/A} \cdot k \right)$$

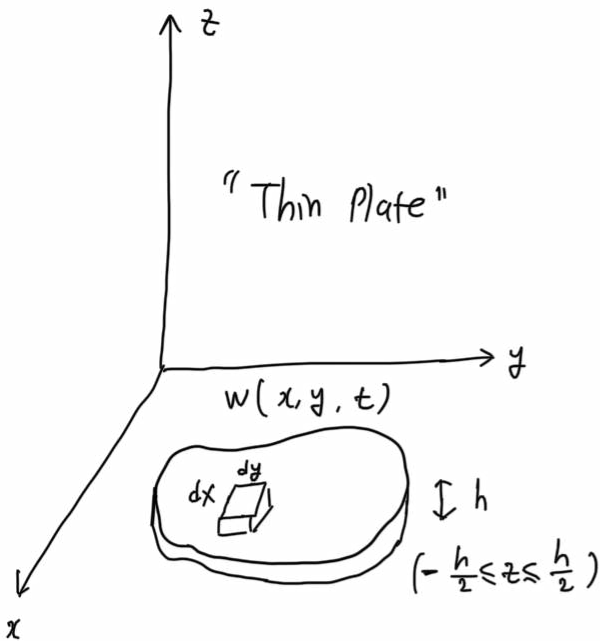
Wave Lecture 22 (20211125)

Waves in plates

kirchhoff plate
Mindlin plate.

- ① Rotational inertia
- ② Shear deformation.

Euler B
Timoshenko



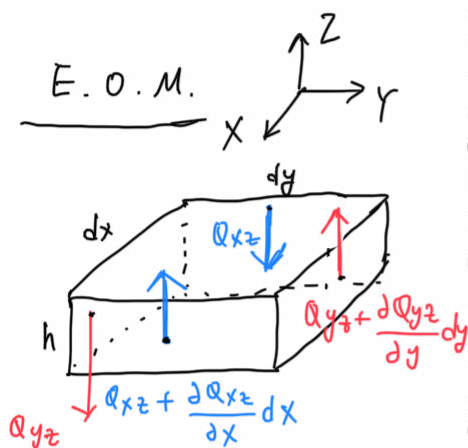
Assumptions

- 1) plate has a "flat" mid-plane denoted as "xy plane"
- 2) Material : Elastic, Isotropic, Homogeneous, and Continuous.
- 3) Transverse deflection $w(x, y, t)$ small ($w \ll 1$)
 $\Rightarrow$ Linear theory.

Only for kirchhoff

- 4) deflection close to a purely translational motion in z -axis (rotational inertia $\times$)
- 5) Line elements perpendicular to the mid-plane remain straight & perpendicular to the midplane during / after deformation.

6) Line elements do not change "Length" ($\epsilon_{zz} = 0$)



Internal bending moment

per length

$$1) M_x dy = - \int_{-\frac{h}{2}}^{\frac{h}{2}} z \sigma_x dy dz$$

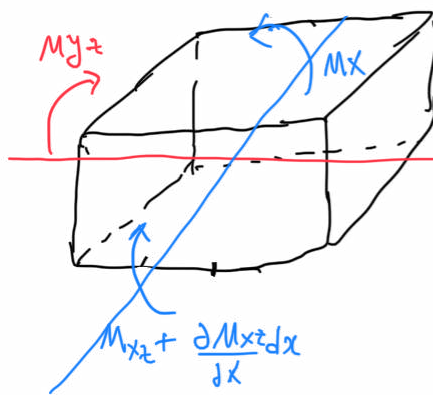
$$2) M_y dx = - \int_{-\frac{h}{2}}^{\frac{h}{2}} z \sigma_y dx dz$$

Internal twisting moment

per length.

$$3) M_{xy} = - \int_{-\frac{h}{2}}^{\frac{h}{2}} \tau_{xy} z dz \quad dy$$

$$4) M_{yx} = + \int_{-\frac{h}{2}}^{\frac{h}{2}} \tau_{yx} z dz \quad dx$$

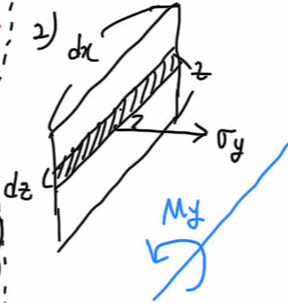
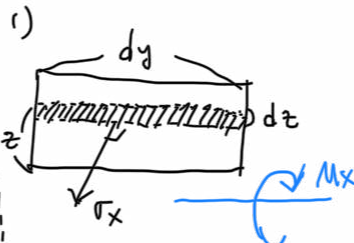


$$M_{xz} \rightarrow M_x$$

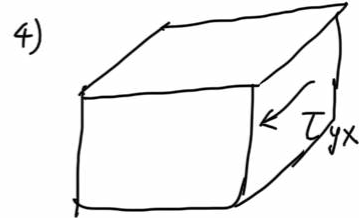
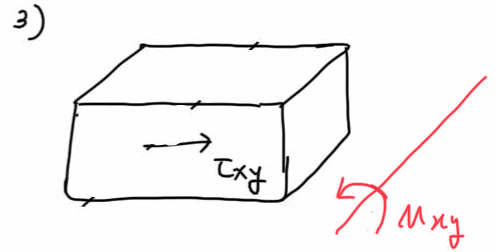
* (Note that M_{xz} (direction of rotation) $\rightarrow$ M_x)

$$M_{yz} + \frac{\partial M_{yz}}{\partial y} dy$$

$$M_{yz} \rightarrow M_y$$

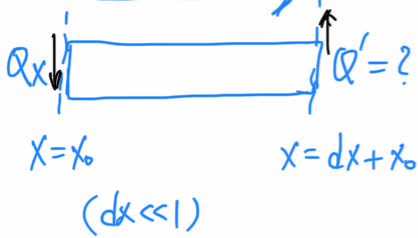


since $\tau_{yx} = \tau_{xy}$
 $\Rightarrow M_{xy} = -M_{yx}$



τ_{yx}

* Note



$$\Rightarrow Q' = Q_x + \frac{\partial Q}{\partial x} \cdot dx$$

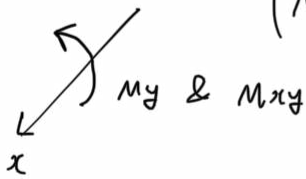
Assuming increase in positive x direction.

① Net force (z)

$$\underbrace{\left(Q_x + \frac{\partial Q_x}{\partial x} dx - Q_x \right) dy + \left(Q_y + \frac{\partial Q_y}{\partial y} dy - Q_y \right) dx}_{\text{per unit length}} = \rho dx dy h \frac{\partial^2 w}{\partial t^2}$$

$$\Rightarrow \frac{\partial Q_x}{\partial x} + \frac{\partial Q_y}{\partial y} = \rho h \frac{\partial^2 w}{\partial t^2} \quad \text{--- ①}$$

② Net moment



$$\left(M_y + \frac{\partial M_y}{\partial y} dy - M_y \right) dx + \left(M_{xy} + \frac{\partial M_{xy}}{\partial x} dx - M_{xy} \right) dy + \left(Q_y + \frac{\partial Q_y}{\partial y} dy + Q_y \right) \frac{dx}{2} dy = J_x = 0.$$



$$\Rightarrow \frac{\partial M_y}{\partial y} + \frac{\partial M_{xy}}{\partial x} + Q_y = 0 \quad \text{--- ②}$$

③ Net moment



$$- \left(M_x + \frac{\partial M_x}{\partial x} dx - M_x \right) dy + \left(M_{yx} + \frac{\partial M_{yx}}{\partial y} dy - M_{yx} \right) dx - \left(Q_x + \frac{\partial Q_x}{\partial x} dx + Q_x \right) \frac{1}{2} dy \cdot dx = J_y = 0$$

$$\Rightarrow - \frac{\partial M_x}{\partial x} + \frac{\partial M_{yx}}{\partial y} - Q_x = 0 \quad \text{--- ③}$$

③ $\rightarrow$ ① $\Rightarrow$ Q equation

② $\rightarrow$ ①

$$\Rightarrow \frac{d}{dx} \left(-\frac{\partial M_x}{\partial x} + \frac{\partial M_{yx}}{\partial y} \right) + \frac{d}{dy} \left(-\frac{\partial M_y}{\partial y} - \frac{\partial M_{xy}}{\partial x} \right) = \rho h \frac{\partial^2 w}{\partial t^2}$$

$$\Rightarrow -\frac{\partial^2 M_x}{\partial x^2} - \frac{\partial^2 M_y}{\partial y^2} - 2 \frac{\partial^2 M_{xy}}{\partial x \partial y} = \rho h \frac{\partial^2 w}{\partial t^2} \quad (\because M_{xy} = -M_{yx})$$

Note : Moment - Curvature Constitutive Eqs.

Just before, we obtained

$$-\frac{\partial^2 M_x}{\partial x^2} - \frac{\partial^2 M_y}{\partial y^2} - 2 \frac{\partial^2 M_{xy}}{\partial x \partial y} = \rho h \frac{\partial^2 w}{\partial t^2}$$

* moment - curvature constitutive equations

$$M_x = - \int_{-\frac{h}{2}}^{\frac{h}{2}} z \sigma_x dz \quad \text{where } \sigma_x = \frac{E}{1-\nu^2} (\epsilon_x + \nu \epsilon_y)$$

$$M_y = - \int_{-\frac{h}{2}}^{\frac{h}{2}} z \sigma_y dz \quad \text{" } \sigma_y = \frac{E}{1-\nu^2} (\epsilon_y + \nu \epsilon_x)$$

$$M_{xy} = - \int_{-\frac{h}{2}}^{\frac{h}{2}} z \tau_{xy} dz \quad \text{" } \tau_{xy} = \frac{E}{2(1+\nu)} \gamma_{xy}$$

under
plane stress
condition

$$\sigma = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & 0 \\ \sigma_{yx} & \sigma_{yy} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

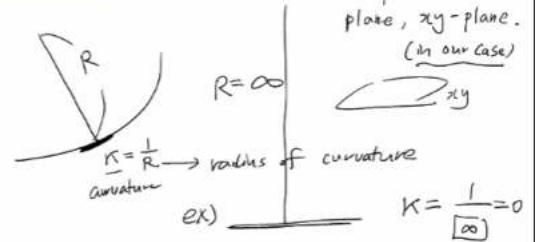
* stress vector
is zero across
a particular
plane, xy-plane.
(in our case)

relations btw

$$M_{x,y,xy} \sim W$$

here, $\epsilon_x \approx -z \frac{\partial^2 w}{\partial x^2}$ (curvature) $\epsilon_y \approx -z \frac{\partial^2 w}{\partial y^2}$ HW

$$\gamma_{xy} \approx -2z \frac{\partial^2 w}{\partial x \partial y}$$



$$\begin{aligned} \text{Then, } M_x &= - \int_{-\frac{h}{2}}^{\frac{h}{2}} z \frac{E}{1-\nu^2} \left(z \frac{\partial^2 w}{\partial x^2} + \nu z \frac{\partial^2 w}{\partial y^2} \right) dz \\ &= \frac{E}{1-\nu^2} \left[\frac{\partial^2 w}{\partial x^2} \int_{-\frac{h}{2}}^{\frac{h}{2}} z^2 dz + \nu \frac{\partial^2 w}{\partial y^2} \int_{-\frac{h}{2}}^{\frac{h}{2}} z^2 dz \right] \\ &= \frac{E h^3}{12(1-\nu^2)} \left(\frac{\partial^2 w}{\partial x^2} + \nu \frac{\partial^2 w}{\partial y^2} \right) \end{aligned}$$

$$\int_{-\frac{h}{2}}^{\frac{h}{2}} z^2 dz = \frac{2}{3} z^3 \Big|_0^{\frac{h}{2}} = \frac{2}{24} h^3 = \frac{h^3}{12}$$

Let $D = \frac{E h^3}{12(1-\nu^2)}$ be "flexural stiffness"

$$M_x = D \left(\frac{\partial^2 w}{\partial x^2} + \nu \frac{\partial^2 w}{\partial y^2} \right)$$

$$M_{xy} = (1-\nu) D \frac{\partial^2 w}{\partial x \partial y}$$

$$M_y = D \left(\nu \frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} \right)$$

$\frac{\partial^2 w}{\partial y^2}$

Let $D = \frac{E h^3}{12(1-\nu^2)}$ be "flexural stiffness"

$\Rightarrow$ EoM: $-D \left(\frac{\partial^4}{\partial x^4} + \frac{\partial^4}{\partial y^4} + 2 \frac{\partial^4}{\partial x^2 \partial y^2} \right) w = \rho h \frac{\partial^2 w}{\partial t^2}$

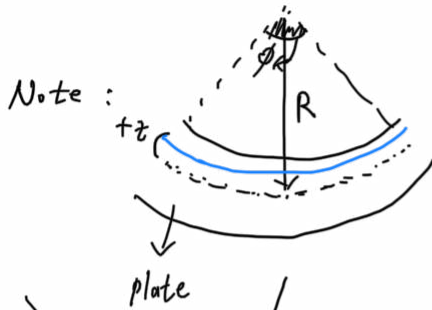
Kirchhoff plate equation $\boxed{-D \nabla^4 w = \rho h \frac{\partial^2 w}{\partial t^2}}$

$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) = \nabla^2 \nabla^2 = \nabla^4$ biharmonic operator

HW: Derive $\epsilon_x, \epsilon_y, \gamma_{xy} \sim w$ relation

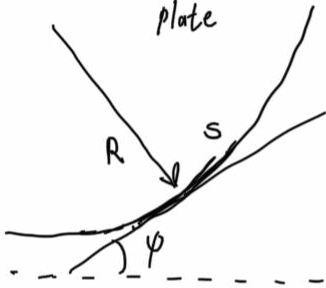
$M_x, M_y, M_{xy} \sim w$ relation

by yourself.



$$\epsilon_x = \frac{\Delta x}{L} = \frac{(R-z)\phi - R\phi}{R\phi} = -\frac{z}{R_x} \quad R = R_x$$

$$\epsilon_y = \frac{\Delta y}{L} = \frac{(R-z)\phi - R\phi}{R\phi} = -\frac{z}{R_y} \quad R = R_y$$



$$d\psi \cdot R = ds \Rightarrow \frac{1}{R} = \frac{\partial \psi}{\partial s}$$

$$\tan \phi = \frac{\partial w}{\partial x} \Rightarrow \sec^2 \psi \frac{\partial \psi}{\partial x} = \frac{\partial^2 w}{\partial x^2}$$

$$ds = \sqrt{1 + \left(\frac{\partial w}{\partial x}\right)^2} dx$$

$$\Rightarrow \frac{\partial \psi}{\partial s} = \frac{\partial \psi}{\partial x} \frac{\partial x}{\partial s} = \frac{\frac{\partial^2 w}{\partial x^2}}{\sec^2 \psi} \cdot \frac{1}{\sqrt{1 + \left(\frac{\partial w}{\partial x}\right)^2}}$$

$$\begin{aligned} \Rightarrow \frac{\partial \psi}{\partial s} &= \kappa_x = \frac{1}{R_x} = \frac{\frac{\partial^2 w}{\partial x^2}}{\left\{1 + \left(\frac{\partial w}{\partial x}\right)^2\right\} \sqrt{1 + \left(\frac{\partial w}{\partial x}\right)^2}} \\ &= \frac{\frac{\partial^2 w}{\partial x^2}}{\left[1 + \left(\frac{\partial w}{\partial x}\right)^2\right]^{3/2}} \end{aligned}$$

Thus, for $(\theta \ll 1 \Leftrightarrow \partial w / \partial x \ll 1)$ case $\Rightarrow \frac{1}{R_x} = \frac{\partial^2 w}{\partial x^2}$ and $\frac{1}{R_y} = \frac{\partial^2 w}{\partial y^2}$.

How about γ_{xy} ?

Wave Lecture 23 (20211130)

In Kirchhoff plate, $-D \nabla^4 w = \rho h \frac{\partial^2 w}{\partial t^2} \Rightarrow \nabla^4 w + \frac{\rho h}{D} \frac{\partial^2 w}{\partial t^2} = 0$

Flexural stiffness

$$\nabla^2 \nabla^2 = \frac{\partial^4}{\partial x^4} + \frac{\partial^4}{\partial y^4} + 2 \frac{\partial^4}{\partial x^2 \partial y^2}$$

Let $w(x, y, t) = W e^{i(g_x x + g_y y - \omega t)}$

where $\vec{g} = (g_x, g_y)$: wave vector.

$$\Rightarrow g^4 W + \frac{\rho h}{D} (-\omega^2) W = 0 \Rightarrow \omega = \sqrt{\frac{D}{\rho h}} g^2 \Rightarrow c_p = \sqrt[4]{\frac{D}{\rho h}} \sqrt{\omega}$$

Let $w(x, y, t) = W e^{i(g_x x + g_y y - \omega t)}$. Then, $\nabla^2 w = (i g_x)^2 w + (i g_y)^2 w = -(g_x^2 + g_y^2) w = -g^2 w$

$$\nabla^4 w = (-g^2) (-g^2) w = g^4 w$$

$$\frac{\partial^2 w}{\partial t^2} = (-i\omega)^2 w = -\omega^2 w$$

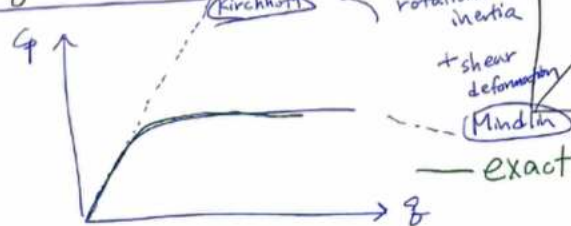
$$g^4 W + \frac{\rho h}{D} (-\omega^2) W = 0$$

$$\Rightarrow g^4 = \frac{\rho h}{D} \omega^2 \Rightarrow \omega = \sqrt{\frac{D}{\rho h}} g^2$$

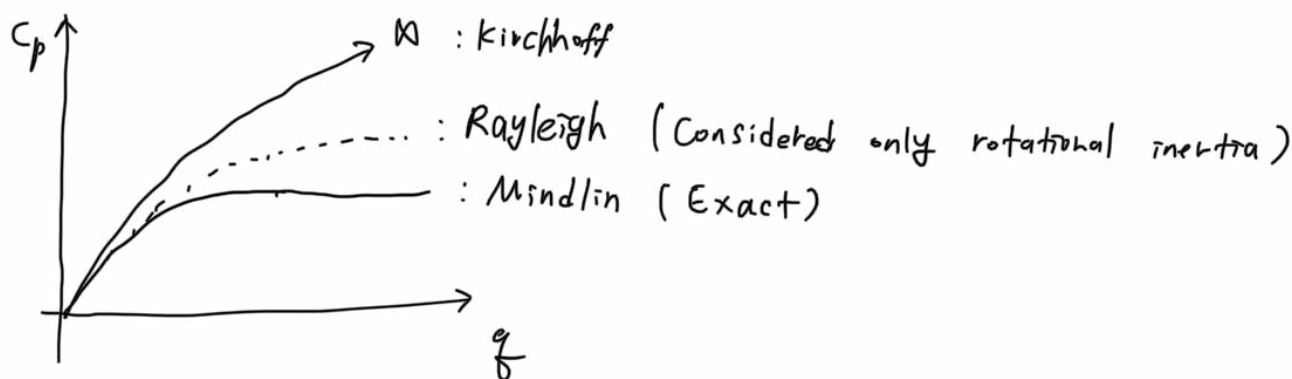
$$\Downarrow$$

$$g = \sqrt[4]{\frac{\rho h}{D}} \sqrt{\omega}$$

$$c_p = \frac{\omega}{g} = \sqrt{\frac{D}{\rho h}} \cdot g = \sqrt[4]{\frac{D}{\rho h}} \sqrt{\omega}$$



rotational inertia + shear deformation
Mindlin
exact
 $\omega \uparrow, c_p \uparrow$
 $\propto \sqrt{\omega}$
w/o bound
 ∞
nonphysical (c.f. E-B)
limitation of K-plate
 $\Downarrow$
Mindlin



* Mindlin plate. (OUT OF SCOPE)

- (iv) rotational inertia considered
- (v)

- (iv) rotational inertia is taken into account.
- (v) Line elements perpendicular to the mid-plane remain straight but not necessarily perpendicular to the mid-plane during and after deformation.

Q: Check Contributions of rotational inertia & shear deformation.

$$\Rightarrow \vec{d} = \underbrace{\nabla \phi}_{\text{Dilatational}} + \underbrace{\nabla \times \psi}_{\text{EquiVolumetric}} \quad \text{Recall Navier's Eq.}$$

$$\rho \ddot{\vec{d}} = (\lambda + \mu) \nabla(\nabla \cdot \vec{d}) + \mu \nabla^2 \vec{d}$$

$$\textcircled{1} \vec{d} = \nabla \phi \quad (\text{curl-free, dilatational}) \quad \textcircled{2} \vec{d} = \nabla \times \psi \quad (\text{div-free, equiVolumetric})$$

$$\rho \ddot{\nabla \phi} = (\lambda + \mu) \nabla(\nabla^2 \phi) + \mu \nabla \nabla^2 \phi$$

$$\rho \nabla \times \ddot{\psi} = (\lambda + \mu) \nabla(\nabla \times \psi) + \mu \nabla^2 \nabla \times \psi$$

$$\Rightarrow \underline{\rho \ddot{\phi} = (\lambda + 2\mu) \nabla^2 \phi} : \underline{\text{P-wave Eq.}} \quad \Rightarrow \underline{\rho \ddot{\psi} = \mu \nabla^2 \psi} : \underline{\text{S-wave Eq.}}$$

B.C.s

① No traction force

$$\sigma_{yy} = \sigma_{yx} = 0$$

② No variation along z axis
(in-plane motion)

$$xy \text{ plane} \rightarrow \underline{\frac{\partial}{\partial z} = 0}$$

$$\vec{d} = \nabla \phi + \nabla \times \psi$$

$$= \begin{pmatrix} \partial \phi / \partial x \\ \partial \phi / \partial y \\ \partial \phi / \partial z \end{pmatrix} + \begin{pmatrix} \frac{\partial \psi_z}{\partial y} - \frac{\partial \psi_y}{\partial z} \\ \frac{\partial \psi_x}{\partial z} - \frac{\partial \psi_z}{\partial x} \\ \frac{\partial \psi_y}{\partial x} - \frac{\partial \psi_x}{\partial y} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{\partial \phi}{\partial x} + \frac{\partial \psi_z}{\partial y} \\ \frac{\partial \phi}{\partial y} - \frac{\partial \psi_z}{\partial x} \\ \frac{\partial \psi_y}{\partial x} - \frac{\partial \psi_x}{\partial y} \end{pmatrix} = \begin{pmatrix} u \\ v \\ w \end{pmatrix} \quad \left\{ \begin{array}{l} \text{Only } \phi, \psi_z \\ \text{are needed} \end{array} \right\}$$

From ①, ②, respectively.

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = \frac{1}{c_1^2} \frac{\partial^2 \phi}{\partial t^2} \quad / \quad \frac{\partial^2 \psi_z}{\partial x^2} + \frac{\partial^2 \psi_z}{\partial y^2} = \frac{1}{c_2^2} \frac{\partial^2 \psi_z}{\partial t^2}$$

Wave Lecture 24 (20211207)

• SAW

$$\rho \ddot{\vec{d}} = (\lambda + \mu) \nabla(\nabla \cdot \vec{d}) + \mu \nabla^2 \vec{d}$$

H.D.T

$$\vec{d} = \nabla \phi + \nabla \times \psi$$

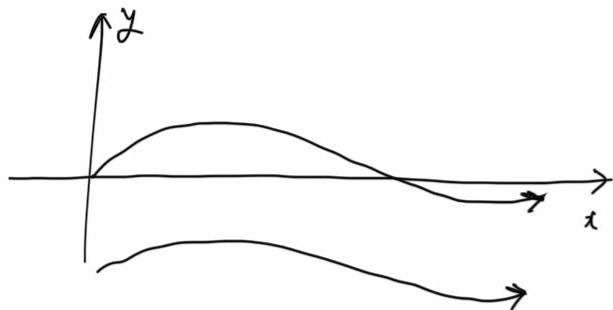
$$\left. \begin{aligned} \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} &= \frac{1}{c_1^2} \frac{\partial^2 \phi}{\partial t^2} \quad (1) \\ \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} &= \frac{1}{c_2^2} \frac{\partial^2 \psi}{\partial t^2} \quad (2) \end{aligned} \right\}$$

B.C.s

$$1) \sigma_{yy} = \lambda \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + 2\mu \frac{\partial v}{\partial y} \quad (3)$$

$$2) \tau_{yx} = G \gamma_{yx} = 2\mu \epsilon_{yx} = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \quad (4)$$

$$\left\{ \begin{aligned} u &= \frac{\partial \phi}{\partial x} + \frac{\partial \psi_2}{\partial y} \\ v &= \frac{\partial \phi}{\partial y} - \frac{\partial \psi_2}{\partial x} \end{aligned} \right.$$



Assume $\phi(x, y, t) = f(y) e^{j(kx - \omega t)}$

$\psi_2(x, y, t) = g(y) e^{j(kx - \omega t)}$

$$(1) \Rightarrow -k^2 f + f'' = -\frac{\omega^2}{c_1^2} f \Rightarrow f'' - \underbrace{\left(k^2 - \left(\frac{\omega}{c_1} \right)^2 \right)}_{= \xi^2} f = 0 \quad \xi^2 > 0 \quad \left(\begin{array}{l} \because \text{if negative,} \\ f = F(\cos, \sin) \\ y \rightarrow -\infty \nRightarrow f = 0 \\ \text{not possible} \end{array} \right)$$

$$(2) \Rightarrow g'' - \underbrace{\left(k^2 - \left(\frac{\omega}{c_2} \right)^2 \right)}_{= \eta^2} g = 0$$

$$f(y) = A e^{-\xi y} + B e^{\xi y}$$

$$g(y) = C e^{-\eta y} + D e^{\eta y}$$

$$\Rightarrow \left\{ \begin{aligned} \phi &= B e^{\xi y} e^{j(kx - \omega t)} \\ \psi_2 &= D e^{\eta y} e^{j(kx - \omega t)} \end{aligned} \right.$$

Note

$$\left\{ \begin{aligned} \xi^2 &= k^2 - \frac{\omega^2}{c_1^2} = k^2 - \frac{\rho \omega^2}{\lambda + 2\mu} \\ \eta^2 &= k^2 - \frac{\omega^2}{c_2^2} = k^2 - \frac{\rho \omega^2}{\mu} \end{aligned} \right.$$

$$\Rightarrow (\lambda + 2\mu) \xi^2 = (\lambda + 2\mu) k^2 - \rho \omega^2$$

$$\mu \eta^2 = \mu k^2 - \rho \omega^2$$

$$\Rightarrow (\lambda + 2\mu) \xi^2 - \lambda k^2 = \mu \eta^2 + \mu k^2 \quad (*)$$

$$\textcircled{3}: (\lambda + 2\mu)(\xi^2 - k^2)B - 2\mu(-k^2B + jkD\eta) = 0$$

$$\Rightarrow [(\lambda + 2\mu)\xi^2 - \lambda k^2]B - 2jkD\mu\eta = 0 \Rightarrow \mu(\eta^2 + k^2)B - 2jk\mu\eta D = 0$$

$\underbrace{\hspace{10em}}_{\therefore (*)}$

$$\textcircled{4} \mu(2\beta jk\xi + \eta^2 D + k^2 D) = 0$$

$$\text{From } \textcircled{3}, \textcircled{4} \Rightarrow \underbrace{\begin{pmatrix} \eta^2 + k^2 & -2jk\eta \\ 2jk\xi & \eta^2 + k^2 \end{pmatrix}}_A \begin{pmatrix} B \\ D \end{pmatrix} = 0 \Rightarrow \det(A) = 0$$

$$\Rightarrow \boxed{(\eta^2 + k^2)^2 - 4k^2\xi\eta = 0}$$

Note that, $\eta^2 = k^2 - \frac{\omega^2}{c_2^2} = k^2\left(1 - \frac{C_R^2}{C_2^2}\right)$ where $C_R = \frac{\omega}{k}$ (phase speed of SAW)

$$\xi^2 = k^2 - \frac{\omega^2}{c_1^2} = k^2\left(1 - \frac{C_R^2}{C_1^2}\right)$$

$$\Rightarrow \left[k^2\left(1 - \frac{C_R^2}{C_2^2}\right) + k^2\right]^2 - 4k^2 \cdot k^2 \sqrt{1 - \frac{C_R^2}{C_1^2}} \sqrt{1 - \frac{C_R^2}{C_2^2}} = 0$$

$$\Rightarrow \boxed{\left(2 - \frac{C_R^2}{C_2^2}\right)^4 = 16\left(1 - \frac{C_R^2}{C_1^2}\right)\left(1 - \frac{C_R^2}{C_2^2}\right)}$$

C_R is independent of frequency (ω)

$$\therefore C_R = C_R(C_1, C_2)$$

Particle Trajectories

Also, $D = \frac{-2jk\xi}{\eta^2 + k^2} \cdot B$

Recall, $\left. \begin{aligned} \phi &= B e^{i\eta y} \cdot e^{j(kx - \omega t)} \\ \psi_z &= D e^{i\eta y} \cdot e^{j(kx - \omega t)} \end{aligned} \right\}$

$$\psi_z = -\frac{2ik\xi}{\eta^2 + k^2} e^{\eta y} e^{i(kx - \omega t)} \quad \checkmark$$

$$\begin{aligned} u &= \frac{\partial \phi}{\partial x} + \frac{\partial \psi_z}{\partial y} = B i k \left[e^{\xi y} - \frac{2 \xi \eta}{\eta^2 + k^2} e^{\eta y} \right] e^{i(kx - \omega t)} \quad \text{a} \\ v &= \frac{\partial \phi}{\partial y} - \frac{\partial \psi_z}{\partial x} = B \xi \left[e^{\xi y} - \frac{2 k^2}{\eta^2 + k^2} e^{\eta y} \right] e^{i(kx - \omega t)} \quad \text{b} \end{aligned}$$

$$u = \frac{\partial \phi}{\partial x} + \frac{\partial \psi}{\partial y} = B_0 k \left[e^{\frac{3y}{2}} - \frac{2}{\gamma^2 + k^2} e^{\gamma y} \right] e^{i(kx - \omega t)}$$

$$\psi = \frac{\partial \phi}{\partial y} - \frac{\partial \psi_2}{\partial x} = B \frac{[e^{zy} - \frac{2k^2}{\eta^2 + k^2} e^{\eta y}]}{\eta_b} e^{i(kx - \omega t)}$$

$$\Rightarrow \left| \frac{u}{a} \right|^2 + \left| \frac{v}{b} \right|^2 = 2. \rightarrow \text{Ellipse.}$$

① oblate

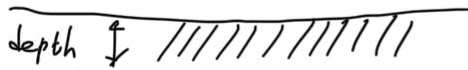
② prolate



① CW

② CCW

- Penetration Depth



$$\frac{u(y)}{u(0)} = \frac{e^{\gamma y} - \frac{2\gamma h}{h^2 + k^2} e^{\delta y}}{1 - \frac{2\gamma h}{h^2 + k^2}} \quad \leftarrow 1$$

$$\frac{V(y)}{V(0)} = \frac{e^{zy} - \frac{2k^2}{\hbar^2 + k^2} e^{\frac{1}{2}y}}{1 - \frac{2k^2}{\hbar^2 + k^2}} \ll 1$$

SAW

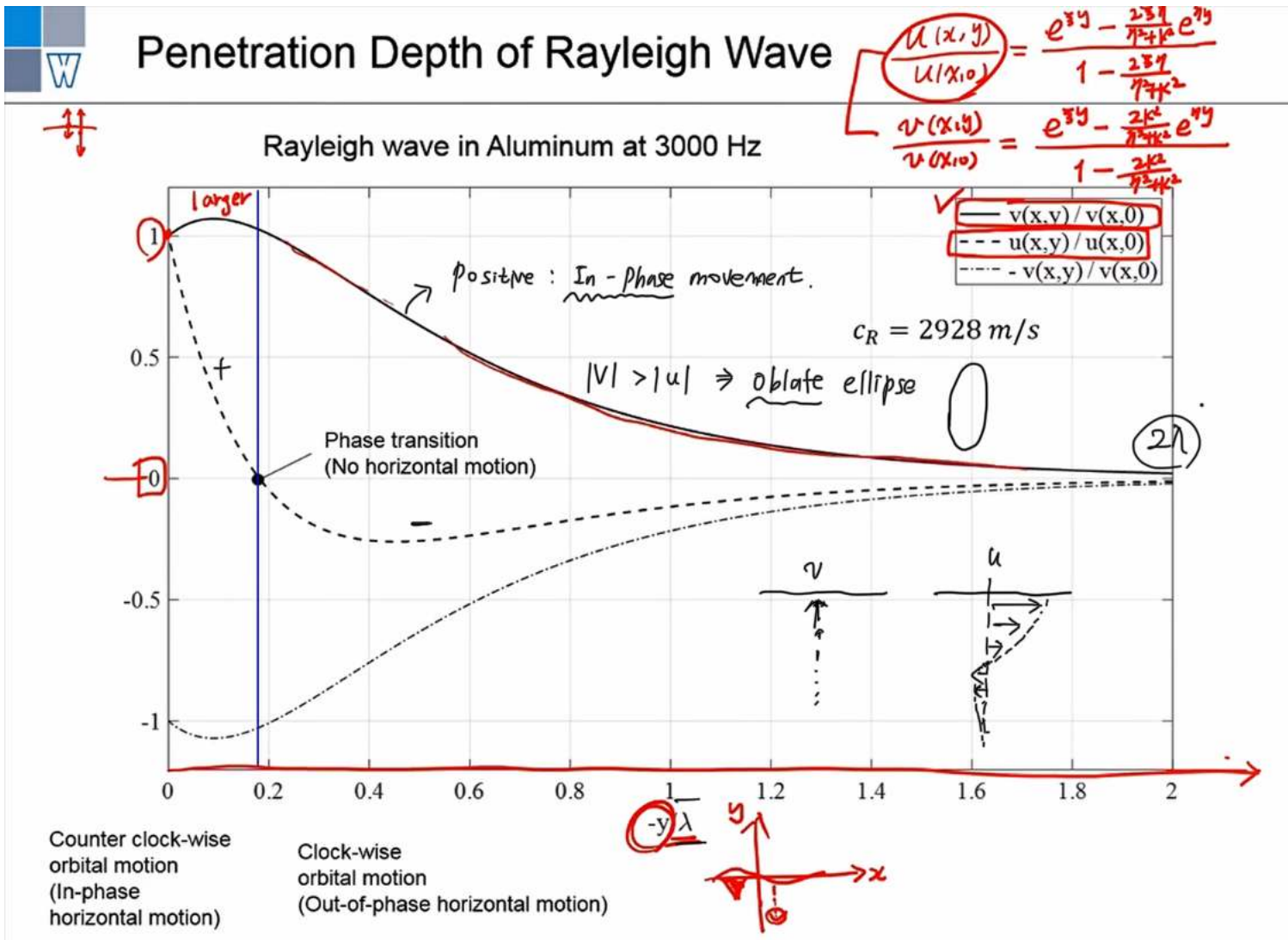
no

HW.

$$\frac{u(y)}{u(0)} = \frac{e^{\xi y} - \frac{2\xi\eta}{\eta^2 + k^2} e^{\eta y}}{1 - \frac{2\xi\eta}{\eta^2 + k^2}}$$

$$\frac{v(y)}{v(0)} = \frac{e^{\xi y} - \frac{2k^2}{\eta^2 + k^2} e^{\eta y}}{1 - \frac{2k^2}{\eta^2 + k^2}}$$

Shap



Aluminium vs Rubber @ 3000 Hz

